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Zeta and L-functions in Number Theory and Combinatorics

Wen-Ching Winnie Li

AMS American Mathematical Society

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Preface

The zeta and L-functions play a central role in number theory. They provide important information of arithmetic nature. For example, the analytic behavior of the Riemann zeta function $\zeta(s)$ on the closed right half-plane $\Re(s) \geq 1$ leads to the estimate of the number of prime numbers up to x by the logarithmic integral $\mathrm{Li}(x)$, and the error of this estimate is controlled by the location of the zeros of $\zeta(s)$ in the critical strip $0 < \Re(s) < 1$. The celebrated Riemann Hypothesis asserts that these zeros should all lie on the line of symmetry $\Re(s) = 1/2$. Under this assumption one can bound the error by $O(x^{1/2} \log x)$. This is the Prime Number Theorem. Furthermore, given a finite Galois extension of $\mathbb{Q}$ with Galois group G, we can partition the prime numbers according to their associated Frobenius conjugacy classes in G. For a conjugacy class C of G, denote by S_C the collection of prime numbers whose associated Frobenius conjugacy classes equal C. The distribution of prime numbers in S_C is described by the Chebotarev Density Theorem, which says that S_C has natural density $|C|/|G|$. In particular, the prime numbers are equidistributed among the sets S_C when the Galois group G is abelian. Specialized to cyclotomic extensions of $\mathbb{Q}$, this is Dirichlet's Theorem on primes in arithmetic progressions. The Chebotarev Density Theorem follows from the holomorphy and nonvanishing of the Artin L-functions attached to nontrivial irreducible representations of G on the half-plane $\Re(s) \geq 1$.

On the geometric side, to a d-dimensional projective smooth irreducible variety V defined over a finite field $\mathbb{F}_q$, Artin and Smith attached the zeta function $Z(V, u)$, which counts points on V with coordinates in finite extensions of $\mathbb{F}_q$. As shown by Grothendieck, $Z(V, u)$ is an alternating product of polynomials $P_i(u)$, $0 \leq i \leq 2d$, arising geometrically; and Deligne proved the Riemann Hypothesis as conjectured by Weil; that is, the zeros of a nonconstant $P_i(u)$ have absolute value $q^{-i/2}$. Like the Riemann zeta function, $Z(V, u)$ has an Euler product over closed points, which play the role of primes of V. When V is a curve, the analytic behavior of $Z(V, u)$ and similarly defined Artin L-functions give rise to analogues of the Prime Number Theorem and the Chebotarev Density Theorem.

These remarkable achievements in number theory are reviewed in Lecture 1. The purpose of this monograph is to provide a systematic and comprehensive account of the developments of these topics in geometry and combinatorics for graduate students and researchers. We shall highlight interactions between number theory and other fields and compare similarities and dissimilarities under different settings. This is done in chronological order. Lecture 2 introduces the first instance, considered by Selberg in the 1950s, on compact Riemann surfaces arising as quotients of the upper half-plane $\mathfrak{H}$ by discrete torsion-free cocompact subgroups Γ of $SL_2(\mathbb{R})$. In this setting the primes of $\Gamma\backslash\mathfrak{H}$ are primitive closed geodesic cycles not

counting the starting points, and the fundamental group Γ plays the role of absolute Galois group. Selberg introduced zeta functions attached to finite-dimensional representations of Γ as suitable products over such primes. These functions have nice analytic properties like $\zeta(s)$, and they satisfy the Riemann Hypothesis for all except possibly finitely many real zeros in the critical strip. The Prime Geodesic Theorem and the Chebotarev Density Theorem for $\Gamma\backslash\mathfrak{H}$ are established by Huber and Sarnak.

Lecture 3 is a digression to finite-dimensional compact Riemannian manifolds and the connection to dynamical systems. We shall see that the distribution of primes is related to the analytic behavior of the associated L-functions of Artin type, which are variations of the Ruelle zeta function in dynamical systems. While this is a much explored topic in dynamical systems, our exposition will stay close to the theme of this monograph.

The remainder of this monograph is devoted to the combinatorial setting. In the 1960s, by interpreting the upper half-plane $\mathfrak{H}$ as the homogeneous space $PGL_2(\mathbb{R})/PO_2(\mathbb{R})$, Ihara extended Selberg's results from the real field $\mathbb{R}$ to a nonarchimedean local field F with q elements in its residue field. The upper half-plane is replaced by a $(q+1)$-regular tree, known as the building of $PGL_2(F)$, on which $PGL_2(F)$ acts. The quotient of the tree by a discrete torsion-free cocompact subgroup of $PGL_2(F)$ is hence a finite $(q+1)$-regular graph X, whose primes are primitive closed geodesic cycles up to starting points, similar to the Riemann surfaces discussed in Lecture 2. The Ihara zeta function $Z(X,u)$ of X counts closed geodesic cycles on X; as such, it can be expressed as a product over the primes of X, like the zeta function of a curve reviewed in Lecture 1. Serre observed that the same definition applies to all finite graphs. In Lecture 4 we study different closed-form expressions of the Ihara zeta function as a rational function in u. Lecture 5 is devoted to the spectral theory for regular graphs. We shall see that, for a $(k+1)$-regular graph X, its zeta function satisfies the Riemann Hypothesis in the sense that its nontrivial poles have the same absolute value (which is $k^{-1/2}$) if and only if X is a Ramanujan graph; that is, its eigenvalues other than $\pm(k+1)$ fall in the spectrum of its universal cover. Ramanujan graphs are extremal expanders with nice properties and wide applications. Explicit constructions of infinite families of $(k+1)$-regular Ramanujan graphs are introduced in Lecture 6. For k equal to a prime power, the construction by Margulis and independently by Lubotzky-Phillips-Sarnak in the 1980s is number-theoretic. The existence of such families for general $k \geq 3$ was established by Marcus-Spielman-Srivastava in 2015 for bipartite graphs using combinatorial and analytical means. It is still an open question to find infinite families of nonbipartite Ramanujan graphs for general k. Lecture 7 deals with Artin L-functions on graphs. It is shown that, as before, good analytic behavior of zeta and L-functions on X leads to the Prime Geodesic Theorem and the Chebotarev Density Theorem on the distribution of primes of graphs.

Graphs are 1-dimensional simplicial complexes. Lecture 8 concerns extensions to higher dimensional simplicial complexes obtained in this century. This is a budding and rapidly evolving research area. Two main themes are considered in this lecture. The first one is the generalization from Ramanujan graphs based on $PGL_2(F)$ to Ramanujan complexes based on $PGL_n(F)$ developed in the early 2000s. There have been several explicit number-theoretic constructions of infinite families of Ramanujan complexes as finite quotients of the building of $PGL_n(F)$.

We shall present the one given by Sarveniazi in 2007. To understand the combinatorial properties and to find applications of these Ramanujan complexes are currently active research areas in mathematics and computer science. The second theme is on zeta and L-functions and prime distributions for 2-dimensional simplicial complexes X_Γ arising as finite quotients by discrete torsion-free cocompact subgroups Γ of the building of $G = PGL_3(F)$ and $PGSp_4(F)$, with the results obtained in the past decade. In this situation for each $i \in \{1, 2\}$, there are two types of i-dimensional simplices and hence two zeta functions counting i-dimensional closed geodesics of X_Γ using simplices of a given type. These zeta functions as well as L-functions of Artin type attached to finite-dimensional irreducible representations of Γ have similar analytic behavior as their counterparts for graphs, and they imply similar and more refined Prime Geodesic Theorems and Chebotarev Density Theorems for primes of X_Γ. In the case that X_Γ is a Ramanujan complex for $PGL_3(F)$, a good error term in each estimate is also obtained. It should be pointed out that a suitable alternating product of these four zeta functions gives rise to the Langlands L-function for $L^2(\Gamma\backslash G)$, which is reminiscent of the zeta function for a surface over a finite field. It is natural to seek similar results for other p-adic groups of Lie type as well as real Lie groups of rank at least 2.

This monograph grew out of the lectures and courses I gave during the years 2014–2017 on various occasions. More materials are supplemented at each stage to broaden the scope. It started with the 10 lectures on combinatorial zeta and L-functions delivered at the NSF-CBMS Regional Research Conference in the Mathematical Sciences, May 12–16, 2014, at the Sundance Resort, Utah. I am indebted to the organizers Jasbir Chahal and Michael Barrus from Brigham Young University for their invitation and hard work, the National Science Foundation, the Conference Board of the Mathematical Sciences, and Brigham Young University for their financial support, and the participants for their enthusiasm and feedback. Special thanks are due to Steve Butler and Alia Hamieh for leading discussion groups and making daily lecture notes available to the participants the next day and to Steve Butler for typing the lecture notes. This manuscript is the skeleton of the Distinguished Lecture Series at the National Tsing Hua University, Taiwan, from May to July of 2015, the one semester graduate course at the Pennsylvania State University in the fall of 2016, and finally a two-month short course at the University of Hong Kong from May to July of 2017. I would like to express my gratitude to the Mathematics Department of the National Tsing Hua University in Taiwan, the Institute of Mathematical Research at the University of Hong Kong, and the Simons Foundation in the USA, which supported the research and writing of this work.

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Wen-Ching W. Li

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LECTURE 1

Number theoretic zeta and L-functions

1.1. The Riemann zeta function

The fundamental theorem of arithmetic states that every integer greater than one can be factored as a product of primes, unique up to ordering. This fact is reflected analytically on the Riemann zeta function defined for $\Re s > 1$ as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots\right) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

It is clear that $\zeta(s)$ is holomorphic and non-vanishing on the half-plane $\Re s > 1$ where both infinite series and infinite product converge absolutely. In [**81**] Riemann showed that the function $\xi(s) := \pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s)$ extends to a meromorphic function on the whole complex plane $\mathbb{C}$, holomorphic everywhere with the exception of simple poles at $s = 0$ and 1, and it satisfies the functional equation $\xi(s) = \xi(1 - s)$. Here $\Gamma(s)$ is defined on the right half-plane $\Re s > 0$ by the integral

$$\Gamma(s) = \int_0^{\infty} e^{-t}\ t^{s-1} dt$$

as a holomorphic function which never vanishes. Using integration by parts repeatedly, one extends $\Gamma(s)$ to $\mathbb{C}$ as a nonvanishing meromorphic function with simple poles at $s = 0, -1, -2, \cdots$. The nonvanishing of $\Gamma(s)$ in turn implies that $\zeta(s)$, being the quotient $\xi(s)/\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)$, has an analytic continuation to the whole s-plane which is holomorphic everywhere except for a simple pole at $s = 1$.

Further, it follows from the functional equation of $\xi(s)$ and the location of simple poles of $\Gamma(s)$ that $\zeta(s)$ has simple zeros at $s = -2, -4, -6, \ldots$. These are called the trivial zeros of $\zeta(s)$. The remaining zeros are called the nontrivial zeros of the Riemann zeta function, which are symmetric with respect to the line $\Re s = \frac{1}{2}$. They fall in the vertical strip $0 < \Re s < 1$ since $\zeta(s)$ is shown to be nonzero on the line $\Re s = 1$ and hence also on the line $\Re s = 0$ by symmetry. Determining the location of the nontrivial zeros remains one the largest open problems in Mathematics.

Conjecture 1 (Riemann hypothesis). *All nontrivial zeros of $\zeta(s)$ lie on the line $\Re s = \frac{1}{2}$.*

The Riemann Hypothesis has deep connections with the distribution of prime numbers. We recall the prime number theorem which describes the asymptotic behaviour of the prime counting function

$$\pi(x) = \#\{p \leq x : p \text{ prime}\}.$$

Write $f(x) \sim g(x)$ as $x \to a$ whenever $\lim_{x \to a} \frac{f(x)}{g(x)} = 1$.

THEOREM 1.1 (Prime number theorem).

$$\pi(x) \sim x/\log x \qquad as\ x \to \infty.$$

Equivalently, the Chebychev function

$$\psi(x) := \sum_{n \le x} \Lambda(n) \sim x \qquad as\ x \to \infty,$$

where Λ is the von Mangoldt function defined by

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime } p \text{ and integer } k \ge 1, \\ 0 & \text{otherwise.} \end{cases}$$

This theorem was proved in 1896 by Hadamard [**32**] and de la Vallée Poussin [**17**], independently.

More interestingly, one has a more precise information of $\psi(x)$ with an estimate of the error term related to the location of the nontrivial zeros of $\zeta(s)$:

$$\psi(x) = x + O(x^{\beta+\varepsilon}) \quad \forall \varepsilon > 0 \quad \text{when } x \text{ is large,}$$

where $\beta = \max\{\Re s : \zeta(s) = 0\}$. If the Riemann Hypothesis holds, then for large x we have

$$\pi(x) = \mathrm{Li}(x) + O\big(x^{1/2}\log x\big),$$

where

$$\mathrm{Li}(x) = \int_2^x \frac{1}{\log t}\, dt = \frac{x}{\log x} + \frac{1!x}{(\log x)^2} + \frac{2!x}{(\log x)^3} + \cdots,$$

and the last equality follows from applying integration by parts repeatedly. Alternatively, this consequence can be stated as

$$\psi(x) = x + O\big(x^{1/2}(\log x)^2\big) \quad \text{as } x \to \infty.$$

1.2. The Dirichlet L-functions

Let N be a positive integer. A Dirichlet character of modulus N is a homomorphism $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \to S^1$. Extend it to a function on $\mathbb{Z}$ by letting $\chi(n) = 0$ for all $(n, N) \ne 1$. As such it satisfies $\chi(m)\chi(n) = \chi(mn)$ for all $n, m \in \mathbb{Z}$. In particular, χ restricted to positive integers is a strongly multiplicative function. Call χ a principal character modulo N if $\chi(n) = 1$ whenever $(n, N) = 1$.

Dirichlet introduced these characters and used them to select arithmetic progressions. Given a Dirichlet character χ, the Dirichlet L-function attached to χ is defined as a variation of the Riemann zeta function by

$$L(\chi, s) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - \chi(p)p^{-s}}.$$

Notice that the factors at $p|N$ are equal to 1. Thus for the principal character χ modulo N, $L(\chi, s)$ is equal to the Riemann zeta function with the factors at $p|N$ removed. A Dirichlet character χ modulo N is called *primitive* if N is its smallest period. If the smallest period of χ is a proper divisor M of N, then χ is imprimitive and it factors through a primitive Dirichlet character χ' modulo M and $L(\chi, s)$ is obtained from $L(\chi', s)$ by removing the factors at primes p dividing N but not M. Hence it suffices to understand the Dirichlet L-functions attached to primitive and non-principal characters.

A Dirichlet L-function shares many properties in common with the Riemann zeta function. For example, $L(\chi, s)$ converges absolutely to a holomorphic function on the half-plane $\Re s > 1$. If χ is a non-principal character, then the series converges conditionally in the strip $0 < \Re s \leq 1$. For a non-principal and primitive character χ, after multiplied by a suitable $\Gamma-$function, $L(\chi, s)$ can be extended to an entire function on the s-plane which satisfies a functional equation relating $L(\chi, s)$ to $L(\chi^{-1}, 1-s)$. Having an Euler product expression implies $L(\chi, s) \neq 0$ for $\Re s > 1$. Further, $L(\chi, s)$ can be shown to be nonvanishing on the line $\Re s = 1$. Similar to the Riemann zeta function, the trivial zeros of $L(\chi, s)$, which arise from the poles of the Γ-factor, lie on the left half-plane $\Re s \leq 0$, and there is a similar conjecture on the location of the nontrivial zeros of $L(\chi, s)$:

CONJECTURE 2 (Riemann hypothesis). *For a primitive Dirichlet character χ, the nontrivial zeros of $L(\chi, s)$, which lie in the critical strip $0 < \Re s < 1$, are on the line $\Re s = \frac{1}{2}$.*

Using these Dirichlet L-functions one can show that the prime numbers are uniformly distributed among arithmetic progressions. To explain this, notice that given a modulus $m > 0$, if the residue r is not coprime to m, then the arithmetic progression $\{r + nm : n \in \mathbb{Z}\}$ contains at most one prime number. Therefore all but finitely many primes are distributed among the $\phi(m)$ arithmetic progressions with residues coprime to m. Here ϕ denotes Euler's totient function. The following theorem, proved by de la Valleé Poussin, says that the set of primes in each of such arithmetic progressions has density $1/\phi(m)$.

THEOREM 1.2 (Dirichlet's theorem on primes in arithmetic progressions). *Let r and m be two positive integers which are coprime. Then*

$$\#\{\text{primes } p \leq x : p \equiv r \mod m\} \sim \frac{1}{\phi(m)} \frac{x}{\log x} \quad \text{as } x \to \infty.$$

1.3. Finite extensions of $\mathbb{Q}$

A *number field* K is a field extension of $\mathbb{Q}$ with finite degree $[K : \mathbb{Q}] =: n$. The role of $\mathbb{Z}$ in $\mathbb{Q}$ is played by the set

$$\mathcal{O}_K = \{x \in K : x \text{ is integral over } \mathbb{Z}\},$$

which is shown to be a subring of K, called the the *ring of integers* in K. Recall that an element x is *integral* over $\mathbb{Z}$ if it satisfies a monic polynomial with coefficients in $\mathbb{Z}$. Like $\mathbb{Z}$, the ring $\mathcal{O}_K$ is a Noetherian integral domain in which every nonzero prime ideal is maximal, and each element in the field K can be written as a quotient of two integral elements. Unlike $\mathbb{Z}$, the ring $\mathcal{O}_K$ is not necessarily a unique factorization domain (UFD).

However, Dedekind showed that each nonzero ideal of $\mathcal{O}_K$ is a finite product of maximal ideals, unique up to order. Since each maximal ideal is a prime ideal, we see that the maximal ideals in $\mathcal{O}_K$ are precisely the nonzero prime ideals of $\mathcal{O}_K$. They are called the "primes" for K.

Each nonzero ideal $\mathfrak{A} \subset \mathcal{O}_K$, viewed as an additive group, has a finite index in $\mathcal{O}_K$, called the norm of $\mathfrak{A}$ and denoted by $N\mathfrak{A} = [\mathcal{O}_K : \mathfrak{A}]$.

With this in place, we define the Dedekind zeta function for K as

$$\zeta_K(s) = \sum_{\substack{0 \neq \mathfrak{A} \subseteq \mathcal{O}_K \\ \text{ideal}}} \frac{1}{(N\mathfrak{A})^s} = \prod_{\mathcal{P} \text{ prime}} \frac{1}{1-(N\mathcal{P})^{-s}},$$

which converges absolutely for $\Re s > 1$. Similar to the Riemann zeta function, the Dedekind zeta function $\zeta_K(s)$ is holomorphic and non-vanishing on $\Re s > 1$, can be extended to a meromorphic function on the whole s-plane which is holomorphic everywhere except for a simple pole at $s = 1$, and satisfies a functional equation relating $\zeta_K(s)$ and $\zeta_K(1-s)$. The trivial zeros of $\zeta_K(s)$ lie in the left half-plane $\Re s \leq 0$ and there is a similar conjecture regarding the non-trivial zeros of the Dedekind zeta function:

CONJECTURE 3 (Generalized Riemann hypothesis). *The zeros of $\zeta_K(s)$ in the critical strip $0 < \Re s < 1$ lie on the line $\Re s = \frac{1}{2}$.*

Concerning the asymptotic behavior of the prime ideals, we have a similar statement.

THEOREM 1.3 (Landau's prime ideal theorem).

$$\#\{\textit{ideal } \mathfrak{A} \subseteq \mathcal{O}_K \mid N(\mathfrak{A}) \leq x\} \sim \frac{x}{\log x} \quad \textit{as } x \to \infty.$$

Equivalently,

$$\sum_{\substack{\textit{ideal } \mathfrak{A} \subseteq \mathcal{O}_K \\ N\mathfrak{A} \leq x}} \Lambda(\mathfrak{A}) \sim x \qquad \textit{as } x \to \infty,$$

where

$$\Lambda(\mathfrak{A}) = \begin{cases} \log(N\mathcal{P}) & \textit{if } \mathfrak{A} = \mathcal{P}^k \textit{ for some prime ideal } \mathcal{P} \textit{ and } k \geq 1; \\ 0 & \textit{else.} \end{cases}$$

The free abelian group generated by the primes of K contains the subgroup of "principal fractional ideals" $\alpha\mathcal{O}_K$ with $\alpha \in K^\times$. The quotient is the *ideal class group* Cl_K of K. Minkowski proved that Cl_K is a finite abelian group. Its order, h_K, is called the *class number* of K. It measures how far off $\mathcal{O}_K$ is from being a principal ideal domain (PID). Since $\mathcal{O}_K$ is a PID if and only if it is a UFD, h_K also measures the failure of $\mathcal{O}_K$ being a UFD.

We now turn to the distribution of primes in an ideal class.

THEOREM 1.4. *Given an ideal class $C \in Cl_K$, one has*

$$\#\{\textit{ideal } \mathfrak{A} \subseteq \mathcal{O}_K \mid N(\mathfrak{A}) \leq x \textit{ and } \mathfrak{A} \in C\} \sim \frac{1}{h_K} \frac{x}{\log x} \quad \textit{as } x \to \infty.$$

In other words,

$$\sum_{\substack{\textit{ideal } \mathfrak{A} \subseteq \mathcal{O}_K \\ N\mathfrak{A} \leq x,\ \mathfrak{A} \in C}} \Lambda(\mathfrak{A}) \sim \frac{x}{h_K} \qquad \textit{as } x \to \infty.$$

Hence the primes are uniformly distributed among the ideal classes.

Let $\mathcal{P}$ be a prime of $\mathcal{O}_K$. Then the quotient $\mathcal{O}_K/\mathcal{P}$ is a finite field of characteristic, say, p. This shows that p and hence $p\mathcal{O}_K$ is contained in $\mathcal{P}$. Furthermore

p is the only prime of $\mathbb{Z}$ contained in $\mathcal{P}$. The primes of $\mathcal{O}_K$ containing p are those occurring in the factorization of the ideal $p\mathcal{O}_K$:

$$p\mathcal{O}_K = (\mathcal{P}_1)^{e_1}(\mathcal{P}_2)^{e_2}\cdots(\mathcal{P}_r)^{e_r}, \tag{1}$$

where $\mathcal{P}_1, \ldots, \mathcal{P}_r$ are distinct primes and all $e_i \geq 1$. The situation is depicted as follows.

$$\begin{array}{ccccc} K & \supset & \mathcal{O}_K & \supset & \mathcal{P}_1, \ldots, \mathcal{P}_r \\ | & & & \diagup & \\ \mathbb{Q} & \supset & \mathbb{Z} & \supset & (p) \end{array}$$

Write p^{f_i} for the cardinality of the finite field $\mathcal{O}_K/\mathcal{P}_i$. Taking norm of both sides of (1) and using the fact that norm is multiplicative, we obtain

$$p^n = N(p\mathcal{O}_K) = (N\mathcal{P}_1)^{e_1}\cdots(N\mathcal{P}_r)^{e_r} = p^{e_1f_1+\cdots+e_rf_r},$$

which implies $n = [K:\mathbb{Q}] = \sum_{i=1}^r e_i f_i$.

We say that p is *unramified* in K if $e_i = 1$ for all $i \in \{1, 2, \cdots, r\}$. Otherwise, we say that p is ramified in K. Only finitely many prime numbers p ramify in K. In fact, p ramifies in K if and only if p divides the discriminant d_K of K. The prime p is said to *split completely* in K if $e_1 = \cdots = e_r = 1$ and $f_1 = \cdots = f_r = 1$, in other words, $r = [K:\mathbb{Q}]$.

Given a number field K, to get a sense of the proportion of prime numbers p which split completely in K, we introduce the concept of Dirichlet density. The Riemann zeta function $\zeta(s)$ has a simple pole at $s = 1$ with residue 1. Thus as $s \to 1^+$, $\zeta(s)$ behaves like $\frac{1}{s-1}$. Taking logarithm of both functions and using the Euler product expression for $\zeta(s)$, we restate the behavior as

$$\sum_{p \text{ prime}} \frac{1}{p^s} \sim \log\frac{1}{s-1} \qquad \text{as } s \to 1^+.$$

A subset S of prime numbers is said to have *Dirichlet density* δ if

$$\sum_{p\in S} \frac{1}{p^s} \sim \delta \ \log\frac{1}{s-1} \qquad \text{as } s \to 1^+,$$

or equivalently,

$$\frac{\sum_{p\in S} \frac{1}{p^s}}{\sum_{p \text{ prime}} \frac{1}{p^s}} \sim \delta \qquad \text{as } s \to 1^+.$$

Therefore a finite set S has Dirichlet density 0, and the set of all prime numbers has Dirichlet density 1. In general, the Dirichlet density δ of S, if it exists, falls between 0 and 1.

THEOREM 1.5. (1) *Let K be a Galois extension of $\mathbb{Q}$ of degree n. Then the set of primes p splitting completely in K has Dirichlet density $1/n$.*

(2) *Two finite Galois extensions K_1 and K_2 of $\mathbb{Q}$ are isomorphic if and only if the set of primes p splitting completely in K_1 agrees with that in K_2.*

The first statement implies that there are infinitely many prime numbers splitting completely in a finite Galois extension of $\mathbb{Q}$, and the second statement says that the finite Galois extensions of $\mathbb{Q}$ are characterized by the set of completely splitting prime numbers. This important fact is also referred to as a form of reciprocity law. The theorem above holds in general when the base field $\mathbb{Q}$ is replaced by a number field F and the prime numbers replaced by the primes in F.

Let $K/\mathbb{Q}$ be a degree n Galois extension. Now we turn our attention to the action of the Galois group on the prime ideals of K. Let p be a rational prime. The Galois group $\mathrm{Gal}(K/\mathbb{Q})$ preserves $p\mathcal{O}_K$ and acts transitively on the maximal ideals $\mathcal{P}_1, \ldots, \mathcal{P}_r \subset \mathcal{O}_K$ lying above p. Hence, we have $e_1 = e_2 = \cdots = e_r = e$ and $f_1 = f_2 = \cdots = f_r = f$. Therefore $n = ref$.

The stabilizer of $\mathcal{P}_i$ in $\mathrm{Gal}(K/\mathbb{Q})$, called a *decomposition group* and denoted by $D_{\mathcal{P}_i/p}$, is given by

$$D_{\mathcal{P}_i/p} = \big\{\sigma \in \mathrm{Gal}(K/\mathbb{Q}) : \sigma(\mathcal{P}_i) = \mathcal{P}_i\big\}.$$

Since $\mathrm{Gal}(K/\mathbb{Q})$ acts transitively on $\mathcal{P}_1, \cdots, \mathcal{P}_r$, as the primes $\mathcal{P}_i$ vary, the subgroups $D_{\mathcal{P}_i/p}$ are conjugate to each other.

Fix $\mathcal{P} = \mathcal{P}_i$ for some $1 \le i \le r$. An automorphism $\sigma \in D_{\mathcal{P}/p}$ induces an automorphism $\bar{\sigma}$ on the quotient $\mathcal{O}_K/\mathcal{P}$ leaving $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ pointwise fixed. The map $\sigma \mapsto \overline{\sigma}$ gives rise to a surjective homomorphism

$$D_{\mathcal{P}/p} \to \mathrm{Gal}\left((\mathcal{O}_K/\mathcal{P})/\mathbb{F}_p\right)$$

whose kernel is called the *inertia subgroup* at $\mathcal{P}$, denoted by $I_{\mathcal{P}/p}$. Thus we obtain the following short exact sequence:

$$0 \to I_{\mathcal{P}/p} \to D_{\mathcal{P}/p} \to \mathrm{Gal}\left((\mathcal{O}_K/\mathcal{P})/\mathbb{F}_p\right) \to 0.$$

Since

$$|D_{\mathcal{P}/p}| = \frac{n}{r} \qquad \text{and} \qquad \left|\mathrm{Gal}\left((\mathcal{O}_K/\mathcal{P})/\mathbb{F}_p\right)\right| = f = \frac{n}{er},$$

the inertia subgroup $I_{\mathcal{P}/p}$ has order e. If p is unramified in K, then $I_{\mathcal{P}/p}$ is trivial and $D_{\mathcal{P}/p}$ is isomorphic to $\mathrm{Gal}\left((\mathcal{O}_K/\mathcal{P})/\mathbb{F}_p\right)$.

For the finite field $\mathcal{O}_K/\mathcal{P}$, its Galois group $\mathrm{Gal}\left((\mathcal{O}_K/\mathcal{P})/\mathbb{F}_p\right)$ is cyclic, generated by the Frobenius automorphism $\mathrm{Frob} : x \mapsto x^p$. The pull-back of Frob in $D_{\mathcal{P}/p}$ is called a Frobenius element $\mathrm{Frob}_{\mathcal{P}/p}$, which is unique up to multiplication by $I_{\mathcal{P}/p}$. Thus $\mathrm{Frob}_{\mathcal{P}/p}$ is well-defined if p is unramified in K. Moreover, in this case the conjugacy class in $\mathrm{Gal}(K/\mathbb{Q})$ of the Frobenius element $\mathrm{Frob}_{\mathcal{P}/p}$ is independent of the choice of $\mathcal{P}$, which we denote by Frob_p.

Example 1.1. Let $K = \mathbb{Q}(i)$. Then $\mathcal{O}_K = \mathbb{Z}[i]$ is the ring of Gaussian integers with discriminant $d_K = -4$. It is known that $\mathcal{O}_K$ is a Euclidean domain and hence is a PID. So the class number $h_K = 1$. It is also clear that $K/\mathbb{Q}$ is Galois with Galois group G consisting of the identity and complex conjugation. We discuss the behaviours of the primes of $\mathbb{Z}$ in $\mathcal{O}_K$.

- 2 is the only prime ramified in K. We have $2\mathcal{O}_K = (1+i)^2 = \mathcal{P}^2$ and the inertia subgroup $I_{\mathcal{P}/2} = G$.
- The prime numbers $p \equiv 1 \mod 4$ split completely in K. This is because such primes can be written as a sum of two squares. For example, $5\mathcal{O}_K = (2+i)(2-i)$ and $13\mathcal{O}_K = (2+3i)(2-3i)$. Hence, a decomposition group $D_{\mathcal{P}/p} = \{id\}$ and we have $\mathrm{Frob}_p = \{\mathrm{id}\}$.

- For prime numbers $p \equiv 3 \mod 4$, $p\mathcal{O}_K$ is a prime of K, and so $D_{\mathcal{P}/p} = G$ and Frob_p is equal to complex conjugation.

The Galois group G of a finite Galois extension K of $\mathbb{Q}$ is finite. Hence G has only finitely many conjugacy classes. On the other hand, all but finitely many prime numbers p are unramified in K, to each such p we have associated the Frobenius conjugacy class Frob_p of G. It is natural to ask how the infinitely many Frob_p distribute among the finitely many conjugacy classes of G. The answer is the following celebrated uniform distribution theorem by Chebotarev.

THEOREM 1.6 (Chebotarev density theorem). *Let K be a finite Galois extension of $\mathbb{Q}$ with $\mathrm{Gal}(K/\mathbb{Q}) = G$. Let C be a conjugacy class of G. Then the set*

$$\{p \text{ prime unramified in } K : \mathrm{Frob}_p = C\}$$

has Dirichlet density $\frac{|C|}{|G|}$.

In fact, the Chebotarev Density Theorem also holds in terms of the stronger natural density, that is,

$$\frac{\#\{p \leq x \text{ prime unramified in } K : \mathrm{Frob}_p = C\}}{\#\{\text{prime } p \leq x\}} \sim \frac{|C|}{|G|} \quad \text{as } x \to \infty.$$

An estimate of the error term is given by Lagarias-Odlyzko in [**56**].

We make two important remarks regarding this theorem. First, if G is abelian, that is, $K/\mathbb{Q}$ is a finite abelian extension, then each conjugacy class is a singleton. In particular, the cyclotomic field $K = \mathbb{Q}(\mu_m)$ with μ_m a primitive m^{th} root of 1 is a finite abelian extension of $\mathbb{Q}$. An element $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$ is uniquely determined by the image $\sigma(\mu_m)$, which is again a primitive m^{th} root of 1, of the form μ_m^r for some integer $r = r(\sigma)$ coprime to m. The map $\sigma \mapsto r(\sigma)$ yields an isomorphism $\mathrm{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$. A prime number p not dividing m is unramified in $\mathbb{Q}(\mu_m)$. For such p, it follows from the definition of the Frobenius element that $\mathrm{Frob}_p(\mu_m) = \mu_m^p$ so that Frob_p is identified with $p \in (\mathbb{Z}/m\mathbb{Z})^\times$. Thus given a residue r coprime to m, we have

$$\{p \nmid m, \mathrm{Frob}_p = r \text{ in } (\mathbb{Z}/m\mathbb{Z})^\times\} = \{p \equiv r \mod m\}.$$

This shows that for $K = \mathbb{Q}(\mu_m)$, the Chebotarev Density Theorem agrees with Dirichlet's Theorem on primes in arithmetic progressions with modulus m. Thus the Chebotarev Density Theorem is a generalization of Dirichlet's Theorem to arbitrary finite Galois extensions of $\mathbb{Q}$.

Second, a prime p splits completely in K if and only if $p\mathcal{O}_K = \mathcal{P}_1 \cdots \mathcal{P}_n$ where $n = [K : \mathbb{Q}]$. This holds if and only if $D_{\mathcal{P}/p} = \{\mathrm{id}\}$, which is equivalent to $\mathrm{Frob}_p = \{\mathrm{id}\}$. So a consequence of the Chebotarev Density Theorem is the following

COROLLARY 1.6.1. *Let $K/\mathbb{Q}$ be a Galois extension of degree n. Then the set of prime numbers splitting completely in K has natural and hence Dirichlet density $\frac{1}{n}$.*

1.4. The Artin L-functions

For a subfield F of $\bar{\mathbb{Q}}$, denote by G_F the absolute Galois group $\mathrm{Gal}(\bar{\mathbb{Q}}/F)$. Let $K/\mathbb{Q}$ be a finite Galois extension with Galois group G. Then G_K is a normal subgroup of $G_\mathbb{Q}$ with the quotient $G_\mathbb{Q}/G_K \cong G$. An n-dimensional representation of G is a homomorphism $\rho : G \to \mathrm{Aut}(V)$ for an n-dimensional vector space V

over $\mathbb{C}$. It is also viewed as a representation of $G_{\mathbb{Q}}$ trivial on G_K. Observe that $\mathrm{Aut}(V) \cong GL_n(\mathbb{C})$. The dimension n is called the *degree* of ρ. Since G is finite, ρ is unitary.

To ρ we associate the Artin L-function $L(\rho, s)$ defined for $\Re s > 1$ by

$$L(\rho, s) = \prod_p \frac{1}{\det(I_n - \rho(\mathrm{Frob}_{\mathcal{P}/p})p^{-s}\big|_{V^{I_{\mathcal{P}/p}}})}.$$

Here, I_n denotes the $n \times n$ identity matrix. For each prime number p, choose a prime ideal $\mathcal{P}$ of $\mathcal{O}_K$ containing p, $V^{I_{\mathcal{P}/p}}$ is the subspace of V fixed by the inertia subgroup $I_{\mathcal{P}/p}$ on which the actions of all Frobenius elements in the decomposition group $D_{\mathcal{P}/p}$ agree. As we change the choice of $\mathcal{P}$ dividing p, the actions of $\rho(\mathrm{Frob}_{\mathcal{P}/p})$ differ by conjugation, hence after taking determinant, the expression is independent of the choice of $\mathcal{P}$ and thus each factor at p is well-defined. When p is unramified in K, which is the case for almost all primes, the inertia subgroup $I_{\mathcal{P}/p}$ is trivial so that $V^{I_{\mathcal{P}/p}} = V$ and we may replace $\mathrm{Frob}_{\mathcal{P}/p}$ by Frob_p to mean any element in the conjugacy class Frob_p.

The function $L(\rho, s)$ is holomorphic on $\Re s > 1$. Brauer showed that $L(\rho, s)$ has a meromorphic continuation to $\mathbb{C}$ which satisfies a functional equation relating $L(\rho, s)$ and $L(\rho^\vee, 1 - s)$, where $\rho^\vee$ denotes the contragradient representation of ρ. Note that when ρ is the trivial representation of G, i.e., the trivial representation $\mathbf{1}_{G_{\mathbb{Q}}}$ of $G_{\mathbb{Q}}$, then $L(\mathbf{1}_{G_{\mathbb{Q}}}, s)$ is nothing but the Riemann zeta function $\zeta(s)$. More generally, by class field theory, a one-dimensional representation ρ of G corresponds to a primitive Dirichlet character $\chi = \chi(\rho)$ so that $L(\rho, s)$ agrees with the Dirichlet L-function $L(\chi, s)$. Thus $L(\mathbf{1}_{G_{\mathbb{Q}}}, s)$ is holomorphic except for one simple pole at $s = 1$ as discussed before, while for a nontrivial one-dimensional ρ, the L-function $L(\rho, s)$ is holomorphic everywhere. Artin conjectured that the same should hold for higher dimensional irreducible representations ρ as well. This is stated in

Conjecture 4 (Artin conjecture). *$L(\rho, s)$ is holomorphic if ρ is irreducible and nontrivial.*

Similar definition, conjecture and results apply to representations of G_F for any number field F. Thus Artin L-functions encompass all zeta and L-functions we have encountered so far. It follows from the definition that $L(\rho_1 \oplus \rho_2, s) = L(\rho_1, s)L(\rho_2, s)$. Furthermore, the Artin L-function is shown to be invariant under induction, that is, for a finite Galois extension K over $\mathbb{Q}$ with Galois group G, if the representation ρ of G is induced from the representation γ of a subgroup H of G, then $L(\rho, s) = L(\gamma, s)$. Apply this to $H = \{\mathrm{id}\}$, the identity subgroup of G, and $\gamma = \mathbf{1}_{\{\mathrm{id}\}}$, the trivial representation of $\{\mathrm{id}\}$. Then $\mathrm{Ind}_{\{\mathrm{id}\}}^G \mathbf{1}_{\{\mathrm{id}\}}$ is the regular representation of G, which decomposes into the direct sum of the irreducible representations ρ of G which occurs with multiplicity equal to its degree $\deg \rho$. By regarding $\mathbf{1}_{\{\mathrm{id}\}}$ as the trivial representation $\mathbf{1}_{G_K}$, we have $\mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}} \mathbf{1}_{G_K} \cong \mathrm{Ind}_{\{\mathrm{id}\}}^G \mathbf{1}_{\{\mathrm{id}\}}$. It follows from the properties of the Artin L-functions discussed above that

$$\begin{aligned} \zeta_K(s) &= L(\mathbf{1}_{G_K}, s) = L(\mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}} \mathbf{1}_{G_K}, s) \\ &= \prod_{\rho \in \mathrm{Irr}(G)} L(\rho, s)^{\deg \rho} = \zeta(s) \prod_{\substack{\rho \in \mathrm{Irr}(G) \\ \rho \neq \mathbf{1}_G}} L(\rho, s)^{\deg \rho}. \end{aligned} \tag{2}$$

Here $\mathrm{Irr}(G)$ denotes the set of irreducible representations of G. We say that $\zeta(s)$ divides $\zeta_K(s)$ if the quotient $\zeta_K(s)/\zeta(s)$ is holomorphic. Thus, $\zeta(s)$ divides $\zeta_K(s)$

if the Artin Conjecture holds. In particular, when G is abelian, all irreducible representations of G are 1-dimensional and the Artin conjecture holds. Therefore $\zeta(s)$ divides $\zeta_K(s)$ for an abelian extension $K/\mathbb{Q}$. In fact it holds for finite Galois extensions of number fields independent of the Artin Conjecture, as shown by H. Aramata and R. Brauer [**13**]. K. Uchida [**102**] and R. W. van der Waall [**103**] independently extended this to the case where K is contained in a solvable normal closure of the base field. The reader is referred to [**107**] for further improvements and discussions. The question remains open for arbitrary number field extensions.

The equation (2) and the Artin Conjecture together imply that if the Generalized Riemann Hypothesis holds for the Dedekind zeta function $\zeta_K(s)$ for K, then the Artin L-functions $L(\rho, s)$ attached to all irreducible representations ρ of $\mathrm{Gal}(K/\mathbb{Q})$ have their nontrivial zeros lie on the line $\Re s = \frac{1}{2}$.

Since the Artin L-function is invariant under induction, the Artin Conjecture is valid for irreducible representations induced from nontrivial degree 1 representations of a subgroup of G. The image of a degree-2 representation of G in $PGL_2(\mathbb{C})$ is a finite group isomorphic to either a cyclic group, or a dihedral group, or the alternating group A_4 on 4 letters, or the symmetric group S_4 on 4 letters, or the alternating group A_5 on 5 letters. These possibilities are called the types of the degree-2 Galois representations. Those of cyclic type are reducible, and those of dihedral type are induced, hence the analytic behavior of their associated Artin L-functions is known, as discussed above. For the remaining three types, the Artin conjecture is shown to hold for odd representations, namely the action of the complex conjugation has trace 0. This was established by Langlands [**57**] for type A_4 and Tunnell [**101**] for type S_4 representations using base change. Taylor et al [**14**, **98**] proved the conjecture for type A_5 representations under some conditions; the unconditional conclusion follows from the settlement of Serre's modularity conjecture [**89**] by Kahre and Wintenberger [**51**] and Kisin [**52**]. The odd degree-2 irreducible representations ρ of $G_{\mathbb{Q}}$ with conductor N are in bijection with weight-1 normalized cuspidal newforms $f = f(\rho)$ of level N such that the Artin L-function $L(\rho, s)$ associated to ρ agrees with the L-function $L(f, s)$ attached to the modular form f. The case of even degree-2 irreducible representations remains open. In general the Artin conjecture has close ties to the Langlands program, which is one of the central themes in number theory in the 21st century.

1.5. Zeta functions for varieties over finite fields

Let V be a smooth, irreducible, projective variety defined over a finite field $\mathbb{F}_q$. For a field k let $V(k)$ be the set of points on V with coordinates in k. For each $n \geq 1$, denote by $N_n(V)$ the cardinality of $V(\mathbb{F}_{q^n})$.

The zeta function for V is defined for $|u| < \frac{1}{q}$ as

$$Z(V, u) := \exp\left(\sum_{n=1}^{\infty} \frac{N_n(V)}{n} u^n\right) = \prod_{v \text{ closed point}} \frac{1}{1 - u^{\deg v}}.$$

We now explain the second expression. The variety V is defined by finitely many homogeneous polynomial equations $f_j = 0$ over $\mathbb{F}_q$. Let $P = (x_0 : x_1 : \cdots : x_r)$ be a point on V with coordinates in $\mathbb{F}_{q^n}$. Let $\sigma \in \mathrm{Gal}\left(\overline{\mathbb{F}}_q/\mathbb{F}_q\right)$. We have $f_j(P) = 0$ and $f_j^{\sigma}(P^{\sigma}) = f_j(P^{\sigma}) = 0$. In other words, if $P \in V(\mathbb{F}_{q^n})$, then so does P^{σ} for all $\sigma \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$. The Galois orbit of P is called a *closed point*, and the number of

points in the orbit is its degree. The second equality can be easily verified by taking logarithmic derivative of both sides. The closed points play the role of "primes".

Now assume that V is a curve defined over $\mathbb{F}_q$. Denote by K the field of $\mathbb{F}_q$-rational functions on V. The constant functions in K are identified with $\mathbb{F}_q$ and K has transcendence degree one over $\mathbb{F}_q$. It is called a function field of one variable or simply a function field when there is no danger of ambiguity. For a closed point v, set $Nv = q^{\deg v}$. Then the zeta function for V with variable $u = q^{-s}$ appears like the Dedekind zeta function for the function field K:

$$Z(V, q^{-s}) = \prod_{v \text{ closed point}} \frac{1}{1 - (Nv)^{-s}} = \zeta_K(s).$$

EXAMPLE 1.2. Let $V = \mathbb{P}^1$ be the projective line over $\mathbb{F}_q$. It is the affine line $\mathbb{A}^1$ together with the point at infinity ∞. Then $N_n(V) = q^n + 1$ and

$$\begin{aligned} Z(\mathbb{P}^1, u) &= \exp\left(\sum_{n=1}^{\infty} \frac{q^n}{n} u^n + \sum_{n=1}^{\infty} \frac{1}{n} u^n\right) \\ &= \exp\left(\log \frac{1}{1-qu} + \log \frac{1}{1-u}\right) = \frac{1}{(1-u)(1-qu)}. \end{aligned}$$

The field K of $\mathbb{F}_q$-rational functions on $\mathbb{P}^1$ is the field $\mathbb{F}_q(x)$. The closed point containing ∞ has degree 1, again denoted by ∞. Let α be a point in $\mathbb{A}^1(\overline{\mathbb{F}_q})$, and let $\mathbb{F}_{q^n}$ be the smallest field containing α. Then the irreducible polynomial of α over $\mathbb{F}_q$ is a monic polynomial $P_\alpha(x)$ of degree n. Note that the roots of $P_\alpha(x)$ are precisely the Galois conjugates of α, so we may identify the closed point of $\mathbb{P}^1$ containing α with the polynomial $P_\alpha(x)$. Conversely, given any monic irrducible polynomial $P(x)$, its roots form a Galois orbit and hence $P(x)$ corresponds to a closed point of $\mathbb{P}^1$. We have shown that the "primes" of $\mathbb{P}^1$ can be identified with the monic irreducible polynomials in the polynomial ring $\mathbb{F}_q[x]$, or equivalently the maximal ideals of $\mathbb{F}_q[x]$, together with the rational function $1/x$ whose zero is the point at ∞. Therefore

$$Z(\mathbb{P}^1, u) = \frac{1}{1-u} \cdot \prod_{P \in \mathbb{F}_q[x] \text{ monic irreducible}} \frac{1}{1 - u^{\deg P}}.$$

Here the first factor corresponds to the closed point ∞.

For a curve V of genus g, its function field K is a finite extension of $\mathbb{F}_q(x)$. F. K. Schmidt and E. Artin [**5**] showed that

$$Z(V, u) = \frac{P(V, u)}{(1-u)(1-qu)} \qquad \text{for } |u| < \frac{1}{q},$$

where $P(V, u)$ is a polynomial in $\mathbb{Z}[u]$ of degree $2g$, and $P(V, 0) = 1$. This rational function expression gives an analytic continuation of $Z(V, u)$. It has two simple poles, at $u = q^{-1}$ and $u = 1$. In terms of s, the poles of $Z(V, q^{-s})$ occur at $s \in 1 + \frac{2\pi\sqrt{-1}}{\log q}\mathbb{Z}$ (corresponding to $u = q^{-1}$) and $s \in 0 + \frac{2\pi\sqrt{-1}}{\log q}\mathbb{Z}$ (corresponding to $u = 1$). There is a functional equation which relates $Z(V, u)$ and $Z(V, \frac{1}{qu})$, and hence $P(V, u)$ and $P(V, \frac{1}{qu})$ in view of the symmetry of the denominator of $Z(V, u)$. Comparing the above expressions for $Z(\mathbb{P}^1, u)$ and $Z(V, u)$, we see clearly that the zeta function of a curve V defined over $\mathbb{F}_q$ is divisible by the zeta function of the projective line over $\mathbb{F}_q$.

Factor

$$P(V,u)=\prod_{j=1}^{2g}(1-\alpha_j u).$$

The Riemann hypothesis asserts that $|\alpha_j|=q^{1/2}$. (As a function in s, it says that the zeros of $Z(V,q^{-s})$ lie on $\Re s=\frac{1}{2}$.) This result is due to Hasse [**36**] for $g=1$ and Weil [**105**] for arbitrary genus g.

The free abelian group generated by the closed points of V is called the group of divisors of V. The degree of a divisor $\sum_v n_v v$, where $n_v\in\mathbb{Z}$ and $n_v=0$ for almost all v, is $\sum_v n_v \deg v$. The subgroup $\mathrm{Div}^0(V)$ of degree zero divisors on V contains all principal divisors which record the zeros and poles of the functions in the function field K of V. The quotient

$$\mathrm{Jac}(V)=\mathrm{Div}^0(V)/\{\text{principal divisors}\}$$

is an abelian variety over $\mathbb{F}_q$ of dimension g, called the Jacobian of V. As shown in Theorem 1, Chap. 5 of [**63**], $P(V,1)=|\mathrm{Jac}(V)(\mathbb{F}_q)|$, which is the class number h_K of the function field K of V.

It is easy to see that

$$\mathrm{Res}_{u=1}\,Z(V,u)=\frac{P(V,1)}{q-1}=\frac{h_K}{q-1}.$$

Note that $q-1$ is the number of roots of unity in K. This is parallel to the residue at $s=0$ of the Dedekind zeta function of the number field K multiplied by the associated Γ-factors, which is equal to $\frac{-h_K R_K}{w_K}$ with w_K being the number of roots of unity in K and R_K the regulator of K.

Now consider the analog of the Prime Number Theorem for the curve V defined over $\mathbb{F}_q$, or equivalently, the function field K of $\mathbb{F}_q$-rational functions on V. We have

THEOREM 1.7 (Prime number theorem for a curve V over $\mathbb{F}_q$). *For integer m large,*

$$\sum_{\substack{v\ \text{closed point}\\ \deg v=m}}1\sim\frac{q^m}{m},$$

and

$$\sum_{\substack{v\ \text{closed point}\\ \deg v<m}}1\sim\frac{q^m}{(q-1)m}.$$

Let K be a function field of one variable with the field of constants $\mathbb{F}_q$, and let K' be a finite Galois extension of K with Galois group G. Then all but finitely many closed points v of K are unramified in K', and for such a v we can define the conjugacy class Frob_v in G as in §3. There is a version of the Chebotarev Density Theorem for function fields.

THEOREM 1.8 (Chebotarev density theorem for function fields). *Let K' be a finite Galois extension of the function field K with the same field of constants as K. Let C be a conjugacy class of the Galois group $G=\mathrm{Gal}(K'/K)$. Then the set of closed points v of K with $\mathrm{Frob}_v=C$ has natural (and hence Dirichlet) density $\frac{|C|}{|G|}$.*

It should be pointed out that the Chebotarev Density Theorem in natural density does not hold if the field of constants of K' is larger than that of K; in this case the theorem holds only in Dirichlet density.

For a finite Galois extension K'/K of function fields with the same field of constants, let ρ be a representation of the Galois group G. Define the Artin L-function $L(\rho, u)$ the same way as in §4. The Artin Conjecture over function fields holds, namely $L(\rho, u)$ is a polynomial in u for ρ irreducible and nontrivial. As before, the degree 1 case follows from class field theory. For higher degree representations, this deep result was proved by Drinfeld [**19**] for $\deg \rho = 2$ and generalized by L. Lafforgue [**55**] to arbitrary degree by establishing the correspondence between irreducible Galois representations of degree n and automorphic representations of $GL(n)$ over the function field K as envisioned by Langlands [**58**]. It follows from an analogue of identity (2) that the zeta function of K divides the zeta function of K'. Further, since the Riemann Hypothesis holds for zeta functions of smooth irreducible projective curves over $\mathbb{F}_q$, it holds for Artin L-functions $L(\rho, u)$ with ρ irreducible and nontrivial. We note that when the field of constants of K' properly contains the field of constants of K, the Artin conjecture holds except for the 1-dimensional representations ρ of G which are unramified twists of the trivial representation of G. In this case $L(\rho, u) = Z(V, cu)$ for a nonzero constant c. Hence it has poles.

Now consider curves over finite fields arising from the modular curve $X_0(N)$ for the congruence subgroup

$$\Gamma_0(N) = \Big\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \ : \ a, b, c, d \in \mathbb{Z}, ad - bc = 1, N|c \Big\}.$$

It is a smooth irreducible projective curve defined over $\mathbb{Q}$ with genus $g_0(N)$. It is known that $X_0(N)$ has good reduction modulo primes p not dividing N and the reduced curve $X_0(N)_{/\mathbb{F}_p}$ has the same genus $g_0(N)$ (cf. [**42**]). Its zeta function is

$$Z\big(X_0(N)_{/\mathbb{F}_p}, u\big) = \frac{P(X_0(N)_{/\mathbb{F}_p}, u)}{(1-u)(1-pu)},$$

in which the numerator can be written as

$$P(X_0(N)_{/\mathbb{F}_p}, u) = \prod_{i=1}^{g_0(N)} (1 - \lambda_i(p)u + pu^2).$$

The values $\lambda_1(p), \ldots, \lambda_{g_0(N)}(p)$ have a deep connection with modular forms as indicated in the following celebrated theorem.

THEOREM 1.9 (Eichler-Shimura [**21**, **93**], Igusa [**42**]). *For primes p which do not divide N, the values $\lambda_1(p), \ldots, \lambda_{g_0(N)}(p)$ are the eigenvalues of the Hecke operator T_p acting on the space of weight 2 cusp forms for $\Gamma_0(N)$.*

As a consequence, each $\lambda_j(p)$, being the sum of two reciprocal zeros of $Z(X_0(N)_{/\mathbb{F}_p}, u)$, satisfies $|\lambda_j(p)| \leq 2\sqrt{p}$. This fact will be used in Lecture 6 when we construct Ramanujan graphs.

Finally consider the general case that the variety V over $\mathbb{F}_q$ has dimension d. The behavior of its zeta function described below was the famous Weil conjectures [**106**] proposed in 1949. Grothendieck [**6**, **30**, **31**] proved that $Z(V,u)$ is a rational function of the form

$$Z(V,u) = \frac{P_1(V,u)P_3(V,u)\cdots P_{2d-1}(V,u)}{P_0(V,u)P_2(V,u)\cdots P_{2d}(V,u)},$$

where

$$P_i(V,u) = \det(I - \Phi^{(i)}u|H^i(V))$$

is a polynomial in $\mathbb{Z}[u]$, obtained as the reciprocal characteristic polynomial of $\Phi^{(i)}$, induced from the Frobenius morphism Φ on V sending a point $P = (x_0 : x_1 : \cdots : x_r) \in V(\overline{\mathbb{F}_q})$ to $\Phi(P) = (x_0^q : x_1^q : \cdots : x_r^q) \in V(\overline{\mathbb{F}_q})$, acting on the ith étale cohomology $H^i(V)$ of V. The Riemann Hypothesis asserts that for $0 \leq i \leq 2d$, the roots of $P_i(V,u)$ have absolute value $q^{-i/2}$, extending what was known for curves. This was proved by Deligne [**18**] for $d \geq 2$.

10.1090/cbms/129/02

LECTURE 2

The Selberg zeta function

2.1. The action of $SL_2(\mathbb{R})$

The group $SL_2(\mathbb{R})$ of 2×2 real matrices with determinant 1 acts on $\mathbb{C} \cup \{\infty\}$ by fractional linear transformations. More precisely,

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R})$$

sends $z \in \mathbb{C} \cup \{\infty\}$ to

$$\gamma \cdot z := \frac{az + b}{cz + d}.$$

As elements in $SL_2(\mathbb{R})$ have positive determinants, the upper half-plane

$$\mathfrak{H} = \{x + iy : x, y \in \mathbb{R}, y > 0\}$$

is preserved by the action of $SL_2(\mathbb{R})$. Observe that the map

$$SL_2(\mathbb{R}) \to \mathfrak{H}$$

given by

$$\gamma \mapsto \gamma \cdot i$$

is surjective since for every $x + iy \in \mathfrak{H}$, we have

$$x + iy = \begin{pmatrix} \sqrt{y} & x\sqrt{y}^{-1} \\ 0 & \sqrt{y}^{-1} \end{pmatrix} \cdot i.$$

The stabilizer of i in $SL_2(\mathbb{R})$ is the special orthogonal group

$$SO_2(\mathbb{R}) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \right\} \cong S^1.$$

Hence we may identify $\mathfrak{H}$ with $SL_2(\mathbb{R})/SO_2(\mathbb{R})$.

2.2. Classification of elements in $SL_2(\mathbb{R})$

The characteristic polynomial of an element $\gamma \in SL_2(\mathbb{R})$ is $\det(xI - \gamma) = x^2 - (\operatorname{tr}\gamma)x + 1 \in \mathbb{R}[x]$. We algebraically classify the elements γ in $SL_2(\mathbb{R})$ into four types.

(1) γ is *elliptic* if and only if its eigenvalues are distinct complex conjugates. This occurs if and only if $|\operatorname{tr}\gamma| < 2$.
(2) γ is *hyperbolic* if and only if it has two distinct real eigenvalues. This happens if and only if $|\operatorname{tr}\gamma| > 2$.
(3) γ is *parabolic* if and only if it has repeated eigenvalues and one-dimensional eigenspace. This occurs if and only if $\operatorname{tr}\gamma = \pm 2$ and $\gamma \neq \pm I_2$ with I_2 the identity 2×2 matrix.

(4) γ is *central* if and only if it has repeated eigenvalues and two-dimensional eigenspace. This occurs if and only if γ is $\pm I_2$.

We examine this classification more closely. Let α be an eigenvalue of $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $SL_2(\mathbb{R})$ with a nonzero eigenvector $\begin{pmatrix} u_\alpha \\ v_\alpha \end{pmatrix}$. Since $\det \gamma = 1$ is the product of the two eigenvalues of γ, we conclude that $\alpha \neq 0$ and $\beta = 1/\alpha$ is also an eigenvalue of γ. In case that $\alpha \neq \beta$, let $\begin{pmatrix} u_\beta \\ v_\beta \end{pmatrix}$ be a nonzero eigenvector of γ with eigenvalue β. It follows from the definition of eigenvectors that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} u_\alpha \\ v_\alpha \end{pmatrix} = \begin{pmatrix} au_\alpha + bv_\alpha \\ cu_\alpha + dv_\alpha \end{pmatrix} = \begin{pmatrix} \alpha u_\alpha \\ \alpha v_\alpha \end{pmatrix},$$

which implies

$$au_\alpha + bv_\alpha = \alpha u_\alpha \qquad \text{and} \qquad cu_\alpha + dv_\alpha = \alpha v_\alpha.$$

So if $v_\alpha \neq 0$, then $z = u_\alpha/v_\alpha \in \mathbb{C}$ and

$$\gamma \cdot z = \frac{az+b}{cz+d} = \frac{au_\alpha + bv_\alpha}{cu_\alpha + dv_\alpha} = \frac{\alpha u_\alpha}{\alpha v_\alpha} = \frac{u_\alpha}{v_\alpha} = z.$$

In other words, $z = u_\alpha/v_\alpha$ is a fixed point of γ if $v_\alpha \neq 0$. If $v_\alpha = 0$, then $u_\alpha \neq 0$, $a = \alpha \neq 0$, and $c = 0$. In this case $z = u_\alpha/v_\alpha = \infty$ and $\gamma \cdot z = a/c = \infty = z$. Therefore $z = u_\alpha/v_\alpha$ is a fixed point of γ in both cases.

Thus in case that γ has two distinct eigenvalues, we have found two distinct fixed points of γ in $\mathbb{C} \cup \{\infty\}$. Now we discuss the number of fixed points γ can have.

Case (I) $c \neq 0$. Then $a/c \neq \infty$ so that ∞ is not a fixed point of γ. All fixed points z of γ lie in $\mathbb{C}$. The relation $\gamma \cdot z = z$ implies that z satisfies the quadratic equation $az+b = cz^2+dz$, hence γ has two fixed points in $\mathbb{C}$ unless the discriminant of the equation $(d-a)^2 + 4bc = (d+a)^2 - 4(ad-bc) = (\operatorname{tr}\gamma)^2 - 4$ vanishes, i.e., γ is parabolic, in which case γ has only one fixed point, which lies in $\mathbb{R}$.

Case (II) $c = 0$. Then ∞ is a fixed point of γ and any fixed point z of γ in $\mathbb{C}$ satisfies $az + b = dz$. If γ is central, then it fixes every point in $\mathbb{C}$; otherwise γ has one fixed point in $\mathbb{R}$ if $a \neq d$ (so that γ is hyperbolic) and no fixed point in $\mathbb{C}$ if $a = d (= \pm 1)$ since $b \neq 0$ (so that γ is parabolic).

Listed below are several equivalent statements on the classification of γ in $SL_2(\mathbb{R})$:

- γ is elliptic
  $\Leftrightarrow |\operatorname{tr}\gamma| < 2$
  $\Leftrightarrow \gamma$ has two distinct eigenvalues α and β which are complex conjugates of each other
  $\Leftrightarrow$ the corresponding eigenvectors $\begin{pmatrix} u_\alpha \\ v_\alpha \end{pmatrix}$ and $\begin{pmatrix} u_\beta \\ v_\beta \end{pmatrix}$ can be chosen to be complex conjugates of each other
  $\Leftrightarrow \gamma$ has two fixed points u_α/v_α and u_β/v_β which are complex conjugates of each other.
- γ is hyperbolic
  $\Leftrightarrow |\operatorname{tr}\gamma| > 2$
  $\Leftrightarrow \gamma$ has two distinct real eigenvalues α and β

$\Leftrightarrow$ the corresponding eigenvectors $\begin{pmatrix} u_\alpha \\ v_\alpha \end{pmatrix}$ and $\begin{pmatrix} u_\beta \\ v_\beta \end{pmatrix}$ are in $\mathbb{R}^2$
$\Leftrightarrow$ γ has two fixed points in $\mathbb{R} \cup \{\infty\}$.

- γ is parabolic
  $\Leftrightarrow$ $|\operatorname{tr}\gamma| = 2$ and $\gamma \neq \pm I_2$
  $\Leftrightarrow$ γ has two equal real eigenvalues and the eigenspace is 1-dimensional
  $\Leftrightarrow$ γ has only one fixed point, which lies in $\mathbb{R} \cup \{\infty\}$.
- γ is central
  $\Leftrightarrow$ $\gamma = \pm I_2$
  $\Leftrightarrow$ γ has two equal real eigenvalues and the eigenspace is 2-dimensional
  $\Leftrightarrow$ every point in $\mathbb{C} \cup \{\infty\}$ is fixed by γ.

This proves the following geometric description of the above algebraic classification.

THEOREM 2.1. *Elements $\gamma \in SL_2(\mathbb{R})$ can be classified as follows:*

(1) *γ is elliptic if and only if γ has two fixed points, one in the upper half-plane and one in the lower half-plane.*
(2) *γ is hyperbolic if and only if γ has two fixed points in $\mathbb{R} \cup \{\infty\}$.*
(3) *γ is parabolic if and only if γ has only one fixed point, which lies in $\mathbb{R} \cup \{\infty\}$.*
(4) *γ is central if and only if γ fixes all points in $\mathbb{C} \cup \{\infty\}$.*

2.3. Compact Riemann surfaces arising as quotients of $\mathfrak{H}$

Let Γ be a discrete torsion-free subgroup of $SL_2(\mathbb{R})$ such that its orbit space $X_\Gamma := \Gamma \setminus \mathfrak{H}$ is a compact Riemann surface of genus g. The upper half-plane $\mathfrak{H}$ is endowed with the hyperbolic metric, under which it has constant negative curvature -1. As a quotient of $\mathfrak{H}$, the Riemann surface X_Γ inherits the metric and the constant curvature -1. Moreover, all Riemann surfaces with constant negative curvature arise this way. Typical examples of such Γ's are the $\mathbb{Z}$-points of the norm 1 subgroup of an indefinite quaternion algebra over $\mathbb{Q}$ as well as its finite index subgroups so that X_Γ consists of the complex points of a Shimura curve.

We explain this in more detail. Recall that a quaternion algebra B over $\mathbb{Q}$ is a 4-dimensional algebra with $\mathbb{Q}$-basis 1, i, j, k subject to the multiplication rules $i^2 = a$, $j^2 = b$, and $k = ij = -ji$ (so that $k^2 = -i^2j^2 = -ab$) for some $a, b \in \mathbb{Q}^\times$. We say that B splits over a field F containing $\mathbb{Q}$ if, as F-algebras, $B(F) = F + Fi + Fj + Fk$ and the 2×2 matrix algebra $M_2(F)$ are isomorphic. Thus if both a and b are squares in $\mathbb{Q}^\times$, then B splits over $\mathbb{Q}$. Further, B does not split over $\mathbb{R}$ if and only if a and b are both negative, in which case B is called a definite quaternion algebra. Otherwise, B is an indefinite quaternion algebra, in which case at least one of a, b is positive. The norm of an element $u = x+yi+zj+wk$ in $B(\mathbb{R})$ is $Nu = x^2 - ay^2 - bz^2 + abw^2$. For B indefinite, $B(\mathbb{R})$ is isomorphic to $M_2(\mathbb{R})$, and this isomorphism imbeds the group B_1 of norm 1 elements in $B(\mathbb{R})$ into $SL_2(\mathbb{R})$. In this way the integral points in B_1 yields a discrete subgroup Γ of $SL_2(\mathbb{R})$, which is cocompact if B does not split over $\mathbb{Q}$. Replacing Γ by a suitable subgroup to avoid torsion elements, if necessary, we obtain an example of Γ of interest.

The assumption on Γ implies that any non-identity element γ in Γ is hyperbolic. Let α and $1/\alpha$ be the real eigenvalues of γ. We study the centralizer of γ in Γ,

denoted $C_\Gamma(\gamma)$. To simplify our computations, choose an element $h \in SL_2(\mathbb{R})$ such that

$$h\gamma h^{-1} = \begin{pmatrix} \alpha & 0 \\ 0 & \frac{1}{\alpha} \end{pmatrix}.$$

This means that the first and second column vectors of h^{-1} are eigenvectors of γ with eigenvalues α and $1/\alpha$, respectively. The centralizer of $h\gamma h^{-1}$ in $h\Gamma h^{-1} \subset SL_2(\mathbb{R})$ consists of elements in $h\Gamma h^{-1}$ of the form

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

As $h\Gamma h^{-1}$ is discrete and torsion-free, the subgroup $C_{h\Gamma h^{-1}}(h\gamma h^{-1})$ is an infinite cyclic group. Hence $C_\Gamma(\gamma) = h^{-1}C_{h\Gamma h^{-1}}(h\gamma h^{-1})h$ is an infinite cyclic group containing $< \gamma >$, the cyclic group generated by γ.

We call γ *primitive* if $C_\Gamma(\gamma) =< \gamma >$. In other words, γ is primitive if and only if $\gamma \neq \beta^r$ for any $\beta \in \Gamma$ and $r > 1$.

2.4. Geodesic cycles in X_Γ

The hyperbolic metric on the upper half-plane $\mathfrak{H}$ is

$$ds = \frac{\sqrt{dx^2 + dy^2}}{y},$$

which is invariant under the action of $SL_2(\mathbb{R})$. The geodesics in $\mathfrak{H}$ consist of the semicircles centered on the real line $\mathbb{R}$ as well as the vertical lines. Any two points in $\mathfrak{H}$ are connected by a unique geodesic path. A non-identity element $\gamma \in \Gamma$ has two fixed points in $\mathbb{R} \cup \{\infty\}$. Through these two fixed points there is a unique geodesic in $\mathfrak{H}$, denoted by $\tilde{C}_\gamma$. Since the two ends of $\tilde{C}_\gamma$ are fixed by γ and γ preserves geodesics in $\mathfrak{H}$, we see that $\tilde{C}_\gamma$ is invariant under γ. Therefore, if z is a point on $\tilde{C}_\gamma$, so is $\gamma \cdot z$.

Let α and β be the two distinct real eigenvalues of γ. Define the length of γ to be

$$\ell(\gamma) = \log\left(\max\left\{\frac{\alpha}{\beta}, \frac{\beta}{\alpha}\right\}\right).$$

PROPOSITION 2.1.1. *Let γ be a non-identity element in Γ. Then for any $z \in \tilde{C}_\gamma$ we have*

$$\operatorname{dist}(z, \gamma \cdot z) = \ell(\gamma).$$

PROOF. Let α and $1/\alpha$ be the two real eigenvalues of γ. We may assume $|\alpha| > 1$ so that $\ell(\gamma) = \log \alpha^2$. Given $z \in \tilde{C}_\gamma$, we show that $\log \alpha^2$ is also the distance between z and $\gamma \cdot z$. Let $h \in SL_2(\mathbb{R})$ be such that

$$h\gamma h^{-1} = \begin{pmatrix} \alpha & 0 \\ 0 & \frac{1}{\alpha} \end{pmatrix} =: \widetilde{\gamma}.$$

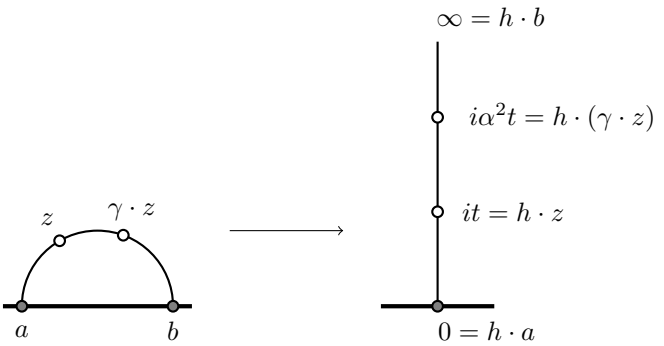
The two fixed points of $\widetilde{\gamma}$ are 0 and ∞. Let $a, b \in \mathbb{R} \cup \{\infty\}$ be the two fixed points of γ. Since $\widetilde{\gamma} \cdot (h \cdot a) = h \cdot (\gamma \cdot a) = h \cdot a$ and $\widetilde{\gamma} \cdot (h \cdot b) = h \cdot (\gamma \cdot b) = h \cdot b$, we may assume $h \cdot a = 0$ and $h \cdot b = \infty$. Then h maps $\tilde{C}_\gamma$ to the verticle line from 0 to ∞, which is the imaginary axis in $\mathfrak{H}$.

To calculate $\operatorname{dist}(z, \gamma \cdot z)$ with $z \in \tilde{C}_\gamma$, we use the conformal mapping h to simplify the computation. We have $h \cdot z = it$ for some $t > 0$ and $h \cdot (\gamma \cdot z) = \widetilde{\gamma} \cdot (h \cdot z) = \widetilde{\gamma} \cdot it = i\alpha^2 t$. Using the hyperbolic metric on the imaginary axis, we compute

$$\operatorname{dist}(z, \gamma \cdot z) = \operatorname{dist}(it, i\alpha^2 t) = \int_t^{\alpha^2 t} \frac{dy}{y} = \log \alpha^2,$$

as desired. □

The above argument is depicted in the following diagram.



A (closed) geodesic cycle in X_Γ lifts to a geodesic path in $\mathfrak{H}$ from a point z to $\gamma \cdot z \neq z$ for some $\gamma \in \Gamma$ not equal to the identity element. To decide which geodesic cycles to count, we note that for a fixed non-identity $\gamma \in \Gamma$, as the point z varies in $\mathfrak{H}$, the minimal distance between z and $\gamma \cdot z$ is attained precisely for $z \in \tilde{C}_\gamma$. In other words, Proposition 2.1.1 implies

$$\min_{z \in \mathfrak{H}} \operatorname{dist}(z, \gamma \cdot z) = \min_{z \in \tilde{C}_\gamma} \operatorname{dist}(z, \gamma \cdot z) = \ell(\gamma).$$

This gives one reason why $\tilde{C}_\gamma$ plays a special role.

The geodesic path in $\mathfrak{H}$ from z to γz with $z \in \tilde{C}_\gamma$ projects to a closed geodesic cycle C_γ in X_Γ starting at $\bar{z}$, the image of z in X_Γ. A cycle has a starting point and an orientation. If we choose a different point z' on $\tilde{C}_\gamma$, then the geodesic path from z' to $\gamma \cdot z'$ also projects to C_γ with the same orientation but a different starting point. Cycles differ only by starting points are said to be *equivalent*. Denote by $[C_\gamma]$ the equivalence class of C_γ; it is C_γ with the starting point ignored. The length of $[C_\gamma]$ is equal to the length of γ by Proposition 2.1.1. The image of $\tilde{C}_\gamma$ in X_Γ wraps around C_γ infinitely many times.

Here is another reason why the geodesics $\tilde{C}_\gamma$, where $\gamma \in \Gamma$ and $\gamma \neq id$, are distinguished. Let z be a point in $\mathfrak{H}$ outside $\tilde{C}_\gamma$. Then the geodesic $\tilde{C}$ in $\mathfrak{H}$ through z and $\gamma \cdot z$ is not preserved by γ, and consequently $\gamma^2 \cdot z$ lies on the geodesic $\gamma \cdot \tilde{C}$, which intersects $\tilde{C}$ at $\gamma \cdot z$. The image in X_Γ of the path from z to $\gamma \cdot z$ along $\tilde{C}$ is a closed geodesic cycle C starting and ending at $\bar{z}$, the image of z in X_Γ. On the other hand, if we change the starting point to $\overline{z'}$ projected from an intermediate point z' on the geodesic path from z to $\gamma \cdot z$, then the closed path along C starting at $\overline{z'}$ lifts to the path in $\mathfrak{H}$ from z' to $\gamma \cdot z$ along $\tilde{C}$ followed by the path from $\gamma \cdot z$ to $\gamma \cdot z'$ along $\gamma \cdot \tilde{C}$, which clearly is not a geodesic from z' to $\gamma \cdot z'$ in $\mathfrak{H}$. This shows that C is a closed geodesic cycle starting at $\bar{z}$, but it is no longer a geodesic cycle starting

at any other point on C. Therefore the $[C_\gamma]$'s are distinguished by the property that they are the only closed geodesic cycles in X_Γ which remain geodesics after changing starting points.

From now on, by closed geodesic cycles in X_Γ we mean the $[C_\gamma]$'s with non-identity $\gamma \in \Gamma$ and we shall not distinguish between $[C_\gamma]$ and C_γ when there is no danger of confusion. For any nonzero integer k, $\tilde{C}_\gamma$ and $\tilde{C}_{\gamma^k}$ coincide since γ and γ^k have the same fixed points. The closed geodesic cycle $[C_{\gamma^{-1}}]$ is $[C_\gamma]$ traveled in reverse direction and $[C_{\gamma^k}]$ with $k > 0$ is $[C_\gamma]$ traveled k times. A closed geodesic cycle in X_Γ is called *primitive* if it is not obtained from traveling along a shorter closed geodesic cycle more than once. Hence $[C_\gamma]$ is primitive if and only if γ is primitive.

Observe that if $\widetilde{\gamma} = h\gamma h^{-1}$ for some $h \in \Gamma$, then h sends the path from z to $\gamma \cdot z$ along $\tilde{C}_\gamma$ to the path from $h \cdot z$ to $h \cdot (\gamma \cdot z) = \widetilde{\gamma} \cdot (h \cdot z)$ along $h \cdot \tilde{C}_\gamma = \tilde{C}_{\widetilde{\gamma}}$. Projecting these paths to X_Γ, we get $[C_\gamma] = [C_{\widetilde{\gamma}}]$. Conversely, suppose $\gamma, \gamma' \in \Gamma$ are such that $C_\gamma = C_{\gamma'}$. Then C_γ lifts to a geodesic path $p(z)$ from z to $\gamma \cdot z$ along $\tilde{C}_\gamma$ and $C_{\gamma'}$ lifts to a geodesic path $p(z')$ from z' to $\gamma' \cdot z'$ along $\tilde{C}_{\gamma'}$. Since they are two liftings in $\mathfrak{H}$ of the same closed geodesic cycle in X_Γ, there is an $h \in \Gamma$ sending $p(z)$ to $p(z')$, and consequently $h \cdot \tilde{C}_\gamma = \tilde{C}_{\gamma'}$. On the other hand, as observed above, $h \cdot \tilde{C}_\gamma = \tilde{C}_{\widetilde{\gamma}}$ with $\widetilde{\gamma} = h\gamma h^{-1}$. We claim that $\gamma' = \widetilde{\gamma}$ so that γ and γ' are conjugate in Γ. To prove the claim, it suffices to show that γ' and $\widetilde{\gamma}$ agree at three points in $\mathfrak{H} \cup \mathbb{R} \cup \{\infty\}$ because then they differ by an element with at least three fixed points, which, by Theorem 2.1, must be the identity as Γ is torsion-free. Let a and b be the two endpoints of $\tilde{C}_\gamma$. Then $h \cdot a$ and $h \cdot b$ are the two endpoints of $h \cdot \tilde{C}_\gamma$, hence they are fixed by γ' and $\tilde{\gamma}$. The third point is $h \cdot z = z'$ because

$$\widetilde{\gamma} \cdot (h \cdot z) = h \cdot (\gamma \cdot z) = \gamma' \cdot z' = \gamma' \cdot (h \cdot z).$$

We have shown that $[C_\gamma] = [C_{\gamma'}]$ if and only if γ and γ' are conjugate in Γ. We summarize the above discussion in

PROPOSITION 2.1.2. *The map sending $[C_\gamma]$ to the conjugacy class $[\gamma]$ of γ in Γ is a length preserving bijection from the set of (equivalence classes of) closed geodesic cycles in X_Γ to the set of conjugacy classes of non-identity elements in Γ such that the primitive closed geodesic cycles correspond to the conjugacy classes of primitive elements.*

As the upper half-plane $\mathfrak{H}$ is simply connected, it is the universal cover of X_Γ with the deck transformation group Γ isomorphic to the fundamental group $\pi_1(X_\Gamma, *)$. If we regard X_Γ as playing the role of a number field K, then $\mathfrak{H}$ plays the role of an algebraic closure $\overline{K}$ and $\pi_1(X_\Gamma, *)$ plays the role of the absolute Galois group $G_K = \mathrm{Gal}(\overline{K}/K)$. The (equivalence classes of) primitive closed geodesic cycles are the primes of X_Γ. To each prime $[C_\gamma]$, the corresponding conjugacy class $[\gamma]$ in Γ plays the role of the "Frobenius" conjugacy class in the absolute Galois group of a number field.

The following diagram helps illustrate this line of thought.

$$\begin{array}{ccc} \overline{K} & \longleftarrow & \mathfrak{H} \\ \Big| & & \Big\downarrow \\ G_K & & \pi_1(X_\Gamma, *) \\ \Big| & & \Big\downarrow \\ K & \longleftarrow & X_\Gamma \end{array}$$

2.5. The Selberg zeta function attached to X_Γ

To a discrete torsion-free co-compact subgroup Γ of $SL_2(\mathbb{R})$, we associated the compact Riemann surface $X_\Gamma = \Gamma\backslash\mathfrak{H}$ of genus g. Introduced by Selberg [**87**] in 1956, the Selberg zeta function of X_Γ counts closed geodesic cycles in X_Γ. More precisely, it is defined as

$$\begin{aligned} Z(X_\Gamma, s) &= \prod_{\substack{[C] \text{ primitive} \\ \text{closed geodesic cycle}}} \prod_{k=0}^{\infty} \left(1 - e^{-\ell(C)(s+k)}\right) \\ &= \prod_{[\gamma] \text{ primitive}} \prod_{k=0}^{\infty} \left(1 - e^{-\ell(\gamma)(s+k)}\right), \end{aligned}$$

in which the second equality follows from the correspondence in Proposition 2.1.2. We shall regard the first expression as a product over the "primes"of X_Γ and the second as over the Frobenius conjugacy classes in the fundamental group Γ attached to the primes of X_Γ. The second expression naturally generalizes to the following zeta-function on X_Γ associated to a d-dimensional unitary representation ρ of Γ, also introduced by Selberg:

$$Z(X_\Gamma, \rho, s) = \prod_{[\gamma] \text{ primitive}} \prod_{k=0}^{\infty} \det\left(I_d - \rho(\gamma) e^{-\ell(\gamma)(s+k)}\right),$$

which reduces to $Z(X_\Gamma, s)$ for ρ the 1-dimensional trivial representation of Γ. These zeta functions converge absolutely on $\Re s > 1$. They are analogous to the Artin L-functions introduced in §4 of Lecture 1. Selberg showed that they can be extended analytically to the whole s-plane as entire functions.

The Laplacian operator with respect to the hyperbolic metric on the upper half-plane $\mathfrak{H}$ is

$$\Delta = \Delta_{\mathfrak{H}} = -y^2\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right).$$

It induces an action on the dense subspace of smooth functions on X_Γ, then extended by continuity to the space $L^2(X_\Gamma) = L^2(\Gamma \setminus \mathfrak{H})$, denoted by Δ_Γ. By abuse of language we shall simply refer to this (and the like) as the action of the Laplacian on the space of L^2-functions of the compact Riemann surface. Since X_Γ is compact, the spectrum of Δ_Γ is discrete. Further, since Δ_Γ is self-adjoint, its eigenvalues are real; and the non-negativity of Δ_Γ implies that the eigenvalues are non-negative. More details concerning the Laplacian operator will be explained in the next section. We arrange the eigenvalues in nondecreasing order:

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots .$$

The eigenvalues $\lambda_0, \lambda_1, \lambda_2, ...$ are called the *spectrum* of the Riemann surface X_Γ. The trivial eigenvalue $\lambda_0 = 0$ is simple with the constant functions being the corresponding eigenfunctions, and the remaining eigenvalues form the nontrivial spectrum. Regard the functions in $L^2(X_\Gamma)$ as functions on $\mathfrak{H}$ invariant under the action of Γ which are square-integrable on the quotient X_Γ. Let ρ be a unitary representation of Γ acting on the d-dimensional $\mathbb{C}$-vector space V endowed with the usual Hermitian product $<,>$. Denote by $L^2(X_\Gamma, \rho, V)$ the space of V-valued functions f on $\mathfrak{H}$ satisfying the following two conditions:

(1) $f(\gamma \cdot z) = \rho(\gamma) f(z)$ for all $\gamma \in \Gamma$ and $z \in \mathfrak{H}$;

(2) f is square-integrable on X_Γ, namely the integral

$$\int_{X_\Gamma} < f(z), f(z) > dz$$

converges.

Denote by $\Delta_\Gamma(\rho)$ the action of Δ on the space $L^2(X_\Gamma, \rho, V)$. For a nontrivial irreducible ρ, the eigenvalues of $\Delta_\Gamma(\rho)$ are all positive:

$$0 < \lambda_0(\rho) \leq \lambda_1(\rho) \leq \cdots .$$

Using the trace formula Selberg established the connection between the zeros of the zeta function $Z(X_\Gamma, \rho, s)$ and the eigenvalues of $\Delta_\Gamma(\rho)$ as follows.

THEOREM 2.2 (Selberg [**87**]). *Let ρ be a finite-dimensional irreducible unitary representation of Γ. Then the zeros of $Z(X_\Gamma, \rho, s)$ are 0, -1, -2,..., and*

$$s = \frac{1}{2} \pm (\frac{1}{4} - \lambda_n(\rho))^{1/2} \quad \textit{for } n \geq 0.$$

The trivial eigenvalue $\lambda_0 = 0$ occurs, with multiplicity one, only for the trivial representation, and it yields two trivial zeros 0 and 1 of $Z(X_\Gamma, s)$. The theorem above implies that the Riemann Hypothesis almost holds for $Z(X_\Gamma, \rho, s)$. More precisely, the nontrivial zeros of $Z(X_\Gamma, \rho, s)$ in the strip $0 \leq \Re s \leq 1$ arising from eigenvalues $\lambda_n(\rho) \geq 1/4$ all lie on the line $\Re s = 1/2$. Since $\lambda_n(\rho) \to \infty$ as $n \to \infty$, they are majority of the zeros in the critical strip. The finitely many eigenvalues $0 < \lambda_n(\rho) < 1/4$ give rise to real zeros which lie in the interval $(0, 1)$ and are symmetric with respect to $\frac{1}{2}$. It is worth pointing out that the universal cover of X_Γ is the upper half-plane $\mathfrak{H}$, on which the Laplacian operator $\Delta_\mathfrak{H}$ has continuous spectrum the interval $[1/4, \infty)$. Thus $Z(X_\Gamma, \rho, s)$ satisfies the Riemann Hypothesis if and only if the nontrivial spectrum of $\Delta_\Gamma(\rho)$ is contained in the spectrum of $\Delta_\mathfrak{H}$ on the universal cover of X_Γ. Selberg conjectured that this should be the case for the zeta functions of congruence subgroups of $SL_2(\mathbb{R})$.

CONJECTURE 5 (Selberg). *For a discrete torsion-free cocompact congruence subgroup Γ of $SL_2(\mathbb{R})$, $\lambda_1(X_\Gamma) \geq 1/4$ so that the Riemann Hypothesis holds for $Z(X_\Gamma, s)$.*

The original conjecture by Selberg in [**88**] was for (principal) congruence subgroups Γ of $SL_2(\mathbb{Z})$. Note that such a Γ is no longer cocompact, but the compactified quotient $X_\Gamma = \Gamma \backslash \mathfrak{H}^*$ has finite volume. On the other hand, a congruence subgroup Γ' of the norm 1 subgroup of an indefinite quaternion algebra over $\mathbb{Q}$ is cocompact in $SL_2(\mathbb{R})$. The eigenvalues of the Laplacian operator on the quotient $X_{\Gamma'}$, by Jacquet-Langlands correspondence [**45**], are among the eigenvalues of the Laplacian on the quotient of a suitable congruence subgroup of $SL_2(\mathbb{Z})$. The above

version of the Selberg conjecture is Selberg's original conjecture combined with the Jacquet-Langlands correspondence. Selberg has established in [**88**] the lower bound $\lambda_1(X_\Gamma) \geq 3/16$ for congruence subgroups Γ of $SL_2(\mathbb{Z})$, hence it also holds for Γ' described above. This lower bound is subsequently improved by several authors.

Hence the obstruction to the Riemann Hypothesis is the existence of small eigenvalues. R. Brooks has shown that given a Riemann surface M with constant negative curvature (such as $X_{\Gamma'}$) and genus at least one, there is a tower of finite coverings M_n of M such that the least positive eigenvalue $\lambda_1(M_n)$ of M_n tends to zero as $n \to \infty$. So there are many X_Γ's (presumably from noncongruence subgroups Γ) for which the Riemann Hypothesis fails.

The zeros of the Selberg zeta function provide information on the distribution of the "primes"of X_Γ. Recall that a closed geodesic cycle $[C_\gamma]$ has length $\ell(C_\gamma) = \ell(\gamma) = \log \alpha^2$, where α is the eigenvalue of γ with absolute value greater than one. Define the norm of $[C_\gamma]$ and $[\gamma]$ to be $N([C_\gamma]) := e^{\ell(C_\gamma)} = \alpha^2 =: N([\gamma])$. Like Dedekind zeta functions for number fields, the distribution of primes of bounded norm is related to the analytic behavior of the function below, called the Ruelle zeta function of X_Γ in dynamical systems:

$$\zeta(X_\Gamma, s) = \prod_{[C] \text{ prime of } X_\Gamma} \frac{1}{1 - N([C])^{-s}} \ .$$

Observe that it is nothing but the quotient of the Selberg zeta function at two different arguments:

$$\zeta(X_\Gamma, s) = \frac{Z(X_\Gamma, s+1)}{Z(X_\Gamma, s)}.$$

It follows from Theorem 2.2 that on the half-plane $\Re s > 0$ the numerator of $\zeta(X_\Gamma, s)$ is holomorphic and nonvanishing, and on the closed half-plane $\Re s \geq 1$, $\zeta(X_\Gamma, s)$ is holomorphic and nonvanishing except for a simple pole at $s = 1$. Further on the critical strip $0 < \Re s < 1$, $\zeta(X_\Gamma, s)$ is holomorphic except for poles at the nontrivial zeros of the Selberg zeta function. In particular, it is holomorphic on an open half plane $\Re s > 1 - t_1$ except $s = 1$. Here t_1 is the largest real zero of $Z(X_\Gamma, s)$ if it exists and $1/2$ otherwise. The analytic behavior of $\zeta(X_\Gamma, s)$ on the half-plane $\Re s \geq 1$ implies the main term of the prime geodesic theorem, as indicated by Selberg and first shown in [**41**] by Huber. The following result of Sarnak provides more information of the error term in terms of the zeros of the Selberg zeta function lying to the right of the line $\Re s = \frac{1}{2}$.

THEOREM 2.3 (Prime geodesic theorem, Sarnak [**84**]). *Let $t_1 \geq \cdots \geq t_k$ be the zeros of $Z(X_\Gamma, s)$ in the interval $(\frac{1}{2}, 1)$, and let $Li(u) = \int_2^u \frac{1}{\log t} dt$ be the logarithmic integral as before. Then for x large, we have*

$$\begin{aligned} & \# \quad \{[C] \text{ primitive closed geodesic cycle in } X_\Gamma : \ N([C]) < x\} \\ & = \quad Li(x) + Li(x^{t_1}) + \cdots + Li(x^{t_k}) + O(x^{3/4}(\log x)^2). \end{aligned}$$

In [**84**] Sarnak also obtained an analogous statement for the Chebotarev density theorem.

Next we proceed to draw a connection between the Selberg zeta function of X_Γ and its spectrum. The two zeros $\frac{1}{2} \pm (\frac{1}{4} - \lambda_n)^{1/2}$ of the Selberg zeta function

$Z(X_\Gamma, s)$ are the roots of $s(1-s) = \lambda_n$. So $Z(X_\Gamma, s)$ is expected to relate to the infinite product

$$\prod_{n=0}^{\infty} \big(\lambda_n - s(1-s)\big).$$

However, this infinite product is not well-defined. To rigidify this, we consider instead the spectral zeta function

$$\zeta(s,w) = \sum_{n=0}^{\infty} \big(\lambda_n - s(1-s)\big)^{-w}.$$

It can be shown that $\zeta(s,w)$ is regular at $w=0$. This allows us to define

$$\det\big(\Delta_\Gamma - s(1-s)\big) := \exp\bigg(-\bigg(\frac{\partial}{\partial w}\zeta(s,w)\bigg)\bigg|_{w=0}\bigg).$$

Note that

$$-\frac{\partial}{\partial w}\big(\lambda_n - s(1-s)\big)^{-w} = \big(e^{-w\log(\lambda_n - s(1-s))}\big)\log\big(\lambda_n - s(1-s)\big),$$

and so formally we have

$$\det\big(\Delta_\Gamma - s(1-s)\big) = \prod_{n=0}^{\infty}\big(\lambda_n - s(1-s)\big).$$

The explicit relation between the Selberg zeta function and the spectrum of X_Γ is given in

THEOREM 2.4 (Sarnak [**83**], Voros [**104**] independently in 1987). *Let Γ be a discrete torsion-free cocompact subgroup of $SL_2(\mathbb{R})$. Then*

$$\det\big(\Delta_\Gamma - s(1-s)\big) = Z(X_\Gamma, s)\bigg(e^{E+s(1-s)}\frac{\Gamma_2(s)^2}{\Gamma(s)}(2\pi)^s\bigg)^{2g-2},$$

where $E = -\frac{1}{4} - \frac{1}{2}\log(2\pi) + 2\zeta'(-1)$, g is the genus of X_Γ, and $\Gamma_2(s)$ is the Barnes double Γ-function satisfying $\Gamma_2(1) = 1$ and $\Gamma_2(s+1) = \frac{\Gamma_2(s)}{\Gamma(s)}$.

Remarks. (1) The factor $e^{E+s(1-s)}\frac{\Gamma_2(s)^2}{\Gamma(s)}(2\pi)^s$ is independent of Γ, and the exponent $2g-2$ gives topological information about X_Γ, corresponding to the Euler characteristic of X_Γ. The left-hand side provides the spectral information of X_Γ. The symmetric of the left hand side between s and $1-s$ gives a functional equation for $Z(X_\Gamma, s)$. Among compact Riemannian manifolds M, this is the only known explicit relation between the zeta function and the spectrum of M.

(2) This theorem has been extended to allow Γ to contain elliptic elements by Efrat [**20**] and X_Γ to have finite volume by Koyama [**53**, **54**].

2.6. Properties of the Laplacian operator

In this section we review some properties of the Laplacian operator on a Riemannian manifold mentioned in the discussions of the previous sections.

Let X be a smooth manifold with a metric

$$ds^2 = \sum_{i,j=1}^{n} g_{ij}\, dx_i dx_j.$$

This metric defines the canonical measure on X by

$$dx = (\det(g_{ij}))^{1/2} dx_1 ... dx_n,$$

with which one defines the inner product

$$\langle f, g \rangle = \int_X f(x) g(x) dx$$

for real-valued functions f, g on X whenever the integral converges. The Laplacian operator Δ_X on X is given by

$$\Delta_X = -(\det(g_{ij}))^{-1/2} \sum_{i,j=1}^{n} \frac{\partial}{\partial x_i} \left(g^{ij} (\det(g_{ij}))^{1/2} \frac{\partial}{\partial x_j} \right),$$

where (g^{ij}) is the inverse matrix of (g_{ij}). Thus Δ_X is a second order elliptic differential operator acting on suitable functions on X. As such, it possesses the following properties:

(1) (symmetric) If $f_1, f_2 \in C_0^\infty(X)$, the space of smooth functions on X with compact support, then

$$\langle \Delta_X f_1, f_2 \rangle = \langle f_1, \Delta_X f_2 \rangle.$$

(2) (nonnegative) $\langle \Delta_X f, f \rangle \geq 0$ for all $f \in C_0^\infty(X)$.

Example 2.1. Let X be the upper half-plane $\mathfrak{H}$ with the hyperbolic metric $ds^2 = \frac{1}{y^2} dx^2 + \frac{1}{y^2} dy^2$ mentioned before. Then $g_{11} = g_{22} = \frac{1}{y^2}$ and $g_{12} = g_{21} = 0$. Hence $g^{11} = g^{22} = y^2$ and $g^{12} = g^{21} = 0$. The formula defining Δ_X in this case reads

$$\Delta_X = -y^2 \left(\frac{\partial}{\partial x} (y^2 \frac{1}{y^2} \frac{\partial}{\partial x}) + \frac{\partial}{\partial y} (y^2 \frac{1}{y^2} \frac{\partial}{\partial y}) \right) = -y^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right),$$

which is the definition familiar to us. The measure on $\mathfrak{H}$ is $\frac{1}{y^2} dx dy$.

Let M be a compact connected Riemannian manifold and $\pi : X \to M$ be a surjective covering map. Put on X the metric that is the pull-back of the metric on M. Then for $f \in C^\infty(M)$, we can either apply the Laplacian Δ_M to f then pull it back to X, or pull it back to X first then apply the Laplacian Δ_X to the resulting function, that is,

$$(\Delta_M f) \circ \pi = \Delta_X (f \circ \pi).$$

But this is NOT a relation between the actions of the operators on the L^2-spaces. For instance, the pull-back of an L^2-function on M may no longer lie in $L^2(X)$. On the other hand, since X has a compact quotient, it is always complete. The Laplacian operator on the space $C_0^\infty(X)$ is extended in a unique way to a self-adjoint non-negative operator from a domain $D(\Delta_X)$ dense in $L^2(X)$ to $L^2(X)$.

In general the spectrum $\sigma(H)$ of a self-adjoint operator H from a dense subspace V of W to W is a disjoint union of two subsets of $\mathbb{R}$: the *point spectrum* $\sigma_p(H)$ and the *continuous spectrum* $\sigma_c(H)$. Here $\sigma_p(H)$ is the set of eigenvalues of H, consisting of $\lambda \in \mathbb{R}$ such that the operator $H - \lambda I$ is not injective. A real number $\lambda \in \sigma_c(H)$ if $H - \lambda I$ is injective, its range is a dense subset of W and not the whole space, and its inverse $(H - \lambda I)^{-1}$ is unbounded. Consequently $(H - \lambda I)^{-1}$ is defined on a dense subset of W and cannot be extended to whole W.

For a self-adjoint operator H_X on X, the bottom of the spectrum $\lambda_0 = \lambda_0(H_X) := \inf \sigma(H_X)$ is a finite value (i.e. $> -\infty$). In particular, $\lambda_0(\Delta_X) \geq 0$. Moreover, if

X is compact, then $\sigma(\Delta_X) = \sigma_p(\Delta_X)$, the multiplicity of each eigenvalue is finite, and zero is always a simple eigenvalue of Δ_X. If X is a globally symmetric space of non-compact type, for instance, $X = \mathfrak{H}$, then $\sigma(\Delta_X) = \sigma_c(\Delta_X) = [\lambda_0, \infty)$ with a positive λ_0. Especially if X is an n-dimensional hyperbolic space of constant curvature -1 (in the sense of curvature tensor), then $\sigma(\Delta_X)$ is the interval $[\frac{(n-1)^2}{4}, \infty)$. In particular, $\sigma(\Delta_{\mathfrak{H}}) = [\frac{1}{4}, \infty)$.

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LECTURE 3

L-functions in geometry

3.1. General setup

The zeta and L-functions we have encountered can be summarized abstractly in the following context. There are three basic ingredients: a countable set $\mathfrak{P}$ of primes, a map $N : \mathfrak{P} \to \mathbb{R}_{>0}$ giving the norm of each prime, and a group Γ such that each prime $p \in \mathfrak{P}$ is associated a conjugacy class $\langle p \rangle$ of Γ. We recall two such examples.

EXAMPLE 3.1. Let $K/\mathbb{Q}$ be a finite Galois extension with Galois group Γ. The set $\mathfrak{P}$ consists of primes p unramified in K, the map N is the norm $N(p) = p$, and the conjugacy class $\langle p \rangle$ is the Frobenius conjugacy class Frob_p. Similar setting holds when the base field is a finite extension of $\mathbb{Q}$, called a number field.

EXAMPLE 3.2. The group Γ is a discrete torsion-free cocompact subgroup of $SL_2(\mathbb{R})$, the set $\mathfrak{P}$ consists of primitive closed geodesic cycles $[C]$ on the Riemann surface $X_\Gamma = \Gamma\backslash\mathfrak{H}$ ignoring the starting point, the conjugacy class $\langle [C] \rangle$ of Γ associated to $[C]$ is the primitive conjugacy class $[\gamma]$ such that $[C] = [C_\gamma]$ as discussed in Lecture 2, §4, and the map N is defined as $N([C_\gamma]) = e^{\ell(\gamma)} = \alpha^2$, where α is the eigenvalue of γ with absolute value > 1.

Obviously one triple $(\mathfrak{P}, N, \Gamma)$ will give rise to many triples simply by taking subsets of $\mathfrak{P}$.

Given a triple $(\mathfrak{P}, N, \Gamma)$, to an n-dimensional unitary representation ρ of Γ we associate the L-function

$$L(\rho, s) = \prod_{p \in \mathfrak{P}} \frac{1}{\det(I_n - \rho(\langle p \rangle)(Np)^{-s})}.$$

In Example 3.1, this is the Artin L-function of ρ with the ramified factors removed, while in Example 3.2 we have

$$L(\rho, s) = \frac{Z(X_\Gamma, \rho, s+1)}{Z(X_\Gamma, \rho, s)}$$

as the quotient of the Selberg zeta functions attached to ρ at two arguments. Observe that in the latter case we can also express the Selberg zeta function attached to ρ in terms of the L-function by

$$Z(X_\Gamma, \rho, s)^{-1} = L(\rho, s)L(\rho, s+1)L(\rho, s+2)\cdots .$$

Given a triple $(\mathfrak{P}, N, \Gamma)$, the collection of $L(\rho, s)$ for all finite-dimensional unitary representations ρ of Γ is called a *package of L-functions*. Such a package is called *nice* if the following properties hold:

(L1) There exists a positive real number h such that each $L(\rho, s)$ in the package converges absolutely to a holomorphic function on the open half-plane $\Re s > h$;
(L2) Each $L(\rho, s)$ has a meromorphic continuation to an open neighborhood of the closed half-plane $\Re s \geq h$;
(L3) Each $L(\rho, s)$ is nonvanishing on $\Re s \geq h$;
(L4) If ρ is an irreducible and nontrivial representation of Γ, then $L(\rho, s)$ is holomorphic on $\Re s \geq h$;
(L5) For the trivial representation $\mathbf{1}_\Gamma$ of Γ, $L(\mathbf{1}_\Gamma, s)$ has a simple pole at $s = h$ and is holomorphic elsewhere on $\Re s \geq h$.

The package of L-functions attached to the triple in both examples above is nice with $h = 1$. This is a well-known fact for Example 3.1, see for instance Corollary 5.47 on p. 143 of the book [**44**] for more detail; for Example 3.2 this was proved by Selberg [**87**] as discussed in Chap 2, §5.

Using the standard argument in analytic number theory, Sunada in [**96**] summarized the fact that if the package of L-functions of the triple $(\mathfrak{P}, N, \Gamma)$ with a finite Γ is nice, then the Chebotarev Density Theorem in natural density would hold, as a consequence of the Tauberian theorem.

THEOREM 3.1 (Sunada). *Suppose that Γ is a finite group and the package of L-functions attached to the triple $(\mathfrak{P}, N, \Gamma)$ is nice with the critical positive h. Then for any conjugacy class C of Γ, we have*

$$\#\{p \in \mathfrak{P} \ : \ \langle p \rangle = C \text{ and } Np \leq x\} \sim \frac{|C|}{|\Gamma|} \frac{x^h}{\log x^h} \quad \text{as } x \to \infty.$$

By summing over all conjugacy classes C of Γ, the above theorem implies

COROLLARY 3.1.1 (Prime geodesic theorem). *Under the same assumption as the theorem above, we have*

$$\#\{p \in \mathfrak{P} \ : \ Np \leq x\} \sim \frac{x^h}{\log x^h} \quad \text{as } x \to \infty.$$

In fact, Corollary 3.1.1 follows from the analytic behavior of the L-function attached to the trivial representation $\mathbf{1}_\Gamma$ described above. Thus Theorem 3.1 can be rephrased in the familiar statement of the Chebotarev density theorem.

COROLLARY 3.1.2 (Chebotarev density theorem in natural density). *Assume the same as in the theorem above. Then for any conjugacy class C of Γ, we have*

$$\frac{\#\{p \in \mathfrak{P} \ : \ \langle p \rangle = C \text{ and } Np \leq x\}}{\#\{p \in \mathfrak{P} \ : \ Np \leq x\}} \sim \frac{|C|}{|\Gamma|} \quad \text{as } x \to \infty.$$

For compact Riemann surfaces with constant negative curvature -1, Theorem 3.1 and its consequences hold with $h = 1$ for any finite Galois cover $X_{\Gamma''} \to X_{\Gamma'}$. Here Γ' is a discrete torsion-free cocompact subgroup of $SL_2(\mathbb{R})$ and Γ'' is a normal subgroup of Γ' of finite index with quotient $\Gamma'/\Gamma'' =: \Gamma$ being the covering group. This was proved in Sarnak's thesis [**84**]. Theorem 3.1 also holds for finite Galois extensions of number fields with Galois group Γ and $h = 1$, as discussed in Lecture 1.

If the condition (L4) is replaced by the weaker condition

(L4)' If ρ is an irreducible and nontrivial representation of Γ, then $L(\rho, s)$ is holomorphic at $s = h$,

then the same argument implies the Chebotarev density theorem in Dirichlet density. In particular, the prime number theorem still holds, but the Chebotarev density theorem may not hold in natural density.

3.2. Anosov flow

A flow on a differentiable manifold M is a differentiable action of the additive group $\mathbb{R}$ on M. More precisely, it is a differentiable map $\phi : M \times \mathbb{R} \to M$ such that its restriction to $t \in \mathbb{R}$, denoted $\phi_t : M \to M$, given by $\phi_t(x) = \phi(x, t)$ is a bijective differentiable map satisfying

(1) $\phi_0 = id$; and
(2) $\phi_s \circ \phi_t = \phi_{s+t}$.

Therefore the flow forms a one parameter subgroup of diffeomorphisms on M. For a fixed point $x \in M$, the map from $\mathbb{R}$ to M given by $t \mapsto \phi(x, t)$ is a differentiable curve on M passing through the point x at $t = 0$. The tangent of this curve at $t = 0$ is a vector in the tangent space at the point x. Hence given a flow ϕ, we obtain a vector field on M, which generates a one-dimensional sub-bundle of the tangent bundle $T(M)$ of M. Moreover, a curve C on M passing through the point x at $s = 0$ is mapped by ϕ_t to a curve $\phi_t(C)$ passing through $\phi_t(x)$ at $s = 0$ and sending the tangent at x along C to the tangent at $\phi_t(x)$ along $\phi_t(C)$. Hence the flow ϕ induces a map on the tangent bundle $T(M)$.

A flow is called an *Anosov flow* if it splits the tangent bundle $T(M)$ of M into three subbundles, each invariant under the flow:

$$T(M) = \mathcal{E}^+ \oplus \mathcal{E}^0 \oplus \mathcal{E}^-,$$

where $\mathcal{E}^+$ is exponentially expanding, $\mathcal{E}^-$ is exponentially contracting, and $\mathcal{E}^0$ is a non-expanding and non-contracting 1-dimensional subbundle spanned by the flow direction. A typical example of an Anosov flow is the flow on the unit tangent bundle of a Riemann surface with constant negative curvature explained below.

As observed in §1 of Lecture 2, the upper half-plane $\mathfrak{H}$ can be identified with $SL_2(\mathbb{R})/SO_2(\mathbb{R})$. The group $SO_2(\mathbb{R})$ is isomorphic to the unit circle S^1. The center of $SL_2(\mathbb{R})$, which consists of $\pm I_2$, acts trivially on $\mathfrak{H}$, so we may identify $\mathfrak{H}$ with $PSL_2(\mathbb{R})/\overline{SO_2(\mathbb{R})}$. Here $\overline{SO_2(\mathbb{R})}$ is $SO_2(\mathbb{R})/\{\pm I_2\}$, hence is isomorphic to $S^1/\pm 1 \cong S^1$. The group $PSL_2(\mathbb{R})$ is the disjoint union of the $\overline{SO_2(\mathbb{R})}$-cosets represented by points z in $\mathfrak{H}$. Imagine the $\overline{SO_2(\mathbb{R})}$-coset represented by z as a circle at z, and regard this circle as the collection of unit vectors in the tangent plane at the point z. This interpretation then identifies $PSL_2(\mathbb{R})$ with the unit tangent bundle $T^1(\mathfrak{H})$ on $\mathfrak{H}$. It is a 3-dimensional real differentiable manifold. Let Γ be a discrete torsion-free cocompact subgroup of $SL_2(\mathbb{R})$. Locally the tangent plane at a point on X_Γ is the same as that at any of its preimages in $\mathfrak{H}$, so we may identify the unit tangent bundle $T^1(X_\Gamma)$ on X_Γ with $\bar{\Gamma}\backslash PSL_2(\mathbb{R})$. Here $\bar{\Gamma}$ is the image of Γ in $PSL_2(\mathbb{R})$.

The tangent space at the identity $I_2 \in PSL_2(\mathbb{R})$ is the Lie algebra $s\ell(2)$ of trace zero 2×2 real matrices. It has a basis consisting of

$$J = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

with the Lie algebra structure given by the bracket product

$$[J, X] = X, \qquad [J, Y] = -Y, \qquad [X, Y] = 2J.$$

The exponential maps sending $t \in \mathbb{R}$ to

$$\phi_t = \exp(tJ) = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix},$$

$$h_t^* = \exp(tX) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad h_t = \exp(tY) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix},$$

respectively define three curves on $PSL_2(\mathbb{R})$ through I_2 at $t = 0$. Their images

$$H_1 := \{\phi_s : s \in \mathbb{R}\},\ H_2 := \{h_s^* : s \in \mathbb{R}\} \text{ and } H_3 := \{h_s : s \in \mathbb{R}\}$$

are three subgroups of $PSL_2(\mathbb{R})$. For $g \in PSL_2(\mathbb{R})$, the coset gH_1 is called a J-curve, gH_2 an X-curve, and gH_3 a Y-curve. These are the unique J-, X- and Y- curves which pass through g at $s = 0$. The tangents at $s = 0$ of the Y-, J- and X- curves on $PSL_2(\mathbb{R})$ respectively determine three left-invariant line bundles $\mathcal{E}^+$, $\mathcal{E}^0$ and $\mathcal{E}^-$ on $PSL_2(\mathbb{R})$. Together, they give the decomposition of the tangent bundle $T(PSL_2(\mathbb{R})) = \mathcal{E}^+ \oplus \mathcal{E}^0 \oplus \mathcal{E}^-$. Modulo the action of $\overline{\Gamma}$ on the base manifold $PSL_2(\mathbb{R})$, we obtain a similar decomposition of $T(T^1(X_\Gamma)) = T(\bar{\Gamma}\backslash PSL_2(\mathbb{R}))$.

Our flow on $PSL_2(\mathbb{R})$ is given by $\{\phi_t\}$, that is, the flow

$$\phi : PSL_2(\mathbb{R}) \times \mathbb{R} \to PSL_2(\mathbb{R})$$

defined by

$$\phi(g, t) = g\phi_t.$$

This readily induces a flow, again denoted by $\{\phi_t\}$, on $\bar{\Gamma}\backslash PSL_2(\mathbb{R})$.

THEOREM 3.2. *Let Γ be a discrete torsion-free cocompact subgroup of $SL_2(\mathbb{R})$. The flow $\{\phi_t\}$ on $T^1(X_\Gamma) = \bar{\Gamma}\backslash PSL_2(\mathbb{R})$ is Anosov.*

PROOF. Since the action of the flow on $T^1(X_\Gamma) = \bar{\Gamma}\backslash PSL_2(\mathbb{R})$ is by right multiplication, it suffices to prove that the flow on $PSL_2(\mathbb{R})$ is Anosov. For this we show that the decomposition $T(PSL_2(\mathbb{R})) = \mathcal{E}^+ \oplus \mathcal{E}^0 \oplus \mathcal{E}^-$ has the required properties, that is, the line bundles $\mathcal{E}^+$, $\mathcal{E}^0$ and $\mathcal{E}^-$ are invariant under the flow $\{\phi_t\}$, $\mathcal{E}^+$ is exponentially expanding, $\mathcal{E}^0$ is neither expanding nor contracting, and $\mathcal{E}^-$ is exponentially contracting, independent of the metric on the manifold.

The invariance of each line bundle under ϕ_t action will follow from $gH_i\phi_t = g\phi_t H_i$ for all $g \in PSL_2(\mathbb{R})$ and $i = 1, 2, 3$. In view of the definition of H_i, this results from the following relations:

(1) $\phi_s\phi_t = \phi_{s+t} = \phi_t\phi_s$,

(2) $h_s^*\phi_t = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} \begin{pmatrix} 1 & se^{-t} \\ 0 & 1 \end{pmatrix} = \phi_t h_{s/e^t}^*$,

(3) $h_s\phi_t = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix} \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ se^t & 1 \end{pmatrix} = \phi_t h_{se^t}$.

Moreover, since the tangent on the X-, J- and Y-curves at $s = 0$ is computed by first taking the derivative with respect to the variable s and then evaluating the resulting function at $s = 0$, we see that (1) implies ϕ_t preserves the length of the tangent vectors in $\mathcal{E}^0$, (2) implies that the length of the tangent vector of an X-curve is multiplied by e^{-t}, hence ϕ_t is exponentially contracting on $\mathcal{E}^-$, while (3) shows that the action of ϕ_t on $\mathcal{E}^+$ is expanding by e^t. □

We define a metric on $PSL_2(\mathbb{R})$ so that all J-curves on $PSL_2(\mathbb{R})$ are geodesics to make $\{\phi_t\}$ a geodesic Anosov flow. To go from $T^1(\mathfrak{H})$ to $\mathfrak{H}$ we simply let $PSL_2(\mathbb{R})$ act on the point i in $\mathfrak{H}$. The image of the J-curve H_1 is $\{\begin{pmatrix} e^{s/2} & 0 \\ 0 & e^{-s/2} \end{pmatrix} \cdot i = e^s i \ : \ s \in \mathbb{R}\}$, which is the imaginary axis in $\mathfrak{H}$. Notice that the two ends of the imaginary axis are the two fixed points of the non-identity elements in H_1. So the image in $\mathfrak{H}$ of the geodesic Anosov flow on $T^1(\mathfrak{H})$ consists of the images of the imaginary axis under the action of $PSL_2(\mathbb{R})$. One checks easily that these are the geodesics on $\mathfrak{H}$ with respect to the hyperbolic metric. The closed geodesic cycles on X_Γ considered in Lecture 2 are precisely the images of the geodesic Anosov flow curves on the unit tangent bundle of X_Γ which are closed in X_Γ. In §4 of Lecture 2 they are called $\tilde{C}_\gamma$ for nonidentity $\gamma \in \Gamma$. Recall that $\tilde{C}_\gamma = h^{-1}H_1 \cdot i$ with $h^{-1} \in SL_2(\mathbb{R})$ whose two columns are two lineaerly independent eigenvectors of γ.

In general, given a smooth manifold M, it is difficult to decide which closed cycles to count. But if there is an Anosov flow on M, then one can define a metric on M so that the Anosov flow is a geodesic flow and we count those closed geodesic cycles which are the closed orbits of the Anosov flow, as exemplified by the Selberg zeta function on compact Riemann surfaces with constant negative curvature -1.

3.3. L-functions and distributions of prime geodesics

Let (M, ϕ_t) be a smooth geodesic Anosov flow on a smooth compact connected Riemannian manifold M. A point $x \in M$ is called *nonwandering* if for every open neighborhood U of x, there is a sequence of real numbers $\{t_n\}$ going to ∞ such that the set $\phi_{t_n}(U) \cap U$ has positive measure. Assume that the nonwandering set of ϕ_t is M. The set $\mathfrak{P}$ of primes consists of all closed orbits of the flow, and for each $p \in \mathfrak{P}$, let

$$N(p) = e^{\ell(p)},$$

where $\ell(p)$ denotes the length of p. Put $\Gamma = \pi_1(M, *)$, the fundamental group of M. A continuous function from the unit circle S^1 to M is called a loop. Elements in Γ are homotopy classes of loops in M starting and ending at a prescribed point. Two homotopy groups with different prescribed points are isomorphic since M is connected. A free loop is a continuous map from S^1 to M without the base-point preserving restriction. Then free homotopy classes of free loops correspond to conjugacy classes in $\pi_1(X, *)$. Denote by $\langle p \rangle$ the conjugacy class corresponding to the free homotopy class of p. We have attached to (M, ϕ_t) a triple $(\mathfrak{P}, N, \Gamma)$.

Let ρ be a finite-dimensional unitary representation of Γ. Define the L-function $L(\rho, s)$ as in §1. The L-function attached to the trivial representation $\mathbf{1}_\Gamma$ is the dynamical L-function defined by Ruelle. The following theorem on properties of $L(\rho, s)$ is proved by Parry and Pollicott for $\rho(\Gamma)$ finite [**78**], and extended by Adachi and Sunada [**2**] to general case.

THEOREM 3.3. *Suppose (M, ϕ_t) is of Anosov type and the flow has topological entropy $h > 0$. Then for any finite-dimensional unitary representation ρ of Γ, the associated L-function $L(\rho, s)$ satisfies (L1)-(L3) in §1 and*

(L'4) *If $\deg \rho \geq 2$ and ρ is irreducible, then $L(\rho, s)$ is holomorphic on $\Re s \geq h$.*

(L'5) *For a character χ, $L(\chi, s)$ has a pole at $s = h + it$ if and only if $\chi(\langle p \rangle) = e^{it\ell(p)} = N(p)^{it}$ for all $p \in \mathfrak{P}$. In this case $L(\chi, s) = L(\mathbf{1}_\Gamma, s - it)$ and every pole on $\Re s = h$ is simple.*

If, in addition, (M, ϕ_t) is topological mixing and a nontrivial character χ has finite order, then $L(\chi, s)$ is holomorphic on $\Re s \geq h$. Thus the package of L-functions is nice. In particular, this is the case for L-functions of the geodesic flow on the unit tangent bundle on a negatively curved manifold.

Here topological mixing means that given open subsets A and B of M, there exists $m > 0$ such that $\phi_t(A) \cap B$ is nonempty for all $t > m$.

Adachi [**1**] proved that the fundamental group $\pi_1(M, *)$ is generated by the homotopy classes of the closed orbits of (M, ϕ_t). Concerning the distribution of these homotopy classes, an analogue of the Dirichlet's theorem on primes in arithmetic progressions holds in the following sense.

The abelianization $\pi_1(M, *)^{ab}$ of the fundamental group $\pi_1(M, *)$ is isomorphic to the homology group $H_1(M, \mathbb{Z})$. To each closed orbit p of (M, ϕ_t), we associated a conjugacy class $\langle p \rangle$ in $\pi_1(M, *)$, whose image in $H_1(M, \mathbb{Z})$ is a singleton. We are interested in the distribution of the closed orbits among the homology classes of M. The reader is referred to the review paper [**96**] by Sunada for the discussion below. Let H be a finite index subgroup of $H_1(M, \mathbb{Z})$. Then any H-coset in $H_1(M, \mathbb{Z})$ contains infinitely many homology classes represented by closed orbits of (M, ϕ_t). To see this, take the group Γ to be the finite group $H_1(M, \mathbb{Z})/H$ and for each prime $p \in \mathfrak{P}$ take the conjugacy class to be the image of $\langle p \rangle$ in $H_1(M, \mathbb{Z})/H$. Then all 1-dimensional representations of $H_1(M, \mathbb{Z})/H$ have finite image so that the package of the L-functions attached to the new triple are nice by Theorem 3.3. Then the desired statement follows from Theorem 3.1. It is an open question for H of infinite index in general. However, it is known for some special cases.

THEOREM 3.4 (Adachi-Sunada [**2**], Katsuda-Sunada [**50**]). *If the geodesic flow on the unit tangent bundle of M is of Anosov type, or if M is non-positively curved and of rank one, then each homology class in $H_1(M, \mathbb{Z})$ contains infinitely many prime closed geodesics.*

In terms of the growth of the number of prime geodesics with respect to length, more is known. Given $\alpha \in H_1(M, \mathbb{Z})$, let

$$\pi(x, \alpha) = \#\{\text{prime geodesics } p \in \mathfrak{P} \quad : \quad \ell(p) < x \text{ and the homology class of } p \text{ is } \alpha\}.$$

If $H_1(M, \mathbb{Z})$ is finite, we conclude from Theorem 3.1 that

$$\pi(x, \alpha) \sim \frac{1}{|H_1(M, \mathbb{Z})|} \frac{e^{hx}}{hx} \quad \text{as } x \to \infty.$$

If $H_1(M, \mathbb{Z})$ is infinite, one can show

$$\lim_{x \to \infty} \frac{\log \pi(x, \alpha)}{x} = h.$$

In this case, there is a conjecture stating the asymptotic behavior of $\pi(x, \alpha)$:

CONJECTURE 6 (Sunada [**96**]). *There exists a positive constant c independent of $\alpha \in H_1(M, \mathbb{Z})$ such that for x large,*

$$\pi(x, \alpha) \sim c \ \frac{e^{hx}}{x^{b_1(M)+1}},$$

where $b_1(M)$ is the first Betti number of M, which is equal to the rank of $H_1(M, \mathbb{Z})$.

Two remarks are in order.

(1) Theorem 3.4 is not true in general. For instance, for the case that $M = \mathbb{R}^2/\mathbb{Z}^2$ is a flat torus so that $H_1(M, \mathbb{Z}) \cong \mathbb{Z}^2$, a class $\alpha \in H_1(M, \mathbb{Z})$ contains a prime geodesic if and only if $\alpha = (a, b)$ is primitive.

(2) Gromov [**29**] considered the following variation. Represent each α in $H_1(M, \mathbb{Z})$ by a closed orbit with shortest length $c(\alpha)$. Let

$$N(x) = \#\{\alpha \in H_1(M, \mathbb{Z}) \ : \ c(\alpha) \leq x\}.$$

Then as $x \to \infty$, we have $N(x) \sim c' x^{b_1(M)}$ for some computable constant c'. In view of this, the above conjecture for infinite $H_1(M, \mathbb{Z})$ can be restated as the existence of a positive constant c'' independent of α such that

$$\pi(x, \alpha) \sim c'' \ \frac{e^{hx}}{hxN(x)} \qquad \text{as } x \to \infty.$$

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LECTURE 4

The Ihara zeta function

4.1. The Ihara zeta function

We begin by noting that

$$GL_2(\mathbb{R}) = GL_2^+(\mathbb{R}) \sqcup GL_2^+(\mathbb{R}) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

where $GL_2^+(\mathbb{R})$ consists of 2×2 real matrices with positive determinant, and

$$GL_2^+(\mathbb{R}) = SL_2(\mathbb{R})\mathcal{Z}_{\mathbb{R}}$$

with $\mathcal{Z}_{\mathbb{R}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} : a \in \mathbb{R}^\times \right\}$ the center of $GL_2(\mathbb{R})$. These simple observations give rise to the following identification of the upper half-plane:

$$\begin{aligned} \mathfrak{H} \approx SL_2(\mathbb{R})/SO_2(\mathbb{R}) &= GL_2^+(\mathbb{R})/SO_2(\mathbb{R})\mathcal{Z}_{\mathbb{R}} \\ &= GL_2(\mathbb{R})/O_2(\mathbb{R})\mathcal{Z}_{\mathbb{R}} \\ &= PGL_2(\mathbb{R})/PO_2(\mathbb{R}), \end{aligned}$$

where

$$O_2(\mathbb{R}) = SO_2(\mathbb{R}) \sqcup SO_2(\mathbb{R}) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$PO_2(\mathbb{R}) = O_2(\mathbb{R})/(O_2(\mathbb{R}) \cap \mathcal{Z}_{\mathbb{R}}) = O_2(\mathbb{R})/\{\pm I_2\}$$

are the standard maximal compact subgroups of $GL_2(\mathbb{R})$ and $PGL_2(\mathbb{R})$, respectively.

Regarding $\mathbb{R}$ as the completion of $\mathbb{Q}$ with respect to the archimedean valuation, Ihara S considered the Selberg zeta function with $\mathbb{R}$ replaced by $\mathbb{Q}_p$, the completion of $\mathbb{Q}$ with respect to the p-adic valuation on $\mathbb{Q}$, and $\mathfrak{H}$ replaced by $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$. Here $\mathbb{Z}_p$ is the ring of integers of $\mathbb{Q}_p$ and $PGL_2(\mathbb{Z}_p)$ is the standard maximal compact subgroup of $PGL_2(\mathbb{Q}_p)$, which is also an open subgroup.

Let Γ be a discrete torsion-free cocompact subgroup of $PGL_2(\mathbb{Q}_p)$. Then $X_\Gamma = \Gamma\backslash PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ is finite. Ihara [**43**] considered the zeta function analogous to the Selberg zeta function:

$$Z(X_\Gamma, u) = \prod_{\substack{\text{conjugacy class } [\gamma] \\ \text{of primitive } \gamma \text{ in } \Gamma}} \frac{1}{1 - u^{\ell(\gamma)}}.$$

Here a nonidentity element γ in Γ has two distinct eigenvalues α and β in $\mathbb{Q}_p^\times$ (defined up to nonzero common multiples in $\mathbb{Q}_p^\times$), and the length of γ is given by $\ell(\gamma) := \max\{\mathrm{ord}_p \frac{\alpha}{\beta}, \mathrm{ord}_p \frac{\beta}{\alpha}\}$. Using this zeta function, Ihara showed that Γ is a free group.

Serre [**90**] observed that $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ has a natural structure as an infinite tree, as we shall explain in Lecture 6, §4, so that X_Γ is a finite graph. Moreover, this tree is the universal cover of X_Γ which has Γ as its fundamental group. This explains why Γ is free. Similar to what we saw in Lecture 2, there is a bijective length preserving correspondence between the set of conjugacy classes of primitive elements in Γ and the set of equivalence classes of primitive closed geodesic cycles on the graph X_Γ so that the Ihara zeta function can be interpreted as counting closed geodesic cycles in X_Γ. As such Ihara's definition readily extends to finite graphs in general.

Let X be a finite connected undirected graph. A cycle on X is a closed path with no backtracking; it has a starting point and an orientation. Two closed paths are *equivalent* if one is obtained from the other by changing the starting point and retaining the orientation. A cycle is called a *geodesic*[1] if all closed paths equivalent to it are backtrackless. It is called *primitive* if it is not obtained from traveling along a shorter cycle more than once. The length $\ell(C)$ of a geodesic cycle C is the number of edges in the path. Denote the equivalence class of C by $[C]$.

For $n \geq 1$, let $N_n(X)$ denote the number of geodesic cycles on X of length n. The Ihara zeta function of X counts the number of geodesic cycles on X. More precisely it is defined as

$$\begin{aligned} Z(X,u) : &= \exp\left(\sum_{n=1}^{\infty} \frac{N_n(X)}{n} u^n\right) \\ &= \prod_{\substack{[C] \\ C \text{ primitive geodesic cycle}}} \frac{1}{1-u^{\ell(C)}}, \end{aligned}$$

which converges absolutely for $|u|$ small. Here the second equality follows from taking logarithmic derivatives of both sides. For a graph, the equivalence classes of primitive geodesic cycles play the role of "primes".

The definition of the zeta function of a graph X is similar to that of a curve V defined over $\mathbb{F}_q$. The table below compares terminologies in the two settings.

Graph X	Curve V defined over $\mathbb{F}_q$
Primitive geodesic cycle C	Point $P \in V(\mathbb{F}_{q^n})$ with n min'l
Equivalence class $[C]$	Closed point v containing P
$\ell(C)$ = no. of geod. cycles in $[C]$	$\deg v$ = no. of points in v
Repeat C for m times	Regard P as a point in $V(\mathbb{F}_{q^{nm}})$

EXAMPLE 4.1. If X is a tree, then $Z(X,u) = 1$ since there are no cycles in a tree. Further, if a graph has some "ends", its zeta function remains unchanged when the ends are removed. Hence the Ihara zeta function depends only on the homotopy class of a graph.

EXAMPLE 4.2. If X is a cycle of length n, then

$$N_m(X) = \begin{cases} 2n & \text{if } m \equiv 0 \mod n, \\ 0 & \text{else.} \end{cases}$$

[1]It is called a tailless geodesic or reduced in some literature.

Thus X has only two primes, each of length n, and

$$Z(X,u) = \frac{1}{(1-u^n)^2}.$$

EXAMPLE 4.3. If X consists of two cycles glued at a vertex, then there are infinitely many equivalence classes of primitive geodesic cycles. They are in one-to-one correspondence with non-periodic reduced words in two variables. In this case, $Z(X,u)$ is an infinite product.

4.2. Ihara's theorem

The zeta function of a graph shares some properties similar to that of a curve over a finite field. The number of neighbors of a vertex x in a graph is called the *degree* of x. A graph is called k-regular if each vertex has degree equal to k. The tree $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ is $(p+1)$-regular and so is its quotient $X_\Gamma = \Gamma\backslash PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ considered by Ihara.

THEOREM 4.1 (Ihara [**43**]). *Suppose X is a $(q+1)$-regular graph on n vertices. Then $Z(X,u)$ is a rational function in u of the form*

$$Z(X,u) = \frac{(1-u^2)^{\chi(X)}}{\det(I - Au + qu^2 I)},$$

where $\chi(X) = \#$ of vertices $- \#$ of edges $= n - \frac{q+1}{2}n = \frac{1-q}{2}n$ is the Euler characteristic of X, and A is the adjacency matrix of X.

Recall that the adjacency matrix of a finite (directed or undirected) graph Y is a square matrix whose rows and columns are indexed by the vertices of Y so that the vv' entry records the number of edges from v to v'. An undirected edge is viewed as an edge with two opposite directions.

Since X is undirected, A is a real symmetric matrix and hence has n real eigenvalues which can be arranged in non-increasing order:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n.$$

The collection of eigenvalues is called the *spectrum of the graph X*.

Using the eigenvalues of the adjacency matrix we write

$$\det(I - Au + qu^2 I) = \prod_{i=1}^{n}(1 - \lambda_i u + qu^2),$$

so that

$$Z(X,u) = \frac{1}{(1-u^2)^{\frac{q-1}{2}n}\prod_{i=1}^{n}(1-\lambda_i u + qu^2)}$$

is the reciprocal of a polynomial.

We point out that the factor $1-u^2$ in the numerator of Ihara's expression of the zeta function is independent of the graph, and its exponent $\chi(X)$ gives topological information of the graph. The denominator provides spectral information of the graph, with each eigenvalue giving rise to two poles of the zeta function. This is parallel to Theorem 2.4 for the Selberg zeta function, which was proved twenty years after Ihara's theorem. The denominator of the zeta function of a $(q+1)$-regular graph, $\det(I - Au + qu^2 I)$, satisfy the symmetry relating u and $1/qu$. If we

set $u = q^{-s}$, then this is a functional equation relating s and $1 - s$ similar to the Selberg zeta function.

If X is not regular, then $Z(X, u)$ is still a rational function in u as described in the following result.

THEOREM 4.2 (Bass [**8**], Stark-Terras [**94**]). *The zeta function of the graph X with vertices $v_1, ..., v_n$ is a rational function of the form*

$$Z(X, u) = \frac{(1 - u^2)^{\chi(X)}}{\det(I - Au + Qu^2)},$$

where

$$Q = \begin{pmatrix} \deg(v_1) - 1 & & & \\ & \deg(v_2) - 1 & & \\ & & \ddots & \\ & & & \deg(v_n) - 1 \end{pmatrix}$$

is the degree matrix of X minus the identity matrix, and the ij-entry of A is the number of edges from v_i to v_j.

4.3. Hashimoto's theorem

Endow two opposite orientations on each edge of X. The out-neighbors of the directed edge $\overrightarrow{vv'}$ are $\overrightarrow{v'v''}$ with $v'' \neq v$. Thus if X is $(q + 1)$-regular, then each directed edge of X has q out-neighbors. Suppose X has r_1 edges, then it has $2r_1$ directed edges. We may form a $(2r_1) \times (2r_1)$ (directed) edge adjacency matrix $T = (t_{ee'})$ whose rows and columns are indexed by the directed edges of X. For $n \geq 1$, the ee diagonal entry of T^n is the sum of $t_{ee_2}t_{e_2e_3} \cdots t_{e_{n-1}e_n}t_{e_ne}$ over all possible directed edges $e_2, ..., e_n$ such that for each pair (ab) of directed edges occurring in the index of the product, b is an out-neighbor of a. Therefore $e \to e_2 \to e_3 \to \cdots \to e_n \to e$ is a geodesic cycle of length n expressed in terms of adjacent directed edges of X. Conversely each geodesic cycle of length n in X starting with the directed edge e contributes 1 to the ee entry of T^n. Hence the ee entry of T^n counts the number of length n geodesic cycles in X starting with the edge e. Consequently we have, for each $n \geq 1$,

$$\operatorname{tr} T^n = N_n(X).$$

This gives rise to the second expression of the zeta function of X.

THEOREM 4.3 (Hashimoto [**33**]). *We have*

$$Z(X, u) = \frac{1}{\det(I - Tu)}.$$

PROOF. Let $\alpha_1, \alpha_2, \ldots, \alpha_{2r_1}$ be the eigenvalues of T. Then

$$\operatorname{tr} T^n = \sum_{j=1}^{2r_1} \alpha_j^n.$$

So we have

$$\begin{aligned} Z(X,u) &= \exp\left(\sum_{n=1}^{\infty}\frac{\operatorname{tr}T^n}{n}u^n\right) = \exp\left(\sum_{n=1}^{\infty}\sum_{j=1}^{2r_1}\frac{\alpha_j^n}{n}u^n\right) \\ &= \exp\left(\sum_{j=1}^{2r_1}\sum_{n=1}^{\infty}\frac{(\alpha_j u)^n}{n}\right) = \exp\left(\sum_{j=1}^{2r_1}\log\frac{1}{1-\alpha_j u}\right) \\ &= \prod_{j=1}^{2r_1}\frac{1}{1-\alpha_j u} = \frac{1}{\det(I-Tu)}. \end{aligned}$$

□

Combining the above two theorems expressing $Z(X,u)$, we obtain the following zeta identity

$$\frac{(1-u^2)^{\chi(X)}}{\det(I-Au+Qu^2)} = \frac{1}{\det(I-Tu)} = Z(X,u). \tag{3}$$

If we can prove the first identity, then Ihara's theorem and its generalization by Bass will be proved.

4.4. A cohomological proof of the zeta identity for a graph

This proof is based on Bass's generalization of Ihara's theorem, rearranged by Hoffman [**38**]. The clue comes from $\chi(X)$ being the number, r_0, of vertices minus the number, r_1, of edges. Rewrite the zeta identity (3) as

$$\frac{(1-u^2)^{r_0}}{(1-u^2)^{r_1}} = \frac{\det(I-Au+Qu^2)}{\det(I-Tu)}.$$

We review some basic concepts in cohomology theory. Consider the cochain complex

$$C^*: \qquad 0 \xrightarrow{d^{-1}} C^0 \xrightarrow{d^0} C^1 \xrightarrow{d^1} C^2 \to \cdots \xrightarrow{d^{n-1}} C^n \xrightarrow{d^n} 0,$$

where each C^i is a free M-module of finite rank (we will use $M = \mathbb{C}[u]$), and each d^i is an M-homomorphism satisfying $d^{i+1}\circ d^i = 0$. The cohomology group $H^i(C^*) = \ker d^i/\operatorname{Im} d^{i-1}$ measures the failure of exactness at C^i of the sequence. By a cochain map $\Phi = \{\Phi^i\}: C^* \to C^*$ we mean a collection of M-homomorphisms $\Phi^i: C^i \to C^i$ for $0 \le i \le n$ such that each square of the diagram

$$\begin{array}{ccccccccccccc} 0 & \longrightarrow & C^0 & \xrightarrow{d^0} & C^1 & \xrightarrow{d^1} & C^2 & \xrightarrow{d^2} & C^3 & \cdots & C^n & \xrightarrow{d^n} & 0 \\ & & \downarrow{\scriptstyle \Phi^0} & & \downarrow{\scriptstyle \Phi^1} & & \downarrow{\scriptstyle \Phi^2} & & \downarrow{\scriptstyle \Phi^3} & & \downarrow{\scriptstyle \Phi^n} & & \\ 0 & \longrightarrow & C^0 & \xrightarrow{d^0} & C^1 & \xrightarrow{d^1} & C^2 & \xrightarrow{d^2} & C^3 & \cdots & C^n & \xrightarrow{d^n} & 0 \end{array}$$

is commutative. Then each homorphism Φ^i induces a map on $H^i(C^*)$.

We shall use the following general fact comparing the effects of a cochain map on the cochain groups and the cohomology groups.

Fact. *We have*

$$\prod_{i=0}^{n}\det\left(\Phi^i \mid C^i\right)^{(-1)^i} = \prod_{i=0}^{n}\det\left(\Phi^i \mid H^i(C^*)\right)^{(-1)^i}$$

provided that none of the determinants in the denominators vanish.

In our case, we set

$$C^0 = \{f : \text{vertices of } X \to \mathbb{C}[u]\},$$
$$C^1 = \{g : \text{directed edges of } X \to \mathbb{C}[u]\}.$$

Then C^0 and C^1 are free $\mathbb{C}[u]$-modules of rank r_0 and $2r_1$, respectively.

Define $d^0 : C^0 \to C^1$ by

$$(d^0 f)(\overrightarrow{vv'}) = uf(v) - f(v').$$

Our complex is

$$C^* : \qquad 0 \xrightarrow{d^{-1}} C^0 \xrightarrow{d^0} C^1 \xrightarrow{d^1} 0.$$

Further, $H^0(C^*) = \ker d^0$ and $H^1(C^*) = C^1/\text{Im } d^0$.

Now we define two maps from C^* to C^*. The first map $\Phi_1 = \{\Phi_1^0, \Phi_1^1\}$ is given as follows:

$$\Phi_1^0 : C^0 \to C^0 \text{ is } I - Au + Qu^2;$$
$$\Phi_1^1 : C^1 \to C^1 \text{ is } (I - Ju) \circ (I - Tu).$$

Here $Jg(\overrightarrow{vv'}) = g(\overrightarrow{v'v})$ for all $g \in C^1$ and all directed edges $\overrightarrow{vv'}$ of X. The second map Φ_2 is simply multiplication by $1 - u^2$ on each C^i. Hence Φ_2 is a cochain map.

Define $\sigma : C^1 \to C^0$ by

$$(\sigma g)(v) = u \sum_{\substack{\text{all edges} \\ \text{out of } v}} g(\overrightarrow{vv'}).$$

We compare the two maps Φ_1 and Φ_2 on C^*.

PROPOSITION 4.3.1. (i) $\Phi_1^0 - \Phi_2^0 = \sigma \circ d^0$,
(ii) $\Phi_1^1 - \Phi_2^1 = d^0 \circ \sigma$.

In other words, we have the following commutative diagram (which means that Φ_1 and Φ_2 are homotopic):

$$\begin{array}{ccc} C^0 & \xrightarrow{d^0} & C^1 \\ \Phi_1^0 - \Phi_2^0 \downarrow & \swarrow \sigma & \downarrow \Phi_1^1 - \Phi_2^1 \\ C^0 & \xrightarrow{d^0} & C^1 \end{array}$$

PROOF. To prove (i), let $f \in C^0$. Then at a vertex v we have

$$\begin{aligned}(\Phi_1^0 - \Phi_2^0)f(v) &= (I - Au + Qu^2)f(v) - (1 - u^2)f(v) \\ &= -u \sum_{\overrightarrow{vv'}} f(v') + (\deg v)u^2 f(v)\end{aligned}$$

and

$$\begin{aligned}\sigma(d^0 f)(v) &= u\sum_{\overrightarrow{vv'}}(d^0 f)(\overrightarrow{vv'}) = u\sum_{\overrightarrow{vv'}}\big(uf(v) - f(v')\big)\\ &= u^2(\deg v)f(v) - u\sum_{\overrightarrow{vv'}} f(v').\end{aligned}$$

This proves (i).

For (ii), first note

$$(I - Ju)(I - Tu) = I - (J + T)u + u^2 JT.$$

Let $g \in C^1$. Then at a directed edge $\overrightarrow{vv'}$ we have

$$\begin{aligned}(\Phi_1^1 - \Phi_2^1)g(\overrightarrow{vv'}) &= (I - Ju)(I - Tu)g(\overrightarrow{vv'}) - (1 - u^2)g(\overrightarrow{vv'})\\ &= -u(J + T)g(\overrightarrow{vv'}) + u^2 J(Tg)(\overrightarrow{vv'}) + u^2 g(\overrightarrow{vv'})\\ &= -u\left(g(\overrightarrow{v'v}) + \sum_{v'' \neq v} g(\overrightarrow{v'v''})\right) + u^2 Tg(\overrightarrow{v'v}) + u^2 g(\overrightarrow{vv'})\\ &= -u\sum_{\substack{\text{all edges}\\ \text{out of } v'}} g(\overrightarrow{v'v''}) + u^2\sum_{w \neq v'} g(\overrightarrow{vw}) + u^2 g(\overrightarrow{vv'})\\ &= -u\sum_{\substack{\text{all edges}\\ \text{out of } v'}} g(\overrightarrow{v'v''}) + u^2\sum_{\substack{\text{all edges}\\ \text{out of } v}} g(\overrightarrow{vw}).\end{aligned}$$

On the other hand,

$$d^0(\sigma g)(\overrightarrow{vv'}) = u(\sigma g)(v) - (\sigma g)(v') = u^2\sum_{\substack{\text{all edges}\\ \text{out of } v}} g(\overrightarrow{vw}) - u\sum_{\substack{\text{all edges}\\ \text{out of } v'}} g(\overrightarrow{v'v''}).$$

This proves (ii). □

An immediate consequence of the above proposition is

COROLLARY 4.3.1. $d^0 \circ (\Phi_1^0 - \Phi_2^0) = d^0 \circ \sigma \circ d^0 = (\Phi_1^1 - \Phi_2^1) \circ d^0$.

Hence $\Phi_1 - \Phi_2$ is a cochain map. Since Φ_2 is a cochain map, so is Φ_1.

As a consequence of Proposition 3.3.1, (i), $\Phi_1^0 - \Phi_2^0$ on $H^0(C^*)$ is 0, namely $\Phi_1^0 = \Phi_2^0$ on $H^0(C^*)$. Similarly, Proposition 4.3.1, (ii) implies $\Phi_1^1 - \Phi_2^1$ on $H^1(C^*)$ is 0, in other words, $\Phi_1^1 = \Phi_2^1$ on $H^1(C^*)$. Applying the **Fact**, we get

$$\begin{aligned}\frac{\det(I - Au + Qu^2)}{\det(I - Ju)\det(I - Tu)} &= \frac{\det(\Phi_1^0 \mid C^0)}{\det(\Phi_1^1 \mid C^1)} = \frac{\det(\Phi_1^0 \mid H^0(C^*))}{\det(\Phi_1^1 \mid H^1(C^*))}\\ &= \frac{\det(\Phi_2^0 \mid H^0(C^*))}{\det(\Phi_2^1 \mid H^1(C^*))} = \frac{\det(\Phi_2^0 \mid C^0)}{\det(\Phi_2^1 \mid C^1)} = \frac{(1 - u^2)^{r_0}}{(1 - u^2)^{2r_1}}.\end{aligned}$$

The zeta identity will follow if we show $\det(I - Ju) = (1 - u^2)^{r_1}$.

By pairing off $\overrightarrow{vv'}$ with $\overrightarrow{v'v}$, we can represent J by r_1 diagonal blocks of the 2×2 matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Therefore

$$\det(I - Ju) = \left(\det \begin{pmatrix} 1 & -u \\ -u & 1 \end{pmatrix} \right)^{r_1} = (1 - u^2)^{r_1},$$

as desired. This completes the proof of the zeta identity (3) for the graph X, and hence Theorems 4.1 and 4.2 follow.

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LECTURE 5

Spectral graph theory

5.1. Eigenvalues of a regular graph

Let X be a finite connected undirected $(q+1)$-regular graph on n vertices, with spectrum

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n.$$

Let f be a non-zero eigenfunction of A with eigenvalue λ. Suppose that $\max_v |f(v)|$, as v runs through all vertices of X, occurs at vertex v_0. Then

$$\lambda f(v_0) = (Af)(v_0) = \sum_{v \text{ adjacent to } v_0} f(v).$$

As v_0 has exactly $q+1$ neighbors and $|f(v)| \leq |f(v_0)|$ for all vertices v in X, we have

$$|\lambda||f(v_0)| \leq \sum_{v \text{ adjacent to } v_0} |f(v)| \leq (q+1)|f(v_0)|,$$

yielding $|\lambda| \leq q+1$.

As the constant functions are eigenfunctions of A with eigenvalue $q+1$, we have

$$q+1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq -(q+1).$$

The multiplicity of $q+1$ as an eigenvalue of A is equal to the number of connected components of X. Thus a $(q+1)$-regular graph X is connected if and only if $\lambda_1 = q+1 > \lambda_2$. Further, the above argument shows that $\lambda_n = -(q+1)$ if and only if X is bipartite (since X is assumed to be connected). Moreover, the eigenvalues of a bipartite graph is symmetric with respect to 0. So if X is bipartite, then the eigenvalue $-(q+1)$ also has multiplicity one.

Call $\pm(q+1)$ the trivial eigenvalues of A (or X), and the remaining eigenvalues nontrivial. We investigate the distribution of the nontrivial eigenvalues.

5.2. The behavior of nontrivial eigenvalues

Denote by $\lambda^-(X)$ the smallest nontrivial eigenvalue of X. It is $\lambda_n(X)$ if X is not bipartite, and it is equal to $\lambda_{n-1}(X) = -\lambda_2(X)$ if X is bipartite. We are interested in the behavior of $\lambda_2(X)$ and $\lambda^-(X)$. Denote by $|X|$ the number of vertices in a graph X.

THEOREM 5.1 (Alon-Boppana). *Let $\{X_j\}$ be a family of connected undirected $(q+1)$-regular graphs such that $|X_j| \to \infty$ as $j \to \infty$. Then*

$$\liminf_{j\to\infty} \lambda_2(X_j) \geq 2\sqrt{q}.$$

There are several proofs of this theorem. The first proof appeared in the paper [**67**] by Lubotzky, Phillips and Sarnak, who proved the above theorem with $\lambda_2(X)$ replaced by $\lambda(X) := \max\{|\lambda_2(X)|, |\lambda^-(X)|\}$. This was shown by counting the number of closed walks of even length $2m$ starting at a fixed point in the infinite $(q+1)$-regular tree, which is the universal cover of a $(q+1)$-regular graph X. Since a tree doesn't contain cycles, this count provides a lower bound for the number of closed walks in X of length $2m$, which is equal to

$$\mathrm{tr}(A^{2m}) = \lambda_1^{2m} + \lambda_2^{2m} + \cdots + \lambda_n^{2m} \leq 2(q+1)^{2m} + (n-2)\lambda(X)^{2m}.$$

Here $\lambda_1, ..., \lambda_n$ are the spectrum of X. This leads to a lower bound for $\lambda(X)$ by taking $2m^{th}$ root and then letting m go to infinity.

The second proof relates eigenvalues to the diameter of the graph.

THEOREM 5.2 (Nilli [**77**]). *Let X be a finite connected undirected $(q+1)$-regular graph with diameter diam$(X) \geq 2b+2$. Then*

$$\lambda_2(X) \geq 2\sqrt{q} - \frac{2\sqrt{q}-1}{b+1}.$$

The diameter of a graph X is the maximum distance between two vertices in X. Note that Nilli's Theorem implies the Alon-Boppana Theorem. Indeed, since $|X_j| \to \infty$ as $j \to \infty$ and all vertices have degree $q+1$, we must have $\mathrm{diam}(X_j) \to \infty$ as $j \to \infty$.

The third approach says that, given any value y slightly less than $2\sqrt{q}$, a fixed positive proportion of the eigenvalues of any $(q+1)$-regular graph are greater than y.

THEOREM 5.3 (Serre [**91**, **92**]). *Given $\varepsilon > 0$, there exists a constant $c = c(\varepsilon, q+1)$ such that for all finite connected undirected $(q+1)$-regular graphs X, the number of eigenvalues λ of X with $\lambda \geq (2-\varepsilon)\sqrt{q}$ is at least $c|X|$.*

Serre's theorem is stronger than the previous two theorems since it implies that for any *fixed* $r \geq 2$, we have

$$\liminf_{j\to\infty} \lambda_r(X_j) \geq 2\sqrt{q}.$$

The above theorems provide a satisfactory description for the largest nontrivial eigenvalue. A natural question is whether there is a similar description for the smallest non-trivial eigenvalues. Namely do we always have

$$\limsup_{j\to\infty} \lambda^-(X_j) \leq -2\sqrt{q} \ ?$$

The answer is no. A counterexample is provided by the line graph Y of a regular graph X: the vertices of Y represent the edges of X and two vertices in Y are adjacent if the corresponding edges in X share a common vertex. The graph Y is again regular and Sachs [**82**] showed that $\lambda^-(Y) \geq -2$. (See also [**65**].) However the following result shows that the question has a positive answer under a mild assumption.

THEOREM 5.4 (Li [**60**] with comments by Serre). *Let $\{X_j\}$ be the same as in Theorem 5.1. Suppose that the length of the shortest odd cycle in X_j tends to ∞ as $j \to \infty$. (A graph with no odd cycles has shortest odd cycle of length ∞.) Then*

$$\limsup_{j\to\infty} \lambda^-(X_j) \leq -2\sqrt{q}.$$

The reader is referred to [**61**] for more detail.

5.3. Ramanujan graphs and the Riemann hypothesis

As discussed in the previous section, in general, a large connected undirected $(q+1)$-regular graph X tends to have $\lambda_2(X) > 2\sqrt{q}$ and $\lambda^-(X) < -2\sqrt{q}$. If this doesn't happen, X is called a Ramanujan graph.

DEFINITION 5.1. A finite connected undirected $(q+1)$-regular graph X is called *Ramanujan* if all nontrivial eigenvalues λ of X satisfy

$$|\lambda| \leq 2\sqrt{q}.$$

Ramanujan graphs are thus spectrally extremal with all nontrivial eigenvalues lying in the interval $[-2\sqrt{q}, 2\sqrt{q}]$.

The universal cover of a $(q+1)$-regular graph is the (infinite) $(q+1)$-regular tree $\mathcal{T}$. The adjacency operator A acts on $L^2(\mathcal{T})$, sending a function f to the function Af whose value at a vertex v is given by

$$(Af)(v) = \sum_{v' \text{ adjacent to } v} f(v').$$

This indicates that the adjacency operator is defined locally. The operator spectrum of A is known to be $[-2\sqrt{q}, 2\sqrt{q}]$, and $2\sqrt{q}$ is called the covering radius for $\mathcal{T}$.

Hence, a finite $(q+1)$-regular graph X is Ramanujan if and only if all of its nontrivial eigenvalues lie in the spectrum of its universal cover.

Let X be a $(q+1)$-regular connected undirected graph with $|X| = n$ and spectrum

$$\lambda_1 > \lambda_2 \geq \cdots \geq \lambda_n$$

as before. Recall that the zeta function of X is

$$Z(X,u) = \frac{1}{(1-u^2)^{\frac{q-1}{2}n} \prod_{j=1}^{n} (1 - \lambda_j u + qu^2)}.$$

For $\lambda_1 = q+1$, we have $1 - \lambda_1 u + qu^2 = 1 - (q+1)u + qu^2 = (1-u)(1-qu)$. If $\lambda_n = -(q+1)$, we have $1 - \lambda_n u + qu^2 = 1 + (q+1)u + qu^2 = (1+u)(1+qu)$. For each nontrivial eigenvalue λ_j, we have $1 - \lambda_j u + qu^2 = (1-\alpha_j u)(1-\beta_j u)$. The poles α_j^{-1} and β_j^{-1} from the nontrivial eigenvalues λ_j are the *nontrivial poles* of $Z(X,u)$. We remark that all nonreal poles of $Z(X,u)$ lie on the circle centered at the origin and with radius $q^{-1/2}$. The real nontrivial poles of $Z(X,u)$ fall in the interval $(-1,-1/q) \cup (1/q,1)$. To see this, suppose λ is a nontrivial eigenvalue with $\alpha = \lambda/2 + \sqrt{\lambda^2 - 4q}/2$ and $\beta = \lambda/2 - \sqrt{\lambda^2 - 4q}/2$ which contribute two nontrivial real poles α^{-1} and β^{-1} of $Z(X,u)$. Then either $0 < \beta \leq \lambda/2 \leq \alpha$ or $\beta \leq \lambda/2 \leq \alpha < 0$ because $\alpha\beta = q > 0$. As $|\lambda| < q+1$, we have $\sqrt{\lambda^2 - 4q}/2 < (q-1)/2$, and none of α, β can be $\pm 1, \pm q$. This gives either $0 < \alpha < (q+1)/2 + (q-1)/2 = q$ or $-q = -(q+1)/2 - (q-1)/2 < \beta < 0$. Since $\alpha\beta = q$, this shows that either $1 < \beta \leq \alpha < q$ or $-q < \beta \leq \alpha < -1$, as claimed. Consequently, $Z(X,u)$ is holomorphic on the disk $|u| < 1/q$.

$Z(X,u)$ is said to satisfy the Riemann Hypothesis if all the nontrivial poles of $Z(X,u)$ have the same absolute value $q^{-1/2}$. Writing $u = q^{-s}$, one sees that $Z(X,q^{-s})$ satisfies the Riemann Hypothesis if and only if all of its nontrivial poles lie on the line $\Re s = \frac{1}{2}$.

Observe that $|\alpha_j| = |\beta_j| = q^{1/2}$ if and only if $|\lambda_j| \leq 2\sqrt{q}$. Therefore X is Ramanujan if and only if $Z(X, u)$ satisfies the Riemann Hypothesis. This is the discrete analogue of what we saw in §5 of Lecture 2 that the Selberg zeta function $Z(X_\Gamma, s)$ of a compact quotient $X_\Gamma = \Gamma\backslash\mathfrak{H}$ of the upper half-plane $\mathfrak{H}$ by a discrete torsion-free subgroup Γ of $SL_2(\mathbb{R})$ satisfies the Riemann Hypothesis if and only if the nontrivial eigenvalues of the Laplacian Δ_Γ on X_Γ fall in the spectrum $[\frac{1}{4}, \infty)$ of the Laplacian Δ $\mathfrak{H}$. As discussed in Lecture 2, the latter is conjectured to hold for congruence subgroups of $SL_2(\mathbb{R})$ by Selberg, and it is known to fail for many Γ's.

It should be pointed out that the zeta function attached to a smooth irreducible projective curve defined over a finite field always satisfies the Riemann Hypothesis, while only the zeta function of a Ramanujan graph satisfies the Riemann Hypothesis, despite the similarity in the definitions of their zeta functions.

5.4. Friedman's theorem

J. Friedman proved Alon's second eigenvalue conjecture, which says that a large random regular graph is very close to being Ramanujan in the following sense.

THEOREM 5.5 (Friedman [**26**]). *Given $\varepsilon > 0$, the probability that a finite connected undirected $(q+1)$-regular graph X has all of its nontrivial eigenvalues λ satisfying $|\lambda| \leq (2+\varepsilon)\sqrt{q}$ goes to 1 as $|X| \to \infty$.*

In other words, the density of $(q+1)$-regular graphs X having a nontrivial eigenvalue λ with $|\lambda| > (2+\varepsilon)\sqrt{q}$ goes to zero as $|X| \to \infty$.

10.1090/cbms/129/06

LECTURE 6

Explicit constructions of Ramanujan graphs

6.1. Expanding constant and spectral gap

Before we delve into the technical details, it is inspiring to shed some light on the significance of Ramanujan graphs from the view point of applications.

There has always been a great interest to construct efficient communication networks where different centers/terminals are connected by cables so as to allow communication between them. However, cables are expensive, so companies have incentive to reduce the usage of cables while still maintain the effectiveness of the communication network. Ramanujan graphs are used to serve this purpose, i.e., strike a balance between costs and efficacy.

Let X be a finite, connected, undirected graph with vertex set V. For two subsets $S, T \subseteq V$ let $E(S,T)$ denote the set of edges between S and T. The *expanding constant* of X is defined as

$$h(X) = \min_{Y \subset V,\ |Y| \leq |V|/2} \frac{|E(Y, V \smallsetminus Y)|}{|Y|}.$$

In practice, Y has information that needs to be sent out. Hence, the more edges that go out of Y, the better this information could be sent. The ratio measures the rate that the information at Y can expand in the network. Notice that the numerator is symmetric in Y and its complement $V \smallsetminus Y$, for the denominator we choose the smaller set between these two. Since we do not know which vertices will send the information we run through all possibilities. The minimum value is a guaranteed expansion rate. The expanding constant is the discrete version of the Cheeger constant in geometry.

Usually, as $|X|$ gets larger, $h(X)$ becomes smaller. Computer scientists are interested in finding an infinite set of graphs that have size going to infinity while still having the expanding constants bounded from below away from zero. A family of graphs $\{X_j\}$ with $|X_j| \to \infty$ is called an *expander* if there is a positive constant c such that $h(X_j) > c$ for all j. Expanders have wide applications in computer science. The reader is referred to the paper [**39**] by Hoory, Linial and Wigderson for a comprehensive survey.

Computing $h(X)$ is difficult in general. It is an NP complete problem. In order to estimate the expanding constant, we need a way to get a handle on $h(X)$. In case X is k-regular, Tanner [**97**] and Alon-Milman [**4**] proved the following lower and upper bounds, respectively, for the expanding constant expressed in terms of the spectral gap, namely the difference between $\lambda_1 = k$ and λ_2:

$$\frac{k - \lambda_2(X)}{2} \leq h(X) \leq \sqrt{2k\big(k - \lambda_2(X)\big)}.$$

This inequality clearly indicates that $h(X)$ is large if and only if the spectral gap is large. Therefore the problem at hand is reduced to finding regular graphs which have a large spectral gap.

The Alon-Boppana Theorem provides an upper bound for the spectral gap

$$k - \lambda_2(X) \leq k - 2\sqrt{k-1} + o(1)$$

for $|X|$ large.

6.2. Some nice features of Ramanujan graphs

For a connected undirected k-regular graph X on n vertices with spectrum

$$k = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_n \geq -k,$$

let $\lambda(X) := \max\{|\lambda_2(X)|, |\lambda^-(X)|\}$ denote the maximum absolute value of the nontrivial eigenvalues of X. Recall that $\lambda^-(X)$ is equal to $\lambda_n(X)$ if X is not bipartite, and equal to $\lambda_{n-1}(X) = -\lambda_2(X)$ if X is bipartite.

The Ramanujan graphs have many nice properties that would make them great for communication networks. We list some of them here:

(1) Ramanujan graphs have large expanding constants. Hence, if there exists an infinite family of k-regular Ramanujan graphs, then they form an optimal expander.

(2) When X is not bipartite, Chung [**15**] gave an upper bound for its diameter in terms of $\lambda(X)$:

$$\operatorname{diam} X \leq \frac{\log(n-1)}{\log(k/\lambda(X))} + 1.$$

The diameter bound for X bipartite is a little worse, as shown by Quenell in [**80**]:

$$\operatorname{diam} X \leq \frac{\log\big((n-2)/2\big)}{\log\big(k/\lambda(X)\big)} + 2.$$

Since the diameter of X measures transmission delay, we see that Ramanujan graphs have short transmission delay.

(3) The mixing lemma, established by Alon-Chung [**3**], says that for any $S, T \subseteq V$ we have

$$\begin{aligned}\left||E(S,T)| - \frac{k}{n}|S||T|\right| &\leq \lambda(X)\sqrt{|S||T|\left(1 - \frac{|S|}{n}\right)\left(1 - \frac{|T|}{n}\right)} \\ &\leq \lambda(X)\sqrt{|S||T|},\end{aligned}$$

where the term $|E(S,T)|$ measures the actual number of edges between S and T, while $\frac{k}{n}|S||T|$ expresses the expected number of edges between S and T (if edges were placed "randomly"). One implication of this lemma is that the edges in a Ramanujan graph are uniformly distributed.

(4) Ramanujan graphs are sparse (i.e., the degree k remains fixed while n gets larger), hence are economical. As communication networks they are fault-tolerant and robust. In other words, if some connections in the network should go bad, the network can still continue to operate.

Thus it is very desirable to use Ramanujan graphs as communication networks. As it is time consuming to compute eigenvalues of a graph, we aim to construct such graphs whose nontrivial eigenvalues are guaranteed to satisfy the required bound.

6.3. Cayley graphs

A graph is a 1-dimensional simplicial complex. As such it is not easy to extract essential information of the graph. However, if there is an algebraic structure on its vertices, then more tools could be used. Initiated by Margulis, a Cayley graph is based on a group and its edges are constructed in a uniform way. Therefore, it admits a lot of symmetry and algebraic methods like representation theory can be employed to find its spectrum. Many important graphs are Cayley graphs.

Let G be a group and let S be a k-element subset of G which generates the whole group. Assume that S is symmetric, that is, $s \in S$ if and ony if $s^{-1} \in S$. The *Cayley graph* $\mathrm{Cay}(G, S)$ is a graph whose vertices are elements of G, and the out-neighbors of $g \in G$ are gs for $s \in S$. This defines a directed graph. But since S is symmetric and $g = (gs)s^{-1}$, we see that g is an out-neighbor of gs and hence the resulting graph is undirected. The Cayley graph is connected since S generates G, and it contains no loop if the identity element is not in S. Further, every vertex is adjacent to exactly k vertices obtained by multiplying by elements in S and hence the graph $\mathrm{Cay}(G, S)$ is k-regular.

Since the adjacency is by right multiplication and G acts on the space $L^2(G)$ of functions on the vertices of $\mathrm{Cay}(G, S)$ by left translation, the G action commutes with the adjacency operator A. Hence we can decompose $L^2(G)$ into a direct sum of irreducible subspaces each of which is invariant by A. In other words, we can represent A via diagonal blocks corresponding to irreducible subspaces and it suffices to find the eigenvalues of A on each block. For example, if G has a 1-dimensional representation χ, then the eigenvalue of A on the space of χ is $\sum_{s\in S} \chi(s)$. In general if ρ is an irreducible representation of G of dimension d, then the eigenvalues of A on the space of ρ are $\frac{1}{d}\sum_{s\in S} \operatorname{tr} \rho(s)$, with multiplicity d; this is a straightforward application of Schur's lemma.

These properties have been used to construct Ramanujan graphs. If a Cayley graph $\mathrm{Cay}(G, S)$ is Ramanujan, then it indicates that the set S is somewhat randomly distributed with respect to the group structure.

The most well-known Ramanujan graph is the Cayley graph on $PSL_2(\mathbb{F}_5)$ with the symmetric generating set

$$S = \left\{ \begin{pmatrix} 1 & \pm 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\} \tag{4}$$

mod 5. This 3-regular graph has 60 vertices; it is the planar graph on the surface of a soccer ball:

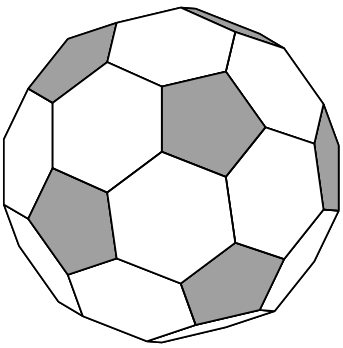


FIGURE 1. Buckyball $C_{60} = \mathrm{Cay}(PSL_2(\mathbb{F}_5), S \mod 5)$

It is called Buckyball C_{60} in chemistry, the most prominent and most studied of the fullerene molecules. A separator theorem of Lipton and Tarjan [**66**] implies that, for a fixed valency k, there are only finitely many planar k-regular Ramanujan graphs. Some other planar Ramanujan fullerenes are C_{80} and C_{84} shown below.[2]

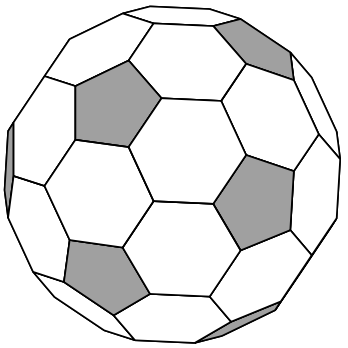


FIGURE 2. C_{80}

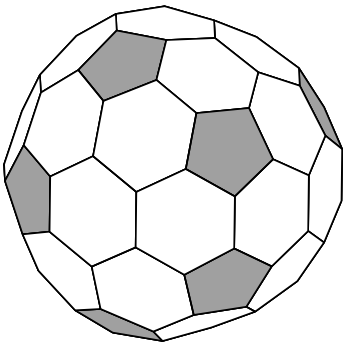


FIGURE 3. C_{84}

The set S given by (4) is known to generate the group $G = PSL_2(\mathbb{Z})$, so we have the 3-regular Cayley graph $\mathrm{Cay}(G, S)$. The Buckyball $\mathrm{Cay}(G \mod 5, S \mod 5)$ is Ramanujan, and so is the Cayley graph $\mathrm{Cay}(G \mod 7, S \mod 7)$, which can be realized on a genus-3 compact surface. While the Cayley graph on G modulo a prime p is no longer Ramanujan in general, Bourgain and Gamburd [**10**] showed that the Cayley graphs $\mathrm{Cay}(G \mod p, S \mod p)$ form a family of expanders with varying primes p. In fact, the same holds if S is replaced by any finite symmetric subset of G which generates a Zariski dense subgroup of G. They also proved in [**11**] that the same holds for G mod p^n, where the prime p is fixed and the exponent n varies. Later, the expanding result is shown to hold by Bourgain, Gamburd and Sarnak [**12**] for G modulo square-free moduli as well.

6.4. $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ as a $(p+1)$-regular tree

To an element $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $GL_2(\mathbb{Q}_p)$ we associate the $\mathbb{Z}_p$-module $L_g = \mathbb{Z}_p\binom{a}{c} \oplus \mathbb{Z}_p\binom{b}{d}$. It is a lattice of rank 2 in $\mathbb{Q}_p^2$. Clearly all rank 2 lattices are obtained this way. Two elements $g, h \in GL_2(\mathbb{Q}_p)$ give rise to the same lattice $L_g = L_h$ if and only if $g^{-1}h \in GL_2(\mathbb{Z}_p)$. Hence $GL_2(\mathbb{Q}_p)/GL_2(\mathbb{Z}_p)$ parameterizes all rank 2 lattices in $\mathbb{Q}_p^2$.

Two lattices L_1 and L_2 are said to be equivalent if $L_1 = \alpha L_2$ for some $\alpha \in \mathbb{Q}_p^\times$. Hence $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ parameterizes the equivalence classes $[L]$ of rank 2 lattices in $\mathbb{Q}_p^2$. They form the vertex set. Two vertices are adjacent if they can be represented by lattices L_1 and L_2 satisfying $L_1 \supsetneq L_2 \supsetneq pL_1 \supsetneq pL_2$. This makes $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ an undirected graph. Note that $L_1/pL_1 \cong \mathbb{F}_p \oplus \mathbb{F}_p$ and L_2/pL_1 is a line in $\mathbb{F}_p^2$. As there are $p+1$ lines in $\mathbb{F}_p^2$, this graph is $(p+1)$-regular. One can show that it is in fact a tree. See, for instance, Serre's book [**90**].

In the table below we list the group-theoretic interpretation of the combinatorial objects we have introduced so far. To define the action of the Hecke operator T_p

[2]Figures 1-3 created by Pascal Vontobel using xfig.

on the L^2-functions f on the vertices of the tree $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$, we first express the double coset

$$PGL_2(\mathbb{Z}_p)\begin{pmatrix}1 & 0\\ 0 & p\end{pmatrix}PGL_2(\mathbb{Z}_p) = \cup_{1\le i\le p+1} g_i PGL_2(\mathbb{Z}_p)$$

as a disjoint union of $p+1$ cosets. Then T_p sends a function f to $T_p(f)$ defined by

$$T_p(f)(gPGL_2(\mathbb{Z}_p)) = \sum_{1\le i\le p+1} f(gg_iPGL_2(\mathbb{Z}_p)).$$

It is straightforward to check that $[L_{gg_i}]$'s are the neighbors of the vertex $[L_g]$ defined above. The action of the Iwahori-Hecke operator on functions on directed edges of the tree $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ is defined similarly, using the double coset $\mathfrak{I}\begin{pmatrix}1 & 0\\ 0 & p\end{pmatrix}\mathfrak{I}$ of the Iwahori subgroup $\mathfrak{I}$. In this case the double coset is a disjoint union of p $\mathfrak{I}$-cosets.

Combinatorial terminology	Group-theoretic terminology
vertices	$PGL_2(\mathbb{Z}_p)$-cosets
(vertex) adjacency operator A	Hecke operator based on $PGL_2(\mathbb{Z}_p)\begin{pmatrix}1 & 0\\ 0 & p\end{pmatrix}PGL_2(\mathbb{Z}_p)$
directed edges	$\mathfrak{I}$-cosets, where $\mathfrak{I}$ is the Iwahori subgroup consisting of $\begin{pmatrix}a & b\\ c & d\end{pmatrix} \in PGL_2(\mathbb{Z}_p)$ with $p \mid c$
directed edge adjacency operator T	Iwahori-Hecke operator based on $\mathfrak{I}\begin{pmatrix}1 & 0\\ 0 & p\end{pmatrix}\mathfrak{I}$

6.5. Ramanujan graphs as finite quotients of $PGL_2(\mathbb{Q}_p)$

The first infinite family of k-regular Ramanujan graphs was constructed by Margulis [**74**] and independently by Lubotzky-Phillips-Sarnak [**67**] using number theory, more precisely, the Ramanujan conjecture for classical modular forms. The Ramanujan graphs constructed by them are finite quotients of the tree $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ for an odd prime p.

We shall follow the approach in [**67**]. The first step is to represent the cosets in $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ by a group Λ and show that the tree is a Cayley graph $\mathrm{Cay}(\Lambda, S)$, as explained below.

Let H be the Hamiltonian quaternion algebra over $\mathbb{Q}$. It has a $\mathbb{Q}$-basis $1, i, j, ij$ satisfying $i^2 = -1$, $j^2 = -1$, and $ij = -ji$. It is ramified exactly at 2 and ∞. Thus H is unramified at an odd prime p, which means that $H(\mathbb{Q}_p) \cong M_2(\mathbb{Q}_p)$. The center of $H^\times$ consists of scalar multiples of 1.

Let D denote $H^\times$ modulo its center so that

$$D(\mathbb{Q}_p) \cong PGL_2(\mathbb{Q}_p).$$

We may identify $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ with $D(\mathbb{Q}_p)/D(\mathbb{Z}_p)$.

The group of units of the ring $H(\mathbb{Z})$ consists of 8 elements ± 1, $\pm i$, $\pm j$, and $\pm ij$. By the well-known theorem of Jacobi, in $H(\mathbb{Z})$ there are $8(p+1)$ elements

$\alpha = a + bi + cj + dij$ where $a, b, c, d \in \mathbb{Z}$ with norm $N(\alpha) = a^2 + b^2 + c^2 + d^2 = p$. If $p \equiv 1 \mod 4$, then exactly one of a, b, c, d is odd. Multiplying by one of the units of $H(\mathbb{Z})$ if necessary, we may assume that a is odd and positive. If $p \equiv 3 \mod 4$, then exactly one of a, b, c, d is even, so we may assume that a is even. Furthermore, if $p \equiv 7 \mod 8$, then by Legendre's three square theorem, p is not a sum of three squares, so we may assume a even and positive. *Assume either* $p \equiv 1 \mod 4$ *or* $p \equiv 7 \mod 8$ so that we can specify S to be the set of $p+1$ norm p elements in $H(\mathbb{Z})$ with a chosen as above. Note that if $\alpha = a + bi + cj + dij$ lies in S, so does $\bar{\alpha} := a - bi - cj - dij$. Further, since $N(\alpha) = \alpha\bar{\alpha} = \bar{\alpha}\alpha = p$ lies in the center of H, the image of S in $D(\mathbb{Z})$, which is also denoted by S, is a symmetric set with cardinality $p+1$.

Lubotzky, Phillips and Sarnak showed that S generates a free group Λ of rank $(p+1)/2$ so that the Cayley graph $\mathrm{Cay}(\Lambda, S)$ is a $(p+1)$-regular tree. Moreover, Λ represents the coset space $D(\mathbb{Q}_p)/D(\mathbb{Z}_p)$ and the two graph structures coincide.

THEOREM 6.1 (Lubotzky-Phillips-Sarnak [**67**]).

$$PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p) \cong \mathrm{Cay}(\Lambda, S)$$

as $(p+1)$*-regular trees.*

Let q be an odd prime other than p. Since the elements in Λ have coordinates in $\mathbb{Z}$ and norm a power of p, which are coprime to q, we may consider Λ modulo q. The kernel $\Lambda(q)$ of this map consists of $\lambda \in \Lambda$ such that, up to multiples in the center of H, $\lambda \equiv 1 \mod q$. This map induces a Cayley graph

$$\mathrm{Cay}(\Lambda \mod q,\ S \mod q) = \mathrm{Cay}(\Lambda(q)\backslash\Lambda,\ S \mod q).$$

In the case that $q \equiv 1 \mod 4$, the field $\mathbb{F}_q$ contains a square root i' of -1. The elements $a + bi + cj + dij$ in $S \mod q$ may be identified with the matrices $\begin{pmatrix} a+bi' & c+di' \\ -c+di' & a-bi' \end{pmatrix}$ in $PGL_2(\mathbb{F}_q)$. Lubotzky, Phillips and Sarnak showed that these matrices generate the group $PSL_2(\mathbb{F}_q)$ or $PGL_2(\mathbb{F}_q)$ according as p in $\mathbb{F}_q$ is a square or not. Thus when $q \equiv 1 \mod 4$, the group $\Lambda(q)\backslash\Lambda$, which is generated by $S \mod q$, can be identified with the more explicit matrix group $PGL_2(\mathbb{F}_q)$ or $PSL_2(\mathbb{F}_q)$ depending on p modulo q being a quadratic non-residue or quadratic residue. This realization facilitates real world implementations. For q large enough, $S \mod q$ has cardinality $p+1$. By varying q we obtain an infinite family of $(p+1)$-regular graphs.

THEOREM 6.2 (Lubotzky-Phillip-Sarnak [**67**]). *For primes* $p \geq 5$, $p \mod 8$ *not congruent to* 3, *and* $q > p^8$, *the graphs* $\mathrm{Cay}(\Lambda \mod q,\ S \mod q)$ *are* $(p+1)$*-regular Ramanujan graphs. Moreover, when* $q \equiv 1 \mod 4$, *the graph* $\mathrm{Cay}(\Lambda \mod q,\ S \mod q)$ *is isomorphic to*

$$\begin{cases} \mathrm{Cay}(PGL_2(\mathbb{F}_q),\ S \mod q) & \text{if } p \text{ is not a square modulo } q; \\ \mathrm{Cay}(PSL_2(\mathbb{F}_q),\ S \mod q) & \text{if } p \text{ is a square modulo } q. \end{cases}$$

Note that the Cayley graph on $PGL_2(\mathbb{F}_q)$ is bipartite, while the graph on $PSL_2(\mathbb{F}_q)$ is not.

Why are these graphs Ramanujan? We use the strong approximation theorem to pass from local to global setting

$$\begin{aligned}\Lambda(q)\backslash\Lambda &= \Lambda(q)\backslash D(\mathbb{Q}_p)/D(\mathbb{Z}_p)\\ &= D(\mathbb{Q})\backslash D(\mathbb{A}_{\mathbb{Q}})/D(\mathbb{R})D(\mathbb{Z}_p)K_q,\end{aligned}$$

where $\mathbb{A}_{\mathbb{Q}}$ is the adèle ring of $\mathbb{Q}$, K_q is a congruence subgroup of $\prod_{\ell\neq p} D(\mathbb{Z}_\ell)$ so that $\Lambda(q) = D(\mathbb{Q}) \cap K_q$. The intersection of $D(\mathbb{Q})$ with the full group $\prod_{\ell\neq p} D(\mathbb{Z}_\ell)$ is the group $D(\mathbb{Z}[\frac{1}{p}])$, which contains $\Lambda(q)$ as a finite index subgroup.

The transition from local to global setting allows the functions on $\Lambda(q)\backslash\Lambda$ to be interpreted as "automorphic forms" on the quaternion group $D(\mathbb{A}_{\mathbb{Q}})$. By Jacquet-Langlands correspondence [**45**], the automorphic forms g on $D(\mathbb{A}_{\mathbb{Q}})$ orthogonal to the eigenfunctions of the adjacency operator A with trivial eigenvalues $\pm(p+1)$ correspond to weight 2 cusp forms f for some congruence subgroup of $SL_2(\mathbb{Z})$, and the action of A on g agrees with the action of the Hecke operator T_p on the corresponding f. Therefore the nontrivial eigenvalues of A are the eigenvalues of T_p on these weight 2 cusp forms f.

The Ramanujan Conjecture on classical cusp forms asserts that the eigenvalues λ of T_p on weight 2 cusp forms satisfy $|\lambda| \leq 2\sqrt{p}$ which is precisely what is needed for our $(p+1)$-regular graphs to be Ramanujan. This conjecture for weight 2 cusp forms was proved by Eichler [**21**] and Shimura [**93**] and completed by Igusa [**42**], as recalled in Theorem 1.9 in §5 of Lecture 1.

The same construction can be applied to any definite quaternion algebras over $\mathbb{Q}$ and quaternion algebras over a function field K with q elements in its field of constants, yielding infinite families of $(q+1)$-regular Ramanujan graphs for any positive prime power q. The construction over a function field K uses the Ramanujan Conjecture for automorphic forms of GL_2 over K proved by Drinfeld [**19**]. Here inner forms of GL_2 are used for the sake of getting finite graphs. Definite quaternion algebras B over $\mathbb{Q}$ are chosen so that, under the local Jacquet-Langlands correspondence, the trivial representation of $B^\times(\mathbb{R})$ corresponds to the weight 2 discrete series representation of $GL_2(\mathbb{R})$ to ensure that the weight of the resulting cusp forms for congruence subgroups of $SL_2(\mathbb{Z})$ under the global Jacquet-Langlands correspondence is 2, which is critical to the Ramanujan bound.

6.6. Hashimoto's class number formula

Mestre and Oesterlé [**75**] considered a similar construction with the Hamiltonian quaternion algebra replaced by other definite quaternion algebras over $\mathbb{Q}$. Fix a prime p and let $\ell \neq p$ be another prime. Let H_ℓ be the definite quaternion algebra over $\mathbb{Q}$ ramified only at ∞ and ℓ. For example, H_2 is the Hamiltonian quaternion algebra H in the previous section. Let D_ℓ be $H_\ell^\times$ modulo its center. Since H_ℓ is unramified at $p \neq \ell$, we have $D_\ell(\mathbb{Q}_p) \cong PGL_2(\mathbb{Q}_p)$. Then $\Gamma_\ell = D_\ell(\mathbb{Z}[\frac{1}{p}])$ is a discrete cocompact subgroup of $D_\ell(\mathbb{Q}_p)$. Hashimoto [**34**] showed that it is torsion free if $\ell \equiv 1 \mod 12$. For such an ℓ, the quotient

$$X_\ell = \Gamma_\ell\backslash D_\ell(\mathbb{Q}_p)/D_\ell(\mathbb{Z}_p) \cong \Gamma_\ell\backslash PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$$

is a $(p+1)$-regular Ramanujan graph. The vertices of X_ℓ are classes of isomorphic supersingular elliptic curves over $\mathbb{F}_\ell$, and two vertices are adjacent if they are represented by elliptic curves p-isogenous to each other. It is Ramanujan because the

space of functions orthogonal to the constant functions corresponds, by Jacquet-Langlands [**45**], to the space $S_2\big(\Gamma_0(\ell)\big)$ of weight 2 cusp forms for $\Gamma_0(\ell)$. By varying $\ell \neq p$ we obtain an infinite family of $(p+1)$-regular Ramanujan graphs.

Further we know that

$$|X_\ell| = 1 + g_0(\ell) = \text{the class number of } H_\ell,$$

where $g_0(\ell)$ is the genus of the modular curve $X_0(\ell)$, which is the completion of the quotient $\Gamma_0(\ell)\backslash\mathfrak{H}$ by adding cusps. It is also equal to $\dim S_2\big(\Gamma_0(\ell)\big)$.

Recall that the zeta function attached to the graph X_ℓ is

$$Z(X_\ell, u) = \frac{(1-u^2)^{\chi(X_\ell)}}{\det(I - A(X_\ell)u + pu^2 I)},$$

where

$$\det(I - A(X_\ell)u + pu^2 I) = \prod_{j=1}^{1+g_0(\ell)} (1 - \lambda_j u + pu^2)$$

and $\lambda_1 > \lambda_2 \geq \cdots \geq \lambda_{1+g_0(\ell)}$ are the eigenvalues of X_ℓ. Here $\lambda_1 = p+1$ is the trivial eigenvalue and the nontrivial eigenvalues $\lambda_2, ..., \lambda_{1+g_0(\ell)}$ are the eigenvalues of the Hecke operator T_p on $S_2\big(\Gamma_0(\ell)\big)$. After removing the term from the trivial eigenvalue $\lambda_1 = p+1$, we get

$$\frac{\det(I - A(X_\ell)u + pu^2 I)}{(1-u)(1-pu)} = \prod_{j=1}^{g_0(\ell)} (1 - \lambda_{j+1} u + pu^2) = P(X_0(\ell)_{/\mathbb{F}_p}, u),$$

where $P(X_0(\ell)_{/\mathbb{F}_p}, u)$ is the numerator of the zeta function of the curve $X_0(\ell)$ modulo p. The last equality follows from the result of Eichler [**21**], Shimura [**93**], and Igusa [**42**], as explained in §5, Lecture 1. This gives a connection between the zeta function of certain graphs and the zeta function of certain curves defined over finite fields.

A maximal tree contained in a graph X is called a *spanning tree*; it uses all the vertices of the graph X. The number of edges in a spanning tree is one less than the number of vertices in X. Each unused edge e of X, when added to the tree, creates a (unique) cycle C_e. These cycles C_e, after growing tails to connect to a chosen base point $*$ of X, generate the fundamental group $\pi_1(X, *)$ of X. In fact $\pi_1(X, *)$ is a free group on these generators and has rank

$$r = \#(\text{edges}) - \big(\#(\text{vertices}) - 1\big) = 1 - \chi(X).$$

The number of spanning trees in X, denoted by $\kappa(X)$, is called *the class number of* X.

THEOREM 6.3 (Hashimoto [**33**, **34**]). *Let X be a finite connected undirected graph whose fundamental group has rank r.*

(1) *The zeta function $Z(X, u)$ has a pole at $u = 1$ of order $r = 1 - \chi(X)$, the rank of $\pi_1(X, *)$, except when X is homotopic to a cycle graph, in which case $r = 1$ and the order is 2.*
(2) *The order of $u = -1$ as a pole of $Z(X, u)$ is*

$$\begin{cases} r-1 & \text{if } X \text{ is not bipartite;} \\ r & \text{if } X \text{ is bipartite and not homotopic to a cycle graph;} \\ 2 & \text{if } X \text{ is bipartite and homotopic to a cycle graph.} \end{cases}$$

(3) *In the case that $u = 1$ is a pole of order r,*

$$(1-u)^{-r} Z(X,u)^{-1}\Big|_{u=1} = 2^r \chi(X)\kappa(X).$$

In particular, if X is $(q+1)$-regular, then

$$\frac{\det(I - A(X)u + qu^2 I)}{(1-u)(1-qu)}\Big|_{u=1} = \kappa(X)|X|.$$

Applying this theorem to $X = X_\ell$ and noticing that the left hand side is equal to $P(X_0(\ell)_{/\mathbb{F}_p}, 1) = \big|\operatorname{Jac}_{X_0(\ell)}(\mathbb{F}_p)\big|$ as discussed in §5 of Lecture 1, we get

$$\kappa(X_\ell)\big(1 + g_0(\ell)\big) = \big|\operatorname{Jac}_{X_0(\ell)}(\mathbb{F}_p)\big|.$$

Recall that $\big|\operatorname{Jac}_{X_0(\ell)}(\mathbb{F}_p)\big|$ is the class number of the function field of $X_0(\ell)$ over $\mathbb{F}_p$. As $\big(1 + g_0(\ell)\big)$ is the class number of H_ℓ and $\kappa(X_\ell)$ is the class number of X_ℓ, the above formula is an identity relating three class numbers.

Assume that the graph X in Theorem 6.3 has n vertices, m edges, and the rank of its fundamental group is $r = m - n + 1 \geq 2$ so that the graph contains infinitely many primes. As a consequence of Theorem 6.3, we prove that the zeta function of X is determined by the primes of X of length $\leq 2n-1$. To see this, let α_i, $1 \leq i \leq 2m$, be the eigenvalues of the directed edge adjacency matrix T of X. As shown in §3, Lecture 4, the zeta function of X is

$$Z(X,u) = \exp\Big(\sum_{\ell=1}^{\infty} \frac{N_\ell(X)}{\ell} u^\ell\Big) = \frac{1}{\det(I - Tu)}$$

since, for each $\ell \geq 1$,

$$N_\ell(X) = \operatorname{tr} T^\ell = \sum_{1 \leq i \leq 2m} \alpha_i^\ell.$$

Let $u_1, u_2, \ldots, u_{2m}$ denote $2m$ indeterminates. The Newton polynomials $\sum_{i=1}^{2m} u_i^\ell$ for $1 \leq \ell \leq 2m$ generate the ring of the symmetric polynomials over $\mathbb{C}$ in $2m$ variables. In particular the numbers $N_\ell(X)$ for all $\ell \geq 2m+1$ are uniquely determined by $N_\ell(X)$ for $1 \leq \ell \leq 2m$, which henceforth determine the Ihara zeta function of X.

Actually, we can do better by using information on the eigenvalues of T. Theorem 6.3 above says that 1 is an eigenvalue with multiplicity r, while -1 has multiplicity $r-1$ if X is not bipartite and multiplicity r if X is bipartite. Hence for a non-bipartite X, the number $N_\ell(X)$ is a symmetric function of degree ℓ on the remaining $2m - 2r + 1 = 2n - 1$ unknown eigenvalues of T; the argument above shows that the Ihara zeta function is determined by $N_\ell(X)$ for $1 \leq \ell \leq 2n-1$. For a bipartite X, it contains no cycles of odd length and the eigenvalues of T are symmetric with respect to 0, so the Ihara zeta function of X is determined by $N_\ell(X)$ for positive even $\ell \leq 2m - 2r = 2n - 2$. The more eigenvalues of T we know, the fewer $N_\ell(X)$ is required. For example, if the graph X is regular, or regular and bipartite, then we can further reduce the upper bound on ℓ by 1 or 2 accordingly.

For integers $\ell \geq 1$, let $M_\ell(X)$ denote the number of primes of X of length ℓ. Then

$$N_\ell(X) = \sum_{d \mid \ell} d \cdot M_d(X).$$

Using the Möbius μ-function, we invert the above expression as

$$M_\ell(X) = \frac{1}{\ell} \sum_{d|\ell} \mu\left(\frac{\ell}{d}\right) N_d(X).$$

This shows that the numbers $N_\ell(X)$ with $\ell \leq 2n-1$ and the numbers $M_\ell(X)$ with $\ell \leq 2n-1$ determine each other. So we can summarize the above discussion in

THEOREM 6.4 (Huang-Li [**40**]). *Suppose that X has n vertices and its fundamental group has rank ≥ 2. Then the Ihara zeta function of X is determined by the number $M_\ell(X)$ of primes of X of length ℓ for $1 \leq \ell \leq 2n-1$. Further, if X is bipartite, its Ihara zeta function is determined by the number $M_\ell(X)$ for all even ℓ with $2 \leq \ell \leq 2n-2$.*

6.7. Biregular bipartite Ramanujan graphs

A bipartite graph is said to be (c,d)-biregular if all the vertices in one bipartition have degree c and those in the other bipartition have degree d.

Let X be a finite (c,d)-biregular bipartite connected undirected graph. It has $\pm\sqrt{cd}$ as its eigenvalues with largest absolute value. (When $c = d$ the graph is regular and the trivial eigenvalues are $\pm d$, as before.) Denote by V_c (resp. V_d) the subset of vertices of X with degree c (resp. d). Then $c\#V_c = d\#V_d$ is the number of edges in X.

PROPOSITION 6.4.1. *With the above notation, 0 is an eigenvalue of X with multiplicity at least $|\#V_c - \#V_d|$.*

PROOF. Without loss of generality, we may assume $c \geq d$ so that $\#V_d \geq \#V_c$. Let f be a nonzero eigenfunction of the adjacency matrix A of X with eigenvalue λ. Denote by $f_{c,\lambda}$ and $f_{d,\lambda}$ the restrictions of f on V_c and V_d, respectively. Then $f(v) \neq 0$ at some vertex v of X. If $\lambda \neq 0$, then $Af(v) = \lambda f(v) \neq 0$ implies that both $f_{c,\lambda}$ and $f_{d,\lambda}$ are nonzero. Further, the function g on X with $g|_{V_c} = f_{c,\lambda}$ and $g|_{V_d} = -f_{d,\lambda}$ is a nonzero eigenfunction with eigenvalue $-\lambda$. The space spanned by f and g is the same space spanned by $f_{c,\lambda}$ and $f_{d,\lambda}$. The space W generated by $f_{c,\lambda}$ and $f_{d,\lambda}$ as λ runs through all positive eigenvalues of A is the space spanned by the eigenfunctions of A with nonzero eigenvalues. It has dimension at most $2\#V_c$. The space of eigenfunctions of A with eigenvalue 0 is orthogonal to W, hence has dimension at least $\#V_d - \#V_c$. In other words, the multiplicity of 0 as an eigenvalue of A is $\geq \#V_d - \#V_c$. □

In view of the above proposition, call $\pm\sqrt{cd}$ and 0 with multiplicity $|\#V_c - \#V_d|$ the trivial eigenvalues of a (c,d)-biregular bipartite graph with the two partitions of vertex set V_c and V_d, and the remaining eigenvalues nontrivial. We are interested in the behavior of its nontrivial eigenvalues, which are symmetric with respect to 0.

Feng-Li [**23**] obtained the following analog of the Alon-Boppana Theorem for biregular bipartite graphs.

THEOREM 6.5 (Feng-Li [**23**]). *Let $\{X_j\}$ be a family of bipartite (c,d)-biregular connected undirected graphs with $|X_j| \to \infty$ as $j \to \infty$. Then*

$$\liminf_{j\to\infty} \lambda_2(X_j) \geq \sqrt{c-1} + \sqrt{d-1}.$$

When $c = d$, the above result reduces to the Alon-Boppana Theorem 5.1 for regular graphs. As with the Alon-Boppana Theorem, this lower bound is the covering radius of the universal cover of (c, d)-biregular bipartite graphs; this was proved in Li-Solé [**65**]. A finite (c, d)-biregular bipartite graph is called *Ramanujan* if all nontrivial eigenvalues λ of the graph satisfy

$$|\lambda| \leq \sqrt{c-1} + \sqrt{d-1}.$$

Again, the Ramanujan graphs are spectrally extremal.

REMARK 6.1. The above definition of Ramanujan biregular bipartite graph is the one used by Marcus, Spielman and Srivastava in [**72**]. From the spectral point of view, if one defines a (c, d)-biregular bipartite connected undirected graph X to be Ramanujan provided that all of its nontrivial eigenvalues λ lie in the spectrum of its universal cover, then λ should satisfy a more restrictive bound

$$|\sqrt{c-1} - \sqrt{d-1}| \leq |\lambda| \leq \sqrt{c-1} + \sqrt{d-1}$$

since for $c \neq d$, the spectrum of the (c, d)-biregular bipartite tree is a disjoint union of two intervals $[-\sqrt{c-1}-\sqrt{d-1}, -|\sqrt{c-1}-\sqrt{d-1}|]\cup[|\sqrt{c-1}-\sqrt{d-1}|, \sqrt{c-1}+\sqrt{d-1}]$. See [**33**, **65**]. From the viewpoint of zeta functions, in order that all nontrivial poles of the zeta function $Z(X, u)$ have the same absolute value (so that the Riemann Hypothesis holds), one would need the more restrictive definition. On the other hand, the counterpart of the Alon-Boppana theorem similar to Theorem 5.4 for the behavior of the least positive eigenvalue of a (c, d)-biregular bipartite graph Y as $|Y| \to \infty$ is not known.

The Ramanujan graphs constructed in the previous sections using number theory have valency $d = q + 1$ with q a prime power. A long standing question has been

QUESTION 6.1. Given an integer $d \geq 3$, does there exist an infinite family of d-regular Ramanujan graphs?

THEOREM 6.6 (Marcus-Spielman-Srivastava [**72**]). *For all $c, d \geq 3$, there exists an infinite family of Ramanujan (c, d)-biregular bipartite graphs.*

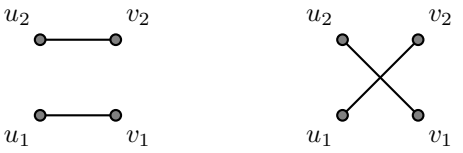
When $c = d$, this answers Question 6.1. in the affirmative. However, all such graphs are bipartite. Whether there are infinite families of non-bipartite d-regular Ramanujan graphs for any $d \geq 3$ and $d - 1$ not a prime power is still an open problem. As we shall see below, the proof of Theorem 6.6 is an "existence" argument. A concrete description of such families of Ramanujan graphs is still lacking.

The strategy of Marcus, Spielman and Srivastava is to show that a finite Ramanujan (c, d)-biregular bipartite graph has a two-fold unramified cover which is also Ramanujan. Hence, if we can find one such graph to start with, then we can continue producing two-fold covers to obtain an infinite family of Ramanujan graphs. Since the complete (c, d)-biregular bipartite graph is Ramanujan, we are done.

Given a finite graph X, a two-fold unramified cover Y of X can be constructed in the following way. We begin with two copies of X and place one copy on top of the other with vertices and edges lined up over each other. We then change some

of the edges. More precisely, for each edge $e = \overline{uv}$ in X, we will make one of the two choices:

(a) We retain the pair of edges in each copy of the graph, as illustrated below on the left. In this case assign the sign $s(e) = 1$.
(b) We cross-connect the edges between the two copies in the graph, as illustrated below on the right. In this case assign the sign $s(e) = -1$.



Note that we cannot only have edges of type (a) as that leaves the graph disconnected; also it is still possible to disconnect with many type (b) edges.

Thus by varying the signing function

$$s : \{\text{edges of } X\} \to \{\pm 1\},$$

we obtain all two-fold unramified covers Y_s, not necessarily all non-isomorphic. The adjacency matrix of Y_s is conjugate to the matrix

$$\begin{pmatrix} A & \\ & A_s \end{pmatrix},$$

where A is the adjacency matrix of X and A_s is obtained from A by retaining the zero entries and replacing a nonzero uv-entry 1 by $s(\overline{uv})$. (If X has multiple edges or loops, add one vertex on each of such edge so that we may assume that X does not have multiple edges.) Observe that the eigenvalues of A_s, being part of the eigenvalues of Y_s, are real. Further, if X is bipartite, so is Y_s, in which case the eigenvalues of A_s are symmetric with respect to 0. If X is (c, d)-biregular, so is Y_s.

Conjecture 7 (Bilu and Linial [**9**]). *Let X be a finite graph. There exists a signing s such that the eigenvalues of A_s are bounded by the covering radius r of the universal cover of X, i.e., lie in $[-r, r]$.*

The conjecture means that, while we cannot do anything to the eigenvalues of X, we can find a nice two-fold cover of X so that all new eigenvalues, namely those of A_s, are as good as they can be. The upper bound of this conjecture was established in [**72**].

Theorem 6.7 (Marcus-Spielman-Srivastava [**72**]). *Given a finite connected undirected graph X, there exists a signing s of edges of X such that the eigenvalues of A_s are in $(-\infty, r]$, where r is the covering radius of the universal cover of X.*

While their bound is only one-sided, if, in addition, X is bipartite, then so is the two-fold cover Y_s. Since the eigenvalues of a bipartite graph are symmetric with respect to 0, if we bound them from one side, then we also bound them from the other side. Thus Y_s is Ramanujan provided that X is.

Remark 6.2. The same argument also proves the existence of a signing s so that all eigenvalues of A_s fall in the interval $[-r, \infty)$, but one cannot conclude the

existence of a signing s so that the eigenvalues of A_s simultaneously satisfy both upper and lower bounds. This is why the bipartite assumption on X is needed.

We sketch the proof of Theorem 6.7 by Marcus-Spielman-Srivastava. Label the edges of X as e_1, e_2,..., e_m. Suppose X has n vertices. The goal is to find a signing s such that all eigenvalues of A_s are $\leq r$. The signing will be chosen one edge at a time. As noted before, for each signing $s \in \{\pm 1\}^m$, the characteristic polynomial of A_s, namely $f_s = \det(xI_n - A_s)$, has degree n and it is real-rooted, that is, all of its roots are real.

A real-rooted polynomial $g(x) = \prod_{1 \leq i \leq n-1}(x - a_i)$ of degree $n-1$ is said to interlace a real-rooted polynomial $f = \prod_{1 \leq j \leq n}(x - b_j)$ of degree n if

$$b_1 \leq a_1 \leq b_2 \leq a_2 \leq \cdots \leq a_{n-1} \leq b_n.$$

Finitely many real-rooted polynomials $f_1, ..., f_k$ are said to have a common interlacing if there is a polynomial g that interlaces each of the f_i. The key idea repeatedly used in the proof is the following

LEMMA 6.7.1. *Let $f_1, ..., f_k$ be degree-n real-rooted polynomials with positive leading coefficients, and let f_ϕ be their sum. If $f_1, ..., f_k$ have a common interlacing, then there exists a polynomial f_i whose largest root is at most the largest root of the sum f_ϕ.*

Ordinarily we do not expect to draw information on the roots of f_i from the information on the roots of their sum f_ϕ; the novelty of the above Lemma is that if the f_i's are real-rooted and have a common interlacing, then something can be said about the roots of some f_i. This idea is also used by Marcus, Spielman and Srivastava to answer the Kadison-Singer problem in von Neumann algebra in the paper [**73**].

Their starting point is the observation that the average of all f_s, that is, $f_\phi = \sum_{s \in \{\pm 1\}^m} f_s$, is the so-called matching polynomial of X, which was shown by Heilmann and Lieb [**37**] to be real-rooted, and by Godsil and Gutman [**27**, **28**] to have all roots $\leq r$. Separating the signings according to the sign of the edge e_1, we let

$$f_{\pm 1} = \sum_{s \in \{\pm 1\}^m,\ s(e_1) = \pm 1} f_s$$

so that $f_\phi = f_{+1} + f_{-1}$. Marcus, Spielman and Srivastava showed that $f_{\pm 1}$ are real-rooted and have a common interlacing. Therefore, by Lemma 6.7.1, one of $f_{\pm 1}$ has all roots $\leq r$; denote it by $f_{s(e_1)}$. The signing of e_1 is determined.

To choose the signing of e_2, write $f_{s(e_1)} = f_{s(e_1),+1} + f_{s(e_1),-1}$, where

$$f_{s(e_1),\pm 1} = \sum_{s' \in \{\pm 1\}^m,\ s'(e_1) = s(e_1),\ s'(e_2) = \pm 1} f_{s'}.$$

The polynomials $f_{s(e_1),\pm 1}$ are shown to be real-rooted and have a common interlacing. Hence by Lemma 6.7.1 again, one of $f_{s(e_1),\pm 1}$ has all roots $\leq r$. This gives $s(e_2)$. Repeat this procedure, finally one arrives at a desired signing.

As outlined above, in this recursive procedure at each step one has to prove two properties of the two new polynomials involved: real-rootedness and having a common interlacing. The following theorem by Fisk reduces common interlacing to real-rootedness.

THEOREM 6.8 (Fisk [**25**]). *Two monic polynomials F and G of the same degree have a common interlacing if $aF + (1-a)G$ are real-rooted for all $0 \le a \le 1$.*

Marcus, Spielman and Srivastava proved a general statement asserting that if the signs of the edges are independently assigned with any probabilities, then the linear combination of all f_s weighted by the probability of s is still real-rooted.

THEOREM 6.9 (Marcus-Spielman-Srivastava [**72**]). *Let $p_1, ..., p_m \in [0,1]$. Then*

$$\sum_{s\in\{\pm 1\}^m} \Big(\prod_{i \text{ with } s(e_i)=1} p_i \Big) \Big(\prod_{j \text{ with } s(e_j)=-1} (1-p_j) \Big) f_s$$

is real-rooted.

Given $s(e_1), ..., s(e_r)$ with $1 \le r \le m-1$, the real-rootedness of

$$f_{s(e_1),\dots,s(e_r)} := \sum_{\substack{s'\in\{\pm 1\}^m \\ s'(e_1)=s(e_1),\dots,\ s'(e_r)=s(e_r)}} f_{s'}$$

results from the above theorem by specifying $p_{r+1} = \cdots = p_m = 1/2$ and $p_j = (1+s(e_j))/2$ for $1 \le j \le r$, and the real-rootedness of $af_{s(e_1),\dots,s(e_r),1} + (1-a)f_{s(e_1),\dots,s(e_r),-1}$ for $0 \le a \le 1$ follows from choosing $p_j = (1+s(e_j))/2$ for $1 \le j \le r$, $p_{r+1} = a$, and $p_{r+2} = \cdots = p_m = 1/2$. This completes the proof of Theorem 6.7.

In [**16**] Cohen considered the case $c = d$ and provided a polynomial time algorithm to compute the expected characteristic polynomials, namely $2^{r-m} f_{s(e_1),\dots,s(e_r)}$ for $0 \le r \le m$, related to this construction, which leads to a deterministic polynomial time algorithm to compute bipartite Ramanujan graphs. It would be highly desirable to give a more explicit description of a bi-regular bipartite two-fold cover Y_s satisfying the conjecture by Bilu and Linial.

Finally we remark that to prove Theorem 6.7 we don't need to determine the signing of each edge of X. Suppose the fundamental group $\pi_1(X, *)$ has rank r. Let T be a spanning tree of X. Since T is simply connected, its lifting in any unramified two-fold cover of X is two copies of disjoint trees isomorphic to T. So we may assume the signing of each edge in T is 1 and it suffices to determine the signing of the remaining r edges outside T.

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LECTURE 7

Artin L-functions and prime distributions for graphs

7.1. Artin L-functions for graphs

Throughout this lecture, X denotes a finite connected undirected graph. Recall that the Ihara zeta function for X is

$$Z(X,u) = \prod_{\substack{[C] \\ C \text{ primitive geodesic cycle}}} \frac{1}{1-u^{\ell(C)}}.$$

Given a geodesic cycle C in X, by joining a path from the base point $*$ to the starting point of C, traveling along C and finally returning to $*$ by following the same path but in reverse direction, we obtain a closed path at $*$ whose homotopy class is an element γ in the fundamental group $\pi_1(X,*)$ of X. Varying the starting point on C yields conjugating elements in the fundamental group. This defines a map from the equivalence class $[C]$ to the conjugacy class $[\gamma]$. Like what we saw for Riemann surfaces, this map is a bijection between the set of equivalence classes $[C]$ of geodesic cycles in X and the set of conjugacy classes $[\gamma]$ of non-identity elements in $\pi_1(X,*)$, which sends classes of primitive cycles to conjugacy classes of primitive elements. See [**33**, **94**] for more details. Define the length of $[\gamma]$ to be the length of the corresponding geodesic cycle $[C]$ so that this correspondence is length preserving. Then we can rewrite the Ihara zeta function as

$$Z(X,u) = \prod_{\substack{[\gamma] \\ \gamma \text{ primitive}}} \frac{1}{1-u^{\ell(\gamma)}}.$$

Let $\rho : \pi_1(X,*) \to \mathrm{Aut}(V)$ be a d-dimensional unitary representation. The Artin L-function attached to ρ is defined as

$$L(X,\rho,u) = \prod_{\substack{[\gamma] \\ \gamma \text{ primitive}}} \frac{1}{\det(I_d - \rho(\gamma)u^{\ell(\gamma)})}.$$

The Artin L-functions for graphs appear simpler than the number-theoretic Artin L-functions because there are no ramifications. Note that for the trivial 1-dimensional representation ρ_0, $L(X,\rho_0,u)$ coincides with $Z(X,u)$.

The Artin L-functions possess properties analogous to those of the Ihara zeta function.

THEOREM 7.1 (Ihara [**43**], Hashimoto [**35**], Stark-Terras [**95**]). *Let ρ be a finite-dimensional unitary representation of $\pi_1(X,*)$. Then there are matrices A_ρ, Q_ρ and T_ρ such that*

$$L(X,\rho,u)=\frac{1}{\det(I-T_\rho u)}=\frac{(1-u^2)^{\chi(X)\deg\rho}}{\det(I-A_\rho u+Q_\rho u^2)}.$$

Here $\chi(X)$ is the Euler characteristic of X.

The matrices A_ρ, Q_ρ and T_ρ are obtained from A, Q, and T that we have dealt with before by replacing each nonzero entry by a suitable $d\times d$ matrix.

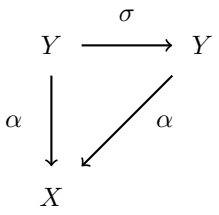
This theorem can also be proved by first establishing the first equality and then applying a cohomological argument to prove the second equality, similar to what we did in §3 and §4 of Lecture 4. Denote by V the d-dimensional ambient space of ρ. In this case, the $\mathbb{C}[u]$-module C^0 (resp. C^1) in §4, Lecture 4 is replaced by the space of $V\otimes_{\mathbb{C}}\mathbb{C}[u]$-valued functions f on the vertices (resp. directed edges) of the universal cover $\tilde{X}$ of X which are compatible with the action of ρ, that is, $f(\gamma x)=\rho(\gamma)f(x)$ for $\gamma\in\pi_1(X,*)$ and x a vertex (resp. directed edge) of $\tilde{X}$. It follows from Theorem 7.1 that $L(X,\rho,u)^{-1}$ is a polynomial of degree equal to $2d$ times the number of edges in X. The Artin conjecture holds for Artin L-functions for graphs.

We remark that the Selberg zeta function $Z(X_\Gamma,\rho,s)$ for a Riemann surface X_Γ of constant negative curvature -1 and a finite-dimensional unitary representation ρ of Γ, as defined in Lecture 2, §5, also admits an expression of the form $\det(I-\mathcal{L}_{\rho,s})$ in terms of the *transfer operator* $\mathcal{L}_{\rho,s}$, analogous to the Hashimoto expression for graphs. This was established by Pohl (cf. [**79**] and references therein) and her coauthors for Γ a congruence subgroup of $SL_2(\mathbb{Z})$. The problem remains open for Γ a discrete torsion-free cocompact subgroup of $PGL_2(\mathbb{R})$, as does the spectral expression, that is, the counterpart of the Ihara expression, for a nontrivial representation ρ. (The case of trivial ρ is known, stated in Theorem 2.4.) Furthermore, the zeros of A_ρ and T_ρ in Theorem 7.1 above are discrete analogues of what physicists called quantum resonances and classical resonances, respectively, attached to a compact Riemann surface with constant negative curvature -1.

7.2. Unramified covers of X

A homomorphism between two graphs maps vertices to vertices, edges to edges and preserves adjacency. A graph homomorphism $Y\xrightarrow{\alpha}X$ is called an *unramified cover* of X if it is a *surjective* local isomorphism. By a local isomorphism we mean that, for all vertices w of Y, α sends the neighbors of w in Y bijectively to the neighbors of $\alpha(w)$ in X. All graphs considered in this lecture are assumed to be connected. Since X is connected, for any vertex v of X, the cardinality of the fiber $\alpha^{-1}(v)$ is independent of the choice of v; it is called the degree of the cover α if it is finite.

An automorphism of an unramified cover $Y \xrightarrow{\alpha} X$ is a graph automorphism σ of Y making the following diagram commute:

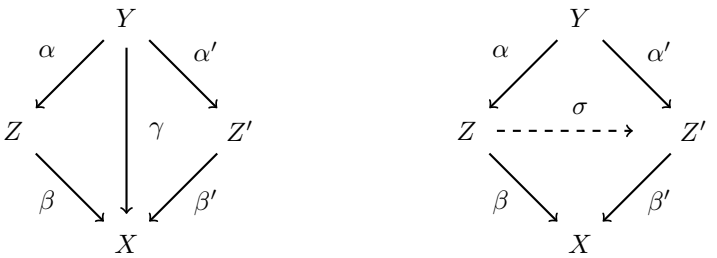


Therefore σ permutes the vertices in a fiber $\alpha^{-1}(v)$ for every vertex v of X. Further it is determined by the action on one fiber since X is connected. The collection of all automorphisms of this cover form a group, denoted by $\mathrm{Aut}(Y \xrightarrow{\alpha} X)$.

The cover $Y \xrightarrow{\alpha} X$ is said to be *Galois* if the group of automorphisms $\mathrm{Aut}(Y \xrightarrow{\alpha} X)$ acts transitively on the fiber $\alpha^{-1}(v)$ at one and hence all vertices v of X; in this case $\mathrm{Aut}(Y \xrightarrow{\alpha} X)$ is the Galois group G_α of α. If $Y \xrightarrow{\alpha} X$ has finite degree d, then it is Galois if and only if $\left|\mathrm{Aut}(Y \xrightarrow{\alpha} X)\right| = d$.

Let $\widetilde{X}$ be a universal cover of X, which is unique up to isomorphism, with the covering map $\widetilde{\alpha} : \widetilde{X} \to X$. The fundamental group $\pi_1(X, *)$ acts on $\widetilde{X}$ as deck transformation. Further, it acts transitively on each fiber $\widetilde{\alpha}^{-1}(v)$. Hence $\widetilde{X} \xrightarrow{\widetilde{\alpha}} X$ is Galois with the Galois group $G_{\widetilde{\alpha}} \cong \pi_1(X, *)$. We may then identify X with $G_{\widetilde{\alpha}} \backslash \widetilde{X}$.

A pair of graph coverings $Y \xrightarrow{\alpha} Z \xrightarrow{\beta} X$ is called an intermediate cover of $Y \xrightarrow{\gamma} X$ if $\gamma = \beta \circ \alpha$. Two intermediate coverings $Y \xrightarrow{\alpha} Z \xrightarrow{\beta} X$ and $Y \xrightarrow{\alpha'} Z' \xrightarrow{\beta'} X$ are said to be isomorphic if there is a graph isomorphism $\sigma : Z \to Z'$ such that $\alpha' = \sigma \circ \alpha$ and $\beta = \beta' \circ \sigma$, in other words, the diagram on the right commutes:



Let $Y \xrightarrow{\gamma} X$ be a Galois cover with Galois group G_γ. We may identify X with $G_\gamma \backslash Y$. For any subgroup $H \subset G_\gamma$, the following pair of natural projections

$$Y \xrightarrow{\alpha'} H \backslash Y \xrightarrow{\beta'} G_\gamma \backslash Y \ (= X)$$

is an intermediate cover of $Y \xrightarrow{\gamma} X$. Moreover, $H \backslash Y \xrightarrow{\beta'} X$ is Galois if and only if H is a normal subgroup of G_γ, and in this case $G_{\beta'} \cong G_\gamma / H$. Conversely, if we have an intermediate cover $Y \xrightarrow{\alpha} Z \xrightarrow{\beta} X$, then there is a unique subgroup $H \subset G_\gamma$ such that $Y \xrightarrow{\alpha} Z \xrightarrow{\beta} X$ is isomorphic to the intermediate cover $Y \xrightarrow{\alpha'} H \backslash Y \xrightarrow{\beta'} G_\gamma \backslash Y = X$. This shows that the Galois theory holds for the Galois cover $Y \xrightarrow{\gamma} X$, that is, there is a bijection between isomorphism classes of intermediate covers of $Y \xrightarrow{\gamma} X$ and

subgroups of G_γ. See [**95**] for details. In particular, taking $Y \xrightarrow{\gamma} X$ to be the universal covering $\widetilde{X} \xrightarrow{\widetilde{\alpha}} X$, we see that $G_{\widetilde{\alpha}} \cong \pi_1(X, *)$ plays the role of an absolute Galois group, and the equivalence classes of intermediate covers of $\widetilde{\alpha}$ play the role of extensions of the base field inside an algebraic closure.

Recall that the zeta function of a finite graph is the reciprocal of a polynomial. Let $Y \xrightarrow{\alpha} X$ be a finite unramified cover. We compare the zeta functions attached to X and Y.

PROPOSITION 7.1.1. *Let $Y \xrightarrow{\alpha} X$ be a finite unramified cover. Then $Z(X,u)^{-1}$ divides $Z(Y,u)^{-1}$ as polynomials.*

This was proved in [**94**] for X a regular graph, and in [**34, 95**] for α a Galois cover. Recall that for finite extensions K/F of number fields, we only know a few occasions where the Dedekind zeta function of F divides that of K, but its counterpart for unramified covers of graphs always holds.

PROOF. By Theorem 4.3,

$$Z(X,u)^{-1} = \det(I - T(X)u) \quad \text{and} \quad Z(Y,u)^{-1} = \det(I - T(Y)u).$$

It suffices to show that the eigenvalues of the directed edge adjacency matrix $T(X)$ of X, counting multiplicities, are also the eigenvalues of that of Y. This is because α is an unramfied cover, hence for each eigenfunction f of $T(X)$ defined on directed edges of X, its pull-back to Y, $f \circ \alpha$, is an eigenfunction of $T(Y)$ with the same eigenvalue as f. □

When the cover α is Galois with the Galois group G_α, we have more information on the factorization of the polynomial $Z(Y,u)^{-1}$. The universal cover $\widetilde{Y}$ of Y is isomorphic to the universal cover of X. Hence the natural covering map $\widetilde{Y} \xrightarrow{\tilde{\beta}} Y$ followed by $Y \xrightarrow{\alpha} X$ is an intermediate cover of $\widetilde{X} \xrightarrow{\widetilde{\alpha}} X$. As α is a Galois cover, there is a normal subgroup H of $G_{\widetilde{\alpha}}$ such that $G_\alpha \cong G_{\widetilde{\alpha}}/H$. A d-dimensional representation ρ of G_α can be viewed as a d-dimensional unitary representation ρ of $G_{\widetilde{\alpha}}$ with kernel containing H. Hashimoto showed that the Artin L-functions for graphs are invariant under induction. Therefore we have

$$\begin{aligned} Z(Y,u) &= \prod_{\rho \in \mathrm{Irr}(G_\alpha)} L(X,\rho,u)^{\deg \rho} \\ &= Z(X,u) \prod_{\substack{\rho \in \mathrm{Irr}(G_\alpha) \\ \rho \text{ nontrivial}}} L(X,\rho,u)^{\deg \rho}, \end{aligned} \tag{5}$$

in which $Z(X,u)$ is the Artin L-function attached to the trivial representation of $G_{\widetilde{\alpha}}$. Since each $L(X,\rho,u)^{-1}$ is a polynomial in u by Theorem 7.1, taking the reciprocal of each L-function, the above gives a polynomial factorization of $Z(Y,u)^{-1}$.

7.3. Distribution of primes for graphs

Let $Y \xrightarrow{\alpha} X$ be a finite unramified Galois cover with Galois group G_α. Suppose $\pi_1(X,*)$ has rank $r \geq 2$ so that X has infinitely many primes. This is equivalent to X not homotopic to a cycle graph or a tree. Since we are interested in the distribution of the primes of X which depends only on the homotopy class of X, we may assume further that each vertex of X has degree ≥ 2.

Let $\Delta_X = \gcd_{[C]}\big(\ell([C])\big)$, where $[C]$ runs through all primes of X, and let $\lambda_X = \max(|\mu|)$, where μ runs over the eigenvalues of the directed edge ajdacency matrix T of X. The Perron-Frobenius Theorem implies that λ_X is an eigenvalue of T. If X is $(q+1)$-regular, then $\lambda_X = q$ as we have observed before. We claim that $\lambda_X > 1$. To see this, recall the two expressions of the zeta function of X by Hashimoto and Ihara as discussed in Lecture 4:

$$\begin{aligned} Z(X,u) &= \frac{1}{\det(I-Tu)} = \frac{1}{\prod\limits_{\mu \text{ eigenvalue of } T}(1-\mu u)} \\ &= \frac{1}{(1-u^2)^{r-1}\det(I-Au+Qu^2)}. \end{aligned}$$

This shows that the product of all μ, up to sign, is equal to the leading coefficient of $Z(X,u)^{-1}$, which is also equal to

$$\det Q = \prod_{v \text{ vertex of } X} (\deg v - 1)$$

up to sign. The assumption $r \geq 2$ implies that at least one vertex of X has degree ≥ 3 so that $\det Q \geq 2$. Thus if $\lambda_X \leq 1$, then all μ satisfy $|\mu| \leq 1$ and so does their product, which in turn would imply $\det Q \leq 1$, a contradiction.

As the poles of $Z(X,u)$ are μ^{-1}, where μ runs through the eigenvalues of T, and λ_X^{-1} is a pole closest to 0, the function $Z(X,u)$ is holomorphic for $|u| < \frac{1}{\lambda_X}$.

THEOREM 7.2 (Prime geodesic theorem for graphs, Hashimoto [**35**]).

$$\sum_{\substack{\text{prime } [C] \\ \ell(C)=n\Delta_X}} 1 \sim \frac{(\lambda_X)^{n\Delta_X}}{n} \qquad \textit{for } n \textit{ large,}$$

$$\sum_{\substack{\text{prime } [C] \\ \ell(C)<n\Delta_X}} 1 \sim \frac{1}{(\lambda_X)^{\Delta_X}-1}\frac{(\lambda_X)^{n\Delta_X}}{n} \qquad \textit{for } n \textit{ large.}$$

Observe that this is parallel to Theorem 1.7, the prime number theorem for closed points on a smooth irreducible projective curve defined over a finite field. By defining the norm $N([C])$ of $[C]$ to be $\lambda(X)^{\ell([C])}$, this theorem is analogous to the prime geodesic theorem, Corollary 3.1.1, for compact Riemannian manifolds disscussed in Lecture 3.

Let C be a primitive geodesic cycle in X starting at v. Choose a vertex $P \in \alpha^{-1}(v)$. The lifting of C in Y starting at P is a path C' ending at $P' \in \alpha^{-1}(v)$. There is a unique $\gamma \in G_\alpha$ such that $P' = \gamma P$. Let f be the order of γ. Then the path C' followed by $\gamma C'$, $\gamma^2 C'$, $\ldots$, $\gamma^{f-1}C'$ gives a primitive geodesic cycle C_1 in Y. There are $\frac{|G_\alpha|}{f}$ primitive geodesic cycles $C_1, C_2, \ldots, C_{|G_\alpha|/f}$ in Y which are the image of C_1 under G_α and which all projects to C^f, the cycle C traveled f times. The prime $[C]$ is said to split completely in Y if $f = 1$, or equivalently, there are $|G_\alpha|$ primes in Y which projects to $[C]$.

The stabilizer of C_1 in G_α is the decomposition group $D_{C_1/C} = \langle\gamma\rangle$, which is cyclic of order f. We call this γ a Frobenius element. If we lift C with another starting point τP for some $\tau \in G_\alpha$, then the Frobenius element is $\tau\gamma\tau^{-1}$. The conjugacy class $[\gamma]$ is the Frobenius conjugacy class at the prime $[C]$, denoted by $\mathrm{Frob}_{[C]}$. Recall that the Galois group G_α is isomorphic to a quotient of $\pi_1(X,*)$.

To the prime $[C]$ we associated the conjugacy class $[\gamma]$ of some primitive element γ in $\pi_1(X,*)$. The Frobenius conjugacy class $\mathrm{Frob}_{[C]}$ is nothing but the image of $[\gamma]$ in G_α.

The following Chebotarev Density Theorem in Dirichlet density for graphs was proved in [**99**, Theorem 22.3].

THEOREM 7.3 (Terras [**99**]). *Let $Y \xrightarrow{\alpha} X$ be a finite Galois unramified cover with Galois group G_α. Suppose the rank of $\pi_1(X,*)$ is at least 2. Let $\mathcal{C}$ be a conjugacy class of G_α. Then the set*

$$S_{\mathcal{C}} = \left\{\text{prime } [C] \text{ of } X : \mathrm{Frob}_{[C]} = \mathcal{C}\right\}$$

has Dirichlet density $\frac{|\mathcal{C}|}{|G_\alpha|}$. In particular, the set of primes of X splitting completely in Y has Dirichlet density

$$\frac{1}{|G_\alpha|} = \frac{1}{\deg(Y \xrightarrow{\alpha} X)}.$$

Here a set S of primes of X is said to have Dirichlet density δ if

$$\sum_{[C]\in S} u^{\ell(C)} \sim \delta \log Z(X,u) \qquad \text{as } u \to (1/\lambda_X)^-,$$

or equivalently,

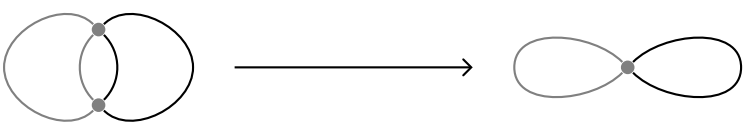
$$\frac{\sum_{[C]\in S} u^{\ell(C)}}{\sum_{[C] \text{ prime}} u^{\ell(C)}} \sim \delta \qquad \text{as } u \to (1/\lambda_X)^-.$$

It is natural to ask whether the Chebotarev Density Theorem would hold in natural density. This was explored by Hashimoto [**35**, Theorem 4(i)]. The following theorem gives a complete answer to this question.

THEOREM 7.4 (Huang-Li [**40**]). *Suppose the rank of $\pi_1(X,*)$ is at least 2. Given a finite Galois cover $Y \xrightarrow{\alpha} X$ with Galois group G_α, the Chebotarev density theorem holds in natural density for one conjugacy class of G_α if and only if it holds for all conjugacy classes of G_α if and only if $\Delta_Y = \Delta_X$, in which case it agrees with the Dirichlet density (Theorem 7.3).*

This result is analogous to the Chebotarev density theorem over function fields (Theorem 1.8) discussed in §5, Lecture 1. We remark that for a finite Galois cover $Y \xrightarrow{\alpha} X$, Terras and Stark [**100**] showed that Δ_Y is either Δ_X or $2\Delta_X$. The proof in [**40**] does not use this information. Before sketching the proof of the theorem, we exhibit below a double unramified cover $Y \to X$ with $\Delta_Y = 2\Delta_X$ and illustrate the key point why the natural density fails to exist for this example adapted from [**40**].

EXAMPLE 7.1. Consider the degree-2 Galois cover $\alpha : Y \to X$, as displayed in the figure below, which preserves the coloring of edges.



Clearly, the fundamental group of X has rank 2, $\Delta_X = 1$ and $\Delta_Y = 2$. The Galois group $G_\alpha = \{\pm id\}$ has two conjugacy classes $\mathcal{C}_+ = \{id\}$ and $\mathcal{C}_- = \{-id\}$. It is easy to see that $S_{\mathcal{C}_+}$ (resp. $S_{\mathcal{C}_-}$) consists of primes of X with even (resp. odd) length, and each set has Dirichlet density 1/2. We claim that neither $S_{\mathcal{C}_+}$ nor $S_{\mathcal{C}_-}$ have natural density. Since the argument is similar, we shall assume that the natural density of $S_{\mathcal{C}_+}$ exists and hence is equal to 1/2, and derive a contradiction.

Since the natural density exists, the following two ratios have the same limit:

$$\begin{aligned} & \lim_{m\to\infty} \frac{\big|\{[C] \in S_{\mathcal{C}_+} \ : \ \ell(C) \le 2m\}\big|}{\big|\{[C] \text{ is a prime of } X \ : \ \ell(C) \le 2m\}\big|} \\ = & \lim_{m\to\infty} \frac{\big|\{[C] \in S_{\mathcal{C}_+} \ : \ \ell(C) \le 2m+1\}\big|}{\big|\{[C] \text{ is a prime of } X \ : \ \ell(C) \le 2m+1\}\big|} = \frac{1}{2}. \end{aligned}$$

As all primes in $S_{\mathcal{C}_+}$ have even length, the subset with length $\le 2m$ agrees with the subset with length $\le 2m+1$, which in turn yields

$$\lim_{m\to\infty} \frac{\big|\{[C] \text{ is a prime of } X \ : \ \ell(C) \le 2m\}\big|}{\big|\{[C] \text{ is a prime of } X : | \ \ell(C) \le 2m+1\}\big|} = 1. \tag{6}$$

On the other hand, the directed edge adjacency matrix T of X is equal to

$$\begin{array}{c} \\ e_1 \\ \overline{e}_1 \\ e_2 \\ \overline{e}_2 \end{array} \begin{array}{c} \begin{array}{cccc} e_1 & \overline{e}_1 & e_2 & \overline{e}_2 \end{array} \\ \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \end{array},$$

where $e_1, \overline{e}_1$ (resp. $e_2, \overline{e}_2$) are the black (resp. gray) edges of X with opposite orientations. We have $\lambda_X = 3$. It then follows from Theorem 7.2 that the limit (6) is equal to 1/3, a contradiction.

Now we sketch the proof of Theorem 7.4. Fix a finite unramified Galois cover $Y \xrightarrow{\alpha} X$ with Galois group G_α. The zeta functions of Y and X are related by (5) in terms of the Artin L-functions $L(X, \rho, u)$ attached to the irreducible representations ρ of G_α. Since Δ_X divides the length of all primes of X, it follows from the definition of the Artin L-function and Theorem 7.1 that each $L(X, \rho, u)^{-1}$ is a polynomial in u^{Δ_X}. Moreover since the primes of Y projects to powers of primes of X, we have $\Delta_X | \Delta_Y$.

For each $1 \le h \le \Delta_Y/\Delta_X$, Hashimoto [**35**] showed that

$$\rho_h(\mathrm{Frob}_{[C]}) = e^{2\pi i h \ell(C)/\Delta_Y}$$

for all primes $[C]$ of X defines a degree-1 representation ρ_h of the Galois group G_α. For each irreducible representation ρ of G_α, its twists by ρ_h, denoted $\rho \otimes \rho_h$, is again an irreducible representation of G_α, and of the same degree as ρ. Further, $L(X, \rho \otimes \rho_h, u) = L(X, \rho, e^{2\pi i h/\Delta_Y} u)$ so that the product over $1 \le h \le \Delta_Y/\Delta_X$ of $L(X, \rho \otimes \rho_h, u)^{-1}$ is a polynomial in u^{Δ_Y}. Rescale the length of edges (to $1/\Delta_X$ instead of 1) so that $\Delta_X = 1$ and Δ_Y is the ratio Δ_Y/Δ_X.

We see from Proposition 7.1.1 that $\lambda_Y \ge \lambda_X$. On the other hand, Hashimoto in [**35**, Lemma 5.1] proved a graph-theoretic analogue of Theorem 3.3 for manifolds. More precisely, he showed that for an irreducible representation ρ of G_α, $L(X, \rho, u)$

is holomorphic and nonvanishing on $|u| \leq 1/\lambda_X$ except for $\rho = \rho_h$ with $1 \leq h \leq \Delta_Y/\Delta_X$, in which case $L(X, \rho_h, u)$ is holomorphic and nonvanishing on $|u| \leq 1/\lambda_X$ except for a simple pole at $u = 1/(\lambda_X e^{2\pi i h/\Delta_Y})$. The poles on the circle $|u| = 1/\lambda_X$ other than $u = 1/\lambda_X$ play the role of "Siegel zeros"as explained in [**100**]. As $Z(Y, u)$ is holomorphic on $|u| < 1/\lambda_Y$ and has a simple pole at $1/\lambda_Y$, this shows that $\lambda_Y = \lambda_X > 1$.

We summarize the above discussion in

PROPOSITION 7.4.1. *Let $Y \xrightarrow{\alpha} X$ be a finite unramified Galois cover with Galois group G_α. Normalize the length of the edges so that $\Delta_X = 1$.*

(1) If $\Delta_Y = \Delta_X$, then for all nontrivial irreducible representations ρ of G_α, the associated Artin L-function $L(X, \rho, u)$ is holomorphic and nonvanishing on an open region containing the closed disc $|u| \leq (\lambda_X)^{-1}$; for ρ trivial, the same holds except for a simple pole at $u = 1/\lambda_X$.

(2) If $\Delta_Y > \Delta_X$, then the analytic behavior stated in (1) holds for all irreducible representations of G_α not equal to ρ_h, $1 \leq h \leq \Delta_Y/\Delta_X$. Further the L-function attached to ρ_h is $L(X, \rho_h, u) = Z(X, ue^{2\pi i h/\Delta_Y})$, which has the same analytic behavior except for a simple pole at $u = 1/(e^{2\pi i h/\Delta_Y}\lambda_X)$.

Note that when $\Delta_Y = \Delta_X$, the analytic behavior stated in Proposition 7.4.1 (1) above is what was called "nice" in §1, Lecture 3 for L-functions in variable s. It implies the Chebotarev density theorem in natural density, as proved by Hashimoto [**35**]. This is a graph-theoretical analogue of Corollary 3.1.2 for manifolds. When $\Delta_Y > \Delta_X$, for any conjugacy class $\mathcal{C}$ of G_α we observe that the lengths of the primes $[C]$ of X with $\mathrm{Frob}_{[C]} = \mathcal{C}$ are congruent modulo Δ_Y/Δ_X and the nonexistence of natural density follows from a similar argument as in the Example 7.1 explained before.

It is worth pointing out that the proof of the Chebotarev density theorem relies on the analytic behavior of the Artin L-functions $L(X, \rho, u)$ for irreducible representations ρ of G_α. For Dirichlet density only information around $u = 1/\lambda_X$ is used, while for natural density one needs information on the circle $|u| = 1/\lambda_X$. In both cases the density theorem holds if the zeta function $Z(X, u)$ is the only function among the $L(X, \rho, u)$ contributing a pole at $u = 1/\lambda_X$ or on $|u| = 1/\lambda_X$, respectively.

7.4. A characterization for finite unramified Galois covers

As indicated in Theorem 1.5, for number fields, a finite Galois extension of $\mathbb{Q}$ is determined by the set of primes of $\mathbb{Q}$ splitting completely in the field. A similar statement also holds for finite Galois covers of a finite graph, namely up to isomorphism, the finite Galois covers $Z \xrightarrow{\alpha} X$ are determined by the set of primes in X which split completely in Z. As a matter of fact, since the graph X is finite, one has a stronger criterion which requires checking the behavior of primes of X of bounded length, as explained below. See [**40**] for more details.

THEOREM 7.5 (Huang-Li [**40**]). *Let $Y \xrightarrow{\alpha} X$ be a Galois cover. Suppose X has n vertices and $\pi_1(X, *)$ has rank ≥ 2. Then two intermediate covers $Y \xrightarrow{\beta} Z \xrightarrow{\gamma} X$ and $Y \xrightarrow{\beta'} Z' \xrightarrow{\gamma'} X$ of α with γ and γ' both finite Galois covers are isomorphic if and only if the set of primes of X of length $\leq (4n-1)d - 1$ which split completely in Z agrees with that in Z'. Here $d = l.c.m.(\deg \gamma, \deg \gamma')$.*

When Y is the universal cover of X, this theorem characterizes all Galois covers of X of finite degree.

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LECTURE 8

Zeta and L-functions of complexes

8.1. The building attached to $PGL_n(F)$

Throughout this lecture F denotes a nonarchimedean local field with q elements in its residue field. Such a field is a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_p((x))$, where q is a power of p. Denote by $\mathcal{O}_F$ the ring of integers of F and by π a uniformizer of $\mathcal{O}_F$, thus the residue field $\mathcal{O}_F/\pi\mathcal{O}_F$ is isomorphic to $\mathbb{F}_q$. Similar to $PGL_2(\mathbb{Q}_p)/PGL_2(\mathbb{Z}_p)$ discussed in §4, Lecture 6, each coset $gPGL_n(\mathcal{O}_F)$ in $PGL_n(F)/PGL_n(\mathcal{O}_F)$ can be interpreted as an equivalence class of rank-n lattices (i.e, $\mathcal{O}_F$-modules) $[L_g]$ in F^n. They are the vertices of the building $\mathcal{B}_n(F)$ attached to $PGL_n(F)$. Two vertices v_1 and v_2 are adjacent if they can be represented by lattices L_1 and L_2 such that $L_1 \supsetneq L_2 \supsetneq \pi L_1 \supsetneq \pi L_2$, as before. Note that $L_1/\pi L_1$ is isomorphic to $(\mathbb{F}_q)^n$ and $L_2/\pi L_1$ is a proper nonzero subspace of $\mathbb{F}_q^n$. Thus a maximal set of mutually adjacent vertices has cardinality n; they form an $(n-1)$-dimensional simplicial complex, called a *chamber*. In other words, vertices v_1,..., v_n form a chamber if they can be represented by lattices $L_1, ..., L_n$ such that

$$L_1 \supsetneq L_2 \supsetneq \cdots \supsetneq L_n \supsetneq \pi L_1,$$

or equivalently,

$$0 \subsetneq L_n/\pi L_1 \subsetneq L_{n-1}/\pi L_1 \subsetneq \cdots \subsetneq L_2/\pi L_1 \subsetneq L_1/\pi L_1 = (\mathbb{F}_q)^n$$

is a complete flag of $\mathbb{F}_q$-subspaces in $(\mathbb{F}_q)^n$. For example, the standard chamber has vertices $[L_{g_i}]$ for $0 \le i \le n-1$, where g_i is the diagonal matrix $\mathrm{diag}(1, ..., 1, \pi, ..., \pi)$ of $\det g_i = \pi^i$. The chambers sharing a fixed vertex correspond bijectively to the complete flags in $(\mathbb{F}_q)^n$. The building $\mathcal{B}_n(F)$, as the union of the chambers, is a contractible $(n-1)$-dimensional simplicial complex. For $0 \le j \le n-1$, the j-dimensional facets of the building $\mathcal{B}_n$ consist of the j-dimensional facets of chambers. The group $PGL_n(F)$ acts on $\mathcal{B}_n(F)$ by left translation, preserving j-dimensional facets for all $0 \le j \le n-1$. Hence we may take discrete torsion-free cocompact subgroups Γ of $PGL_n(F)$ with $\mathrm{ord}_\pi \det \Gamma \subset n\mathbb{Z}$ and obtain finite $(n-1)$-dimensional complexes $X_\Gamma = \Gamma\backslash\mathcal{B}_n(F)$ as before.

Each vertex $[L_g]$ has a type, given by $\mathrm{ord}_\pi \det g \mod n$. Adjacent vertices have different types. In particular, the vertices in a chamber exhaust all possible types exactly once. The 1-skeleton of $\mathcal{B}_n(F)$ is the graph consisting of the vertices and edges in the building; the vertices are partitioned according to their types and the adjacency relation makes it an n-partite graph. There are $n-1$ Hecke operators A_i, $i = 1, ..., n-1$, based on the double cosets $PGL_n(\mathcal{O}_F)g_iPGL_n(\mathcal{O}_F)$; the action of A_i on $f \in L^2(PGL_n(F)/PGL_n(\mathcal{O}_F))$ is given by

$$A_i f(x) = \sum_{\substack{y \text{ adjacent to } x, \\ type\ y = (type\ x)+i \mod n}} f(y).$$

Equivalently, we may express the double coset $PGL_n(\mathcal{O}_F)g_iPGL_n(\mathcal{O}_F)$ as a disjoint union $\cup_{j\in I_i} g_{i,j}PGL_n(\mathcal{O}_F)$ of right $PGL_n(\mathcal{O}_F)$-cosets indexed by a finite set I_i, then

$$A_i f(gPGL_n(\mathcal{O}_F)) = \sum_{j\in I_i} f(gg_{i,j}PGL_n(\mathcal{O}_F)),$$

similar to what happened for the case $n = 2$ discussed in §4, Lecture 6. The operators A_i and A_{n-i} are transpose of each other, and the A_i's commute. Consequently the induced operators A_i on a finite quotient X_Γ are simultaneously diagonalizable.

Each $(n-2)$-dimensional simplex is contained in $q+1$ chambers. Each vertex x of $\mathcal{B}_n(F)$ has

$$q_{n,i} = \frac{(q^n-1)\cdots(q-1)}{(q^i-1)\cdots(q-1)(q^{n-i}-1)\cdots(q-1)}$$

neighbors of type equal to $(type\ x) + i \mod n$. We say that $\mathcal{B}_n(F)$ is $(q+1)$-regular. Topologically the building $\mathcal{B}_n(F)$ is simply connected, hence it serves as the universal cover of its finite left quotients, which are finite simplicial complexes of dimension $n-1$.

8.2. Spectral theory of regular complexes from $\mathcal{B}_n(F)$

MacDonald [**70, 71**] proved that, for $1 \le \ell \le n-1$, the spectrum of A_ℓ on $\mathcal{B}_n(F)$ is $q^{\ell(n-\ell)/2}\Omega_{n,\ell}$, where

$$\Omega_{n,\ell} = \{\sigma_\ell(z_1, ..., z_n) : z_1, ..., z_n \in S^1, z_1\cdots z_n = 1\}$$

and $\sigma_\ell(z_1, ..., z_n) = \sum_{1\le i_1<\cdots<i_\ell\le n} z_{i_1}\cdots z_{i_\ell}$ is the ℓth elementary symmetric polynomial in n-variables. Note that $\Omega_{2,1}$ is the interval $[-2, 2]$. For $n \ge 3$, the spectrum $\Omega_{n,\ell}$ of A_ℓ is no longer real; it is a region in the complex plane invariant under multiplication by the nth roots of unity.

The following Alon-Boppana type theorem holds for finite (left) quotients of the bulding $\mathcal{B}_n(F)$. Let $\{X_j\}$ be a family of $(q+1)$-regular finite quotients of $\mathcal{B}_n(F)$ with $|X_j| \to \infty$ as $j \to \infty$. The operators A_ℓ for $1 \le \ell \le n-1$ defined above act on functions on vertices of X_j. The trivial eigenvalues of A_ℓ are $q_{n,\ell}e^{2\pi ir/n}$ for $r = 1, ..., n$.

THEOREM 8.1 (Li [**62**]). *Assume that each X_j contains a ball isomorphic to a ball in $\mathcal{B}_n(F)$ with radius approaching ∞ as $j \to \infty$. Then, for each $1 \le \ell \le n-1$, the closure of the collection of eigenvalues of $A_\ell(X_j)$ for all $j \ge 1$ contains the region $q^{\ell(n-\ell)/2}\Omega_{n,\ell}$.*

When $n = 2$, this reduces to the Alon-Boppana Theorem 5.1 and its counterpart Theorem 5.4 for graphs.

8.3. Ramanujan complexes as finite quotients of $\mathcal{B}_n(F)$

In view of the definition of Ramanujan graphs for regular and bi-regular bipartite graphs as discussed in Lectures 5 and 6, a finite quotient X of $\mathcal{B}_n(F)$ is called a Ramanujan complex if, for $1 \le \ell \le n-1$, all nontrivial eigenvalues of A_ℓ on X fall in $q^{\ell(n-\ell)/2}\Omega_{n,\ell}$, the spectrum of A_ℓ on $\mathcal{B}_n(F)$. Theorem 8.1 says that a Ramanujan complex is spectrally extremal. Like graphs, there are explicitly constructed families of Ramanujan complexes.

THEOREM 8.2. *For q equal to a prime power and $n \geq 2$, there exist explicitly constructed infinite families of $(q+1)$-regular $(n-1)$-dimensional Ramanujan complexes arising as quotients of $\mathcal{B}_n(F)$.*

The explicit constructions rely on the Ramanujan conjecture established for certain representations of GL_n or the multiplicative group of division algebras over function fields. We sketch the basic approach, which is similar to the explicit construction for Ramanujan graphs by Lubotzky-Phillips-Sarnak explained in §5, Lecture 6. Let $K = \mathbb{F}_q(x)$ be a rational function field such that $F = K_v$ is the completion of K at a degree-1 place v not equal to the place at infinity ∞. Let H be a central simple division algebra of dimension n^2 over K, totally ramified at ∞ and unramified at v. Put $D = H^\times/\text{center}$. Then $D(K_v) \cong PGL_n(F)$ so that the building $\mathcal{B}_n(F)$ is isomorphic to $D(K_v)/D(\mathcal{O}_v)$. By strong approximation theorem, for a compact open subgroup $\mathcal{K}$ of $\prod_{w\neq v,\infty} D(\mathcal{O}_w)$, the double coset space

$$X_{\mathcal{K}} = D(K)\backslash D(\mathbb{A}_K)/D(K_\infty)D(\mathcal{O}_v)\mathcal{K}$$

can be expressed locally at v as

$$X_{\mathcal{K}} = \Gamma_{\mathcal{K}}\backslash D(K_v)/D(\mathcal{O}_v) = \Gamma_{\mathcal{K}}\backslash\mathcal{B}_n(F).$$

Here $\Gamma_{\mathcal{K}} = D(K)\cap\mathcal{K}$ is a discrete cocompact subgroup of $\mathcal{B}_n(F)$. Shrinking $\mathcal{K}$ if necessary we may also assume $\Gamma_{\mathcal{K}}$ is torsion-free and $\text{ord}_\pi \det\Gamma_{\mathcal{K}} \equiv 0 \mod n$ so that it preserves the type of each vertex. As such, $X_{\mathcal{K}}$ is a finite $(q+1)$-regular $(n-1)$-dimensional simplicial complex.

The space of functions on vertices of $X_{\mathcal{K}}$ are automorphic forms on $D(\mathbb{A}_K)$ right invariant by $D(K_\infty)D(\mathcal{O}_v)\mathcal{K}$. It contains the constant functions and their twists by n^{th} roots of unity. They are common eigenfunctions of A_ℓ for $1 \leq \ell \leq n-1$ with trivial eigenvalues $q_{n,\ell}e^{2\pi ir/n}$ for $r = 1, ..., n$. By Jacquet-Langlands correspondence the orthogonal complement $V_{\mathcal{K}}$ of these functions can be interpreted as certain automorphic forms on $PGL_n(\mathbb{A}_K)$ with the same eigenvalues for the Hecke operators A_ℓ for $1 \leq \ell \leq n-1$. As Lafforgue [**55**] proved the Ramanujan conjecture for cuspidal representations for $GL_n(\mathbb{A}_K)$, so we just have to choose appropriate $\mathcal{K}$ so that only cuspidal automorphic forms occur in $V_{\mathcal{K}}$ in order that $X_{\mathcal{K}}$ is Ramanujan. By varying the congruence subgroups $\mathcal{K}$, an infinite family of $(n-1)$-dimensional Ramanujan complexes is obtained.

It should be pointed out that, when $n = 2$, the building $\mathcal{B}_2(F)$ is a $(q+1)$-regular tree and any choice of a compact open subgroup $\mathcal{K}$ of $\prod_{w\neq v,\infty} D(\mathcal{O}_w)$ such that $\Gamma_{\mathcal{K}}$ is torsion-free gives rise to a Ramanujan graph. For $n \geq 3$, Lubotzky, Samuels and Vishne [**69**] showed that the same method yields Ramanujan complexes if n is a prime. When n is a composite, more care is required in order to avoid the residual spectrum of PGL_n to occur in the space $V_{\mathcal{K}}$ since such representations do not satisfy the Ramanujan conjecture. They showed that there are indeed infinitely many $\mathcal{K}$ giving rise to complexes that are not Ramanujan.

There are three explicit constructions of Ramanujan complexes in the literature, by Li [**62**], Lubotzky-Samuels-Vishne [**68**], and Sarveniazi [**86**] from 2004 to 2007. At the time when their works were done, the Jacquet-Langlands correspondence for the multiplicative groups of division algebras of dimension n^2 over function fields was established only for primes n. The proof of the correspondence for all n appeared only in 2016 by Badulescu and Roche [**7**]. Li's construction [**62**] avoided this problem by using the results of Laumon-Rapoport-Stuhler [**59**],

where the Ramanujan conjecture was shown to hold for automorphic representations of $D(\mathbb{A}_K)$ having a local component at a place $w \neq v, \infty$ being Steinberg. Her construction did not use Lafforgue's result [**55**] either, but required the division algebras to ramify at least at 4 places. The other two constructions assumed the validity of the Jacquet-Langlands correspondence and used Lafforgue's result, but could be applied to division algebras totally ramified only at two places. The method in [**68**] closely parallels that in [**67**] by finding a group to represent the vertices of $\mathcal{B}_n(F)$. So did [**86**], which established this fact in a different and elegant way. To show variations of viewpoints, we sketch below Sarveniazi's approach [**86**].

Denote by H the simple central division algebra over $K = \mathbb{F}_q(t)$ of dimension n^2 so that

$$H(K) = \mathbb{F}_{q^n}(t) + \mathbb{F}_{q^n}(t)\tau + \cdots + \mathbb{F}_{q^n}(t)\tau^{n-1},$$

where $\tau^n = t$ and $\tau\alpha = \alpha^q\tau$ for $\alpha \in \mathbb{F}_{q^n}$. Then H is totally ramified at $t = \infty$ with invariant $-1/n$ and at $t = 0$ with invariant $1/n$, and unramified elsewhere.

The group of norm 1 elements in $\mathbb{F}_{q^n}$ is

$$N_1 = \{\alpha \in \mathbb{F}_{q^n} : N_{\mathbb{F}_{q^n}/\mathbb{F}_q}(\alpha) = 1\} = \{\beta^{q-1} : \beta \in \mathbb{F}_{q^n}^\times\}.$$

Here the second expression follows from the Hilbert Theorem 90. It has cardinality

$$|N_1| = \frac{q^n - 1}{q - 1} = q^{n-1} + q^{n-2} + \cdots + q + 1.$$

Observe that, up to multiplication by elements in N_1, $1 - \alpha\tau$ with $\alpha \in N_1$ are the elements in $H(K)$ of reduced norm $1 - t$.

Think of τ as the Frobenius automorphism on $\mathbb{F}_{q^n}$ sending x to x^q and regard a polynomial $\sum_{0\leq i\leq n-1} a_i\tau^i$ with $a_i \in \mathbb{F}_{q^n}$ as an $\mathbb{F}_q$-linear endomorphism on $\mathbb{F}_{q^n}$ which sends x to $\sum_{0\leq i\leq n-1} a_i x^{q^i}$. View $\mathbb{F}_{q^n}$ as an n-dimensional vector space over $\mathbb{F}_q$. Given a complete flag of $\mathbb{F}_q$-vector spaces

$$0 \subset < v_1 > \subset < v_1, v_2 > \subset \cdots \subset < v_1, v_2, ..., v_n > = \mathbb{F}_{q^n},$$

we shall find unique elements $1 - \alpha_1\tau$, $1 - \alpha_2\tau$,..., $1 - \alpha_n\tau$ with $\alpha_i \in N_1$ such that for $1 \leq i \leq n$,

$$< v_1, ..., v_i > \text{ is the kernel of } (1 - \alpha_i\tau)\cdots(1 - \alpha_1\tau) \text{ on } \mathbb{F}_{q^n}.$$

First we note that for any $\beta^{1-q} \in N_1$, the kernel of $1-\beta^{1-q}\tau$ is 1-dimensional, equal to $< \beta >$. This is because $x \in \mathbb{F}_{q^n}^\times$ is such that $(1 - \beta^{1-q}\tau)(x) = x - \beta^{1-q}x^q = 0$ if and only if $x^{q-1} = \beta^{q-1}$, which means that x and β differ by a multiple in $\mathbb{F}_q^\times$. Hence we choose $\alpha_1 = v_1^{1-q}$ so that $< v_1 >$ is the kernel of $1 - \alpha_1\tau$. The image of v_2 under $1 - \alpha_1\tau$ is $v_2 - \alpha_1 v_2^q =: \beta_2$ is a nonzero element in $\mathbb{F}_{q^n}$. Letting $\alpha_2 = \beta_2^{1-q}$, we see that v_2 is annihilated by $(1 - \alpha_2\tau)(1 - \alpha_1\tau)$ so that the kernel of $(1 - \alpha_2\tau)(1 - \alpha_1\tau)$ is precisely $< v_1, v_2 >$. Inductively, suppose we have found $\alpha_1, ..., \alpha_i$ in N_1 so that the kernel of $(1 - \alpha_i\tau)\cdots(1 - \alpha_1\tau)$ is $< v_1, ..., v_i >$ for $1 \leq i < N$, then $\alpha_{i+1} = \beta_{i+1}^{1-q}$ with $\beta_{i+1} = (1 - \alpha_i\tau)\cdots(1 - \alpha_1\tau)(v_{i+1})$ has the desired property that the kernel of $(1-\alpha_{i+1}\tau)\cdots(1-\alpha_1\tau)$ is $< v_1, ..., v_{i+1} >$. Note that the α_i's are uniquely determined by the complete flag.

Since the product $(1-\alpha_n\tau)\cdots(1-\alpha_1\tau) = 1 + \cdots + a_n\tau^n$ is a polynomial in τ of degree n which is the zero map on $\mathbb{F}_{q^n}$, and so is $1-\tau^n = 1-t$. The difference of the two, if nonzero, would yield a nontrivial polynomial relation of degree $\leq n - 1$

in the Frobenius of $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$, contradicting the linear independence of 1, τ,..., τ^{n-1}. Hence we have

$$(1-\alpha_n\tau)\cdots(1-\alpha_1\tau) = 1-t.$$

This shows that a complete flag of $\mathbb{F}_{q^n}$ gives rise to a unique factorization of $1-t$ as a product of n terms of the form $1-\alpha\tau$ with $\alpha \in N_1$. Conversely, any such factorization yields a complete flag with $< v_1, ..., v_i >$ being the kernel of $(1-\alpha_i\tau)\cdots(1-\alpha_1\tau)$. Hence there is a bijection between the complete flags of $\mathbb{F}_{q^n}$ and the factorizations of $1-t$ as a product of n terms of the form $1-\alpha\tau$ with $\alpha \in N_1$.

Now let F be the completion of $K = \mathbb{F}_q(t)$ at the place $t = 1$. Put $D = H^\times/\text{center}$. As H is unramified at $t = 1$, we have $D(F) \cong PGL_n(F)$ and the building $\mathcal{B}_n(F) = D(F)/D(\mathcal{O}_F)$. Let S consist of $1-\alpha_1\tau$, $(1-\alpha_2\tau)(1-\alpha_1\tau)$,..., $(1-\alpha_{n-1}\tau)\cdots(1-\alpha_1\tau)$, where all α_i are in N_1, and each element of S divides $1-t$. To show that S is symmetric, it suffices to prove that given $u = (1-\alpha_i\tau)\cdots(1-\alpha_1\tau)$ in S there is an element $v = (1-\alpha_n\tau)\cdots(1-\alpha_{i+1}\tau)$ in S such that the product vu is $1-t$. Since $1-t$ lies in the center of H, this would imply v and u are inverse to each other in $D(F)$. To find v, write $< v_1 >= ker(1-\alpha_1\tau)$, $< v_1, v_2 >= ker(1-\alpha_2\tau)(1-\alpha_1\tau)$, ..., $< v_1, ..., v_i >= ker\ u$ as we did before. This gives a partial flag $0 \subset < v_1 > \subset < v_1, v_2 > \subset \cdots \subset < v_1, ..., v_i >$ in $\mathbb{F}_{q^n}$, which we complete by adding a choice of subspaces

$$< v_1, ..., v_i, v_{i+1} > \subset < v_1, ..., v_{i+1}, v_{i+2} > \subset \cdots \subset < v_1, ..., v_n >= \mathbb{F}_{q^n}.$$

As shown before, there are unique $\alpha_{i+1}, ..., \alpha_n$ in N_1 such that this complete flag corresponds to the factorization

$$(1-\alpha_n\tau)\cdots(1-\alpha_{i+1}\tau)(1-\alpha_i\tau)\cdots(1-\alpha_1\tau) = 1-t.$$

Then $v = (1-\alpha_n\tau)\cdots(1-\alpha_{i+1}\tau)$ lies in S and $vu = 1-t$, as desired.

Let Λ be the group generated by S. Sarveniazi showed that the 1-skeleton of $\mathcal{B}_n(F)$ is the Cayley graph $Cay(\Lambda, S)$. To see this, recall that all chambers containing a fixed vertex x correspond bijectively to the complete flags in $(\mathbb{F}_q)^n$. By viewing $\mathbb{F}_{q^n}$ as an n-dimensional vector space over $\mathbb{F}_q$, these chambers correspond bijectively to the factorizations of $1-t$ as a product of n factors of the form $1-\alpha\tau$ with $\alpha \in N_1$, as observed before. In other words, given a factorization

$$(1-\alpha_n\tau)(1-\alpha_{n-1}\tau)\cdots(1-\alpha_1\tau) = 1-t,$$

we obtain a chamber with vertices $x_1 = x$, $x_2 = (1-\alpha_1\tau)x_1$, $x_3 = (1-\alpha_2\tau)(1-\alpha_1\tau)x_1 = (1-\alpha_2\tau)x_2$, ..., $x_n = (1-\alpha_{n-1}\tau)x_{n-1}$, and $x_1 = (1-\alpha_n\tau)x_n$.

In this setting the Hecke operators A_ℓ, $1 \le \ell \le n-1$, can be neatly expressed as sending $f \in L^2(\Lambda)$ to

$$A_\ell f(x) = \sum f((1-\alpha_\ell\tau)\cdots(1-\alpha_1\tau)x),$$

where the sum is over all elements $(1-\alpha_\ell\tau)\cdots(1-\alpha_1\tau)$ in S which have reduced norm $(1-t)^\ell$.

Sarveniazi constructed infinite families of finite $(q+1)$-regular Ramanujan complexes by taking the building modulo varying monic irreducible polynomials $f(t) \in \mathbb{F}_q[t]$ of degree d and coprime to t and $t-1$. To simplify our exposition, assume n divides d so that the residue field of $f(t)$ contains the field $\mathbb{F}_{q^n}$. (This resembles the choice of $q \equiv 1 \mod 4$ in the construction of Ramanujan graphs by

Lubotzky-Phillips-Sarnak discussed in §5, Lecture 6.) In other words, the 1-skeleton of the finite $(n-1)$-dimensional Ramanujan simplicial complex is

$$Cay(\Lambda \mod f(t),\ S \mod f(t)).$$

Like in [**67**], he further identified the vertex set as $PGL_n(\mathbb{F}_{q^d})$ if $1-t$ is not an rth power modulo $f(t)$ for any $r|n$ and $r>1$, and $PSL_n(\mathbb{F}_{q^d})$ if $1-t$ is an nth power modulo $f(t)$. If $1-t$ is congruent to an rth power mod $f(t)$ for some $r|n$ and $r>1$, then the group is an index-r subgroup of $PGL_n(\mathbb{F}_{q^d})$ containing $PSL_n(\mathbb{F}_{q^d})$. The element $1-\alpha\tau$ in S is identified with the following $n\times n$ matrix:

$$\begin{pmatrix} 1 & -\alpha & & & & & \\ & 1 & -\alpha^q & & & & \\ & & \cdot & & & & \\ & & & \cdot & & & \\ & & & & \cdot & & \\ & & & & & 1 & -\alpha^{q^{n-2}} \\ -\alpha^{q^{n-1}}t & & & & & & 1 \end{pmatrix} \in PGL_n(\mathbb{F}_{q^d}).$$

The combinatorial meaning and possible applications of Ramanujan complexes have been active research topics in recent years. For instance, it was discovered that explicit constructions of Ramanujan complexes also lead to optimal logical gates for quantum computers on a single qubit [**85**].

8.4. Zeta and L-functions of finite quotients of $\mathcal{B}_3(F)$

Let Γ be a discrete torsion-free cocompact subgroup of $PGL_3(F)$ such that $\operatorname{ord}_\pi \det\Gamma \equiv 0 \bmod 3$. Then left translation by Γ on the building $\mathcal{B}_3(F)$ preserves the types of vertices. The quotient

$$X_\Gamma = \Gamma\backslash\mathcal{B}_3(F) = \Gamma\backslash PGL_3(F)/PGL_3(\mathcal{O}_F)$$

is a finite $(q+1)$-regular 2-dimensional simplicial complex. Our goal is to define the zeta function $Z(X_\Gamma,u)$ of X_Γ such that it has the following nice properties possessed by zeta functions of graphs:

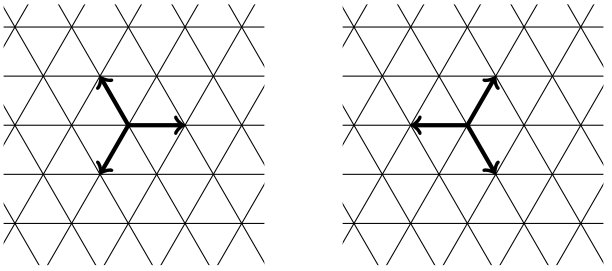
(1) It is a rational function with a closed form expression;

(2) It provides topological and spectral information of X_Γ;

(3) It satisfies the Riemann Hypothesis if and only if X_Γ is Ramanujan.

The first task is to decide what kind of "closed geodesics"to count. The 2-dimensional building $\mathcal{B}_3(F)$ is obtained by gluing together its "apartments "along certain chambers. Each apartment is a Euclidean plane tiled by equilateral triangles, which are chambers. The Euclidean metric on each apartment in turn yields the metric on the building $\mathcal{B}_3(F)$ and thus on the quotient X_Γ. Recall that adjacent vertices have different types. The type of a directed edge from vertex x to vertex y is defined to be $type(y)-type(x)$. A directed path using edges of the same type i is

called a path of type i; the same path traveled in opposite direction has type $3-i$. Out of a vertex in an apartment there are three directed edges of type 1 (bold in the left figure) and three of type 2 (bold in the right figure) as depicted below:



edges of type 1 edges of type 2

A geodesic cycle C of X_Γ is a closed 1-dimensional geodesic path in X_Γ. It has a starting vertex and orientation. It lifts to a geodesic path C' starting at a vertex in $\mathcal{B}_3(F)$, which can be shown to lie in an apartment $\mathcal{A}$ of $\mathcal{B}_3(F)$. Hence it is a straight line in $\mathcal{A}$ starting at a vertex v and ending at a vertex γv for some non-identity element $\gamma \in \Gamma$. To achieve our goals, it turns out that we should only consider those C contained in the 1-skeleton of X_Γ so that C' is a straight line in $\mathcal{A}$ using edges of the same type $i \in \{1,2\}$. Furthermore, a geodesic cycle C is required to remain a geodesic cycle after changing the starting vertex. This is equivalent to v, γv and $\gamma^2 v$ being collinear in $\mathcal{A}$. Such a C is called a geodesic cycle of X_Γ of type i; the same cycle traveled in opposite direction is a geodesic cycle of type $3-i$. As before, C is primitive if it is not obtained by repeating a shorter geodesic cycle more than once. Two geodesic cycles are equivalent if they differ by starting points. Denote by $[C]$ the equivalence class of C. As before, the primes of X_Γ are equivalence classes of primitive geodesic cycles. The geometric length $\ell(C)$ of C is the number of edges in C.

For $n \geq 1$, let $N_n(X_\Gamma)$ denote the number of geodesic cycles in X_Γ of type 1 and geometric length n; it is also equal to that of type 2. For $i = 1, 2$ the type i edge zeta function of X_Γ is defined as

$$Z_{1,i}(X_\Gamma, u) = \exp\big(\sum_{n=1}^{\infty} \frac{N_n(X_\Gamma)}{n} u^n\big) = \prod_{[C] \text{ prime of type } i} \frac{1}{1-u^{\ell(C)}}.$$

Combined, they define the (edge) zeta function for X_Γ:

$$Z(X_\Gamma, u) = Z_{1,1}(X_\Gamma, u) Z_{1,2}(X_\Gamma, u^2) = \prod_{i=1}^{2} \prod_{[C] \text{ prime of type } i} \frac{1}{1-u^{\ell(C)i}}.$$

Given a type 1 edge $v_1 \to v_2$, its type 1 neighbors are defined to be the type 1 edges $v_2 \to v_3$ in $\mathcal{B}_3(F)$ such that the vertices v_1, v_2, v_3 do not form a chamber. Similar to Theorem 4.3, $Z_{1,1}(X_\Gamma, u)^{-1}$ is a polynomial in u which can be expressed in terms of the type 1 edge adjacency matrix T_E of X_Γ:

$$Z_{1,1}(X_\Gamma, u) = \frac{1}{\det(I - T_E u)}.$$

The stabilizer of the type 1 edge from $[L_{I_3}]$ to $[L_{\mathrm{diag}(1,1,\pi)}]$ in the standard chamber is the parahoric subgroup

$$E = \{ \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in PGL_3(\mathcal{O}_F) : \pi | g, \pi | h \}.$$

Thus the type 1 edges in $\mathcal{B}_3(F)$ are parameterized by $PGL_3(F)/E$. The operator T_E can be expressed as a parahoric Hecke operator based on the E-double coset $E\,\mathrm{diag}(1,1,\pi)E$. As type 2 edges are the opposite of type 1 edges, the type 2 adjacency matrix is the transpose T_E^t of T_E, and

$$Z_{1,2}(X_\Gamma, u) = \frac{1}{\det(I - T_E^t u)}.$$

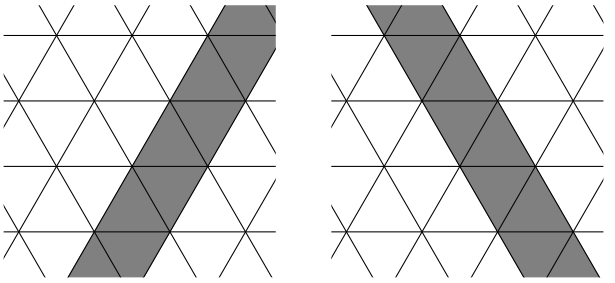
Therefore

$$Z(X_\Gamma, u) = \frac{1}{\det(I - T_E u)\det(I - T_E^t u^2)}.$$

It is much more challenging to find an expression for $Z(X_\Gamma, u)$ analogous to Ihara's theorem for graphs, providing topological information involving the Euler characteristic

$$\chi(X_\Gamma) = \#\text{vertices} - \#\text{edges} + \#\text{chambers}$$

of X_Γ and the spectral information involving the eigenvalues of the Hecke operators A_1 and A_2 acting on functions on vertices of X_Γ. Since X_Γ is 2-dimensional, this will also involve backtrackless sequences of adjacent chambers (i.e. sharing edges) of X_Γ, called *galleries*. We shall only count closed geodesic galleries in X_Γ whose boundaries are geodesic cycles in X_Γ of type 2, as shown in the two examples below.



two geodesic galleries in $\mathcal{B}_3(F)$

Define primitive and equivalent closed geodesic galleries similar to those of cycles. Call the equivalence classes of such primitive closed geodesic galleries 2-dimensional primes of type 2. Note that the same gallery traveled in reverse direction has boundary of type 1, hence the 2-dimensional primes of type 1 are the opposite of those of type 2. The length of a gallery is the number of chambers in the sequence. Denote by $M_n(X_\Gamma)$ the number of closed geodesic galleries in X_Γ of length n. Define the gallery zeta function of X_Γ to be

$$Z_{2,2}(X_\Gamma, u) = \exp\Big(\sum_{n=1}^{\infty} \frac{M_n(X_\Gamma)}{n} u^n\Big) = \prod_{2-\dim' l \text{ prime } [C] \text{ of type } 2} \frac{1}{1 - u^{\ell(C)}}.$$

The Iwahori subgroup

$$B = \{ \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in PGL_3(\mathcal{O}_F) : \pi|d, \pi|g, \pi|h\}$$

is the stabilizer of the three vertices of the standard chamber, hence it stabilizes a chamber together with an orientation, called a *directed chamber*, which is expressed as an ordered triple of the three vertices of the chamber up to cyclic permutation. So $PGL_3(F)/B$ parameterizes all directed chambers in the building $\mathcal{B}_3(F)$. Define the out-neighbors of a directed chamber (v_1, v_2, v_3) to be the directed chambers (v_2, v_3, v_4) in $\mathcal{B}_3(F)$ with $v_4 \neq v_1$. Denote by T_B the adjacency matrix of directed chambers of X_Γ. It is an Iwahori Hecke operator based on the B-double coset $B \begin{pmatrix} 0 & 1 & 0 \\ \pi & 0 & 0 \\ 0 & 0 & \pi \end{pmatrix} B$. Then, expression like Theorem 4.3 holds for the gallery zeta function as well:

$$Z_{2,2}(X_\Gamma, u) = \frac{1}{\det(I - T_B u)}.$$

Now we are ready to state our main theorem for the zeta function of a finite quotient of $\mathcal{B}_3(F)$.

THEOREM 8.3 (Kang-Li [**46**], Kang-Li-Wang [**48**], Kang-Li [**47**]). *The edge zeta function for X_Γ is a rational function in u which can be expressed in two ways:*

$$Z(X_\Gamma, u) = \frac{1}{\det(I - T_E u)\det(I - T_E^t u^2)} \tag{2}$$

$$= \frac{(1-u^3)^{\chi(X_\Gamma)}}{\det(I - A_1 u + A_2 q u^2 - q^3 u^3)\det(I + T_B u)}. \tag{3}$$

Here $\chi(X_\Gamma)$ is the Euler characteristic of X_Γ, A_1 and A_2 are Hecke operators, T_E is a parahoric Hecke operator, and T_B is an Iwahori Hecke operator introduced above.

This theorem has three proofs, revealing different interpretations of the identity. In [**46**] the numbers $N_n(X_\Gamma)$ and $M_n(X_\Gamma)$ were computed and compared in terms of contributions from elements in Γ up to conjugation. The identity was first discovered from these computations. The second proof [**48**] is representation-theoretic. The authors computed the eigenvalues of the operators T_E, T_B, A_1 and A_2 and compared them. The third proof [**47**] is cohomological, analogous to that for graphs but more involved.

The zeta function $Z(X_\Gamma, u)$ is said to satisfy the Riemann Hypothesis if all nontrivial zeros of $\det(I - A_1 u + A_2 q u^2 - q^3 u^3)$ from the nontrivial eigenvalues of A_1 and A_2 have the same absolute value q^{-1}. This happens if and only if X_Γ is Ramanujan. It was shown in [**48**, Theorem 2] that the Ramanujan condition has two more equivalent statements in terms of the operators T_E and T_B, respectively.

THEOREM 8.4 (Kang-Li-Wang [**48**]). *The following statements are equivalent:*

(1) X_Γ is Ramanujan;

(2) The nontrivial zeros of $\det(I - A_1 u + A_2 q u^2 - q^3 u^3)$ have the same absolute value q^{-1};

(3) The nontrivial zeros of $\det(I - T_E u)$ have absolute values q^{-1} and $q^{-1/2}$;

(4) The nontrivial zeros of $\det(I - T_B u)$ have absolute values 1, $q^{-1/2}$, and $q^{-1/4}$.

This is reminiscent of the Riemann Hypothesis for the zeta function $Z(V,u)$ attached to a smooth irreducible projective surface V defined over a finite field $\mathbb{F}_q$ discussed in §5 of Lecture 1. Recall that the numerator and denominator of $Z(V,u)$ are decomposed into products $P_1(V,u)P_3(V,u)$ and $P_0(V,u)P_2(V,u)P_4(V,u)$, respectively, where $P_i(V,u) = \det(I - \Phi^{(i)}u)$ is the determinant of the induced Frobenius operator $\Phi^{(i)}$ acting on the finite-dimensional vector space, the ith étale cohomology $H^i(V)$ of V, and the roots of $P_i(V,u) \neq 1$ all have the same absolute value $q^{-i/2}$. The above theorem is extended to $PGL_n(F)$ by First in [**24**].

Let $K = \mathbb{F}_q(T)$ be a rational function field and let v be a degree-1 place of K so that the completion K_v of K at v is isomorphic to $F = \mathbb{F}_q((T))$. Similar to what we saw for curves, in [**59**] Laumon, Rapoport and Stuhler proved the existence of certain moduli surfaces V over K which have good reduction at v and the existence of suitably chosen discrete torsion-free cocompact subgroups $\Gamma(V)$ of $PGL_3(F)$ arising from certain division algebras of dimension 9 over K unramified at v so that in the zeta function $Z(X_{\Gamma(V)},u)$ of the complex $X_{\Gamma(V)} = \Gamma(V)\backslash PGL_3(F)/PGL_3(\mathcal{O}_F)$, the factor

$$\frac{\det(I - A_1(X_{\Gamma(V)})u + A_2(X_{\Gamma(V)})qu^2 - q^3u^3)}{\prod_{j=1}^{3}(1-\mu_3^j u)(1-\mu_3^j qu)(1-\mu_3^j q^2u)}$$

is equal to the factor $P_2(V,u)$ occurring in the zeta function of V mod v. Here μ_3 is a primitive cubic root of unity and the factor in the denominator comes from trivial eigenvalues of A_1 and A_2.

Another connection with number theory is that the factor

$$\frac{1}{\det(I - A_1(X_\Gamma)u + A_2(X_\Gamma)qu^2 - q^3u^3)}$$

occurring in the zeta function $Z(X_\Gamma,u)$ of X_Γ is a Langlands L-function. To explain this, we begin with the definition of Langlands L-functions. A reductive group G over F has a Langlands dual group $G^\vee$ over $\mathbb{C}$. Each irreducible unramified representation σ of $G(F)$ is induced from an unramified character of the maximal torus of $G(F)$. To σ we associate a conjugacy class $\{c(\sigma)\}$ of $G^\vee(\mathbb{C})$. The Langlands L-function of σ with respect to a chosen representation r of $G^\vee(\mathbb{C})$ is defined as

$$L(\sigma,r,u) = \frac{1}{\det(I - r(c(\sigma))u)}.$$

Given a discrete torsion-free cocompact subgroup Γ of $G(F)$, the Langlands L-function of $\Gamma\backslash G(F)$ with respect to r is

$$L(\Gamma\backslash G(F),r,u) = \prod_\sigma L(\sigma,r,u),$$

where σ runs through all unitary irreducible unramified representations of $G(F)$ occurring in the space $L^2(\Gamma\backslash G(F))$, counting multiplicity.

The Langlands dual group of $PGL_3(F)$ is $SL_3(\mathbb{C})$. We choose r to be the standard representation r_{std} of $SL_3(\mathbb{C})$, namely the identity map from $SL_3(\mathbb{C})$ to itself, viewed as a 3-dimensional representation of $SL_3(\mathbb{C})$. A unitary unramified irreducible representation σ occurring in $L^2(\Gamma\backslash PGL_3(F))$ is a principal series representation induced from three unramified characters χ_1, χ_2, and χ_3

of $F^\times$ with $\chi_1\chi_2\chi_3 = 1$. The conjugacy class attached to such a σ is the set $\{\chi_1(\pi), \chi_2(\pi), \chi_3(\pi)\}$, called the Satake parameter of σ. Thus

$$L(\sigma, r_{\text{std}}, u) = \frac{1}{(1-\chi_1(\pi)u)(1-\chi_2(\pi)u)(1-\chi_3(\pi)u)}.$$

One verifies that the Langlands L-function for $\Gamma\backslash PGL_3(F)$ is

$$L(\Gamma\backslash PGL_3(F), r_{\text{std}}, qu) = \frac{1}{\det(I - A_1(X_\Gamma)u + A_2(X_\Gamma)qu^2 - q^3u^3)}.$$

The group Γ is the fundamental group of X_Γ. Let ρ be a d-dimensional unitary representation of Γ. The Artin L-function for 1-dimensional complexes of X_Γ is defined as

$$L_1(X_\Gamma, \rho, u) = \prod_{[C] \text{ prime of type } 1} \frac{1}{\det(I_d - \rho(\text{Frob}_{[C]})u^{\ell(C)})}.$$

Here, given a prime $[C]$, lift C to a path C' in $\mathcal{B}_3(F)$ starting at a vertex P, then the end point of C' is γP for some $\gamma \in \Gamma$. The conjugacy class of γ is denoted $\text{Frob}_{[C]}$, called the Frobenius conjugacy class at the prime $[C]$. Unlike the case of graphs, there is no good algebraic description of such γ. More precisely, conjugacy classes of primitive elements in Γ do give rise to $\text{Frob}_{[C]}$ at primes $[C]$; on the other hand, there are $\text{Frob}_{[C]}$ which are not conjugacy classes of primitive elements. See [**46**] for more detail. Similarly one defines the Artin L-function $L_2(X_\Gamma, \rho, u)$ for 2-dimensional complexes of X_Γ as a product over equivalence classes of primitive closed geodesic galleries whose boundaries are geodesic cycles of type 2. Then each Artin L-function is a rational function in u and expressions similar to the zeta functions attached to X_Γ hold for Artin L-functions. More precisely, we have

THEOREM 8.5 (Kang-Li [**47**]). *Let ρ be a d-dimensional unitary representation of Γ. Then*

(1) *For $i = 1, 2$, the Artin L-function $L_i(X_\Gamma, \rho, u)$ is invariant under induction.*
(2) *There exist operators $T_E(\rho)$ and $T_B(\rho)$ such that*

$$L_1(X_\Gamma, \rho, u) = \frac{1}{\det(I - T_E(\rho)u)}, \quad L_2(X_\Gamma, \rho, u) = \frac{1}{\det(I - T_B(\rho)u)},$$

and

$$(1-u^3)^{\chi(X_\Gamma)d} L(\text{Ind}_\Gamma^{PGL_3(F)} \rho, r_{\text{std}}, qu) = \frac{L_1(X_\Gamma, \rho, u)L_1(X_\Gamma, \rho, u^2)}{L_2(X_\Gamma, \rho, -u)}.$$

Here the Langlands L-function is the product of the Langlands L-functions of the irreducible representations, counting multiplicity, occurring in the induced representation $\text{Ind}_\Gamma^{PGL_3(F)} \rho$ of $PGL_3(F)$. This theorem is proved using cohomological method, similar in spirit to the argument in §4, Lecture 4 but a lot more complicated.

8.5. Zeta functions of finite quotients of the building $\Delta(F)$ of $Sp_4(F)$

Let V be a 4-dimensional vector space over F equipped with the nondegenerate alternating bilinear form $\langle\ ,\rangle$, whose values on the standard basis $\{e_1, e_2, f_1, f_2\}$ of V are given by

$$\langle e_1, f_2\rangle = \langle e_2, f_1\rangle = 1$$

and

$$\langle e_i, f_i\rangle = \langle e_i, e_i\rangle = \langle f_i, f_i\rangle = 0 \qquad \text{for } i \in \{1, 2\}.$$

The linear transformations on V preserving the bilinear form $\langle\ ,\rangle$ constitute the symplectic group $Sp_4(F)$. Its associated building $\Delta(F)$ is a contractible 2-dimensional simplicial complex. Each apartment of $\Delta(F)$ is an Euclidean plane, tiled by isosceles right triangles. Each isosceles right triangle is a chamber, the two vertices on its hypotenuse are special vertices, and the third vertex is nonspecial. Depicted in Figure 4 is the standard apartment $\mathcal{A}$ in which the vertices are described using the lattice model.[3] More precisely, a vertex marked by $[a, b, c, d]$ represents the equivalence class of the rank-4 lattice with a basis $\pi^a e_1$, $\pi^b e_2$, $\pi^c f_1$, and $\pi^d f_2$. The special vertices are at the crossing of solid lines and the remaining vertices are nonspecial.

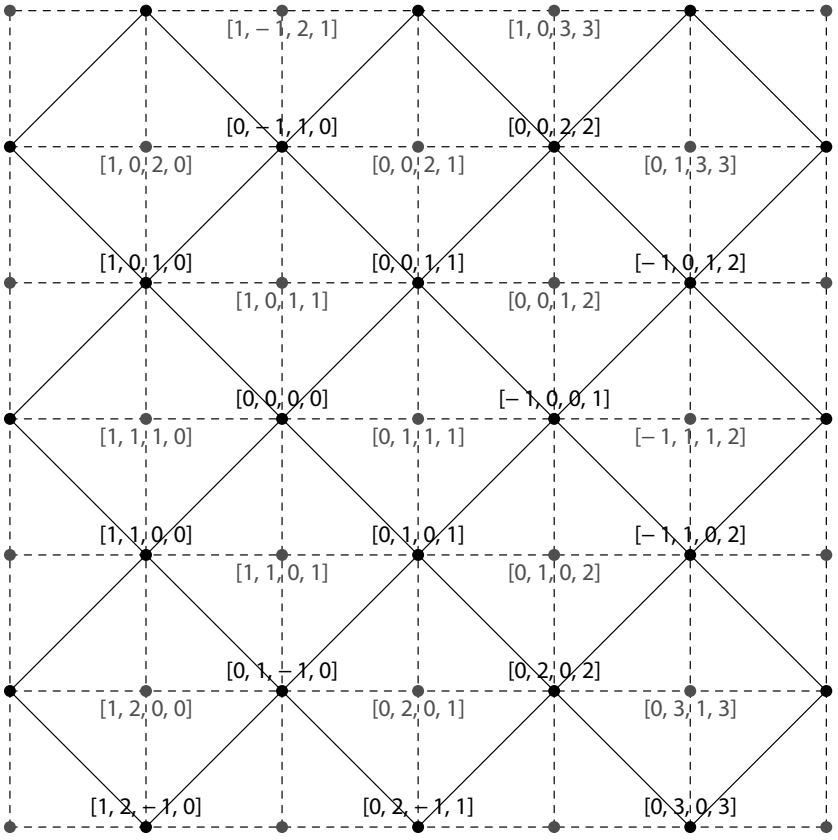


FIGURE 4. The standard apartment $\mathcal{A}$ of $\Delta(F)$

[3]It is Fig. 1 in [**22**], used by permission of Oxford University Press.

The group of symplectic similitudes $GSp_4(F)$ preserves the bilinear form $\langle\ ,\rangle$ up to scalar multiples. Modulo its center, the quotient group $PGSp_4(F)$ acts transitively on the special vertices of $\Delta(F)$ and preserves nonspecial vertices. As the stabilizer of the special vertex $[0,0,0,0]$ is $PGSp_4(\mathcal{O}_F)$, the cosets in $PGSp_4(F)/PGSp_4(\mathcal{O}_F)$ parameterize the special vertices on the building $\Delta(F)$. Out of a special vertex, there are two kinds of directed edges: those between two special vertices are of type 1 and those between a special and a nonspecial vertices are of type 2. The type 1 edges are parameterized by the cosets of the Siegel congruence subgroup P_1 of $PGSp_4(\mathcal{O}_F)$ and the type 2 ones by the cosets of the Klingen congruence subgroup P_2 of $PGSp_4(\mathcal{O}_F)$. The cosets of the Iwahori subgroup $I = P_1 \cap P_2$ parameterize the directed chambers of $\Delta(F)$. For example, on the standard apartment $\mathcal{A}$, the 4 vectors out of $v = [0,0,0,0]$ ending at $[0,0,1,1]$, $[1,0,1,0]$, $[1,1,0,0]$, $[0,1,0,1]$ are type 1 edges, and the 4 vectors ending at $[0,1,1,1]$, $[1,0,1,1]$, $[1,1,1,0]$, $[1,1,0,1]$ are type 2 edges.

Two vertex adjacency operators A_1 and A_2 act on special vertices: A_1 is based on the double coset $PGSp_4(\mathcal{O}_F)\,\mathrm{diag}(1,1,\pi,\pi)PGSp_4(\mathcal{O}_F)$ and A_2 is q^2+1 times the identity operator plus the operator based on the double coset $PGSp_4(\mathcal{O}_F)\,\mathrm{diag}(1,\pi,\pi,\pi^2)PGSp_4(\mathcal{O}_F)$. For $i=1,2$, the type i edge adjacency operator L_{P_i} are based on the double cosets $P_1\,\mathrm{diag}(1,1,\pi,\pi)P_1$ and $P_2\,\mathrm{diag}(1,\pi,\pi,\pi^2)P_2$, respectively. We shall be concerned with two directed chamber adjacency operators L_I based on the double coset ItI with $t = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -\pi & 0 & 0 \\ 0 & 0 & 0 & \pi \end{pmatrix}$ and L'_I based on the double coset $It'I$ with $t' = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \pi^{-1} \\ \pi & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$. See [**22**, **49**] for more details.

Let Γ be a discrete torsion-free cocompact subgroup of $PGSp_4(F)$ such that $\mathrm{ord}_\pi \det\Gamma \equiv 0 \mod 4$. Then Γ preserves the types of vertices and $X_\Gamma = \Gamma\backslash\Delta(F)$ is a finite 2-dimensional simplicial complex. Define 1-dimensional primes of type i and the zeta function of X_Γ counting the closed geodesic cycles contained in the 1-skeleton of X_Γ using edges of the same type as in the previous section:

$$Z(X_\Gamma, u) = \prod_{i=1}^{2} \prod_{1-\mathrm{dim'l\ prime}\ [C]\ \mathrm{of\ type}\ i} \frac{1}{1-u^{\ell(C)i}}.$$

Similar to Theorem 8.3 for finite quotients of the building $\mathcal{B}_3(F)$, we have two closed form expressions for the zeta function of X_Γ in terms of the various operators introduced above.

THEOREM 8.6 (Fang-Li-Wang [**22**]). *The zeta function $Z(X_\Gamma, u)$ defined above is a rational function in u with two expressions:*

$$\begin{aligned} Z(X_\Gamma, u) &= \frac{1}{\det(I - L_{P_1}u)} \frac{1}{\det(I - L_{P_2}u^2)} \\ &= \frac{(1-u^2)^{\chi(X_\Gamma)}(1-q^2u^2)^{-(q^2-1)N(\Gamma)}}{\det(I - A_1u + qA_2u^2 - q^3A_1u^3 + q^6Iu^4)\det(I - L_Iu)}. \end{aligned}$$

Here $\chi(X_\Gamma) = \#vertices - \#edges + \#chambers$ in X_Γ is the Euler characteristic of X_Γ, and $2N(\Gamma)$ is the number of special vertices in X_Γ.

The dual group of $PGSp_4(F)$ is $\mathrm{Spin}_5(\mathbb{C}) \cong Sp_4(\mathbb{C})$, which has a degree 4 spin representation r_{spin} and a degree 5 standard representation r_{std}. A unitary irreducible representation σ of $PGSp_4(F)$ occurring in $L^2(\Gamma\backslash PGSp_4(F))$ is induced from $\chi_1 \times \chi_2 \rtimes \tau$, where χ_1, χ_2, and τ are unramified characters of $F^\times$ satisfying $\chi_1\chi_2\tau^2 = 1$. The Langlands L-function of σ with respect to r_{spin} is

$$L(\sigma, r_{\mathrm{spin}}, u) = \frac{1}{(1-\chi_1\tau(\pi)u)(1-\chi_2\tau(\pi)u)(1-\chi_1\chi_2\tau(\pi)u)(1-\tau(\pi)u)}$$

and that with respect to r_{std} is

$$\begin{aligned} & L(\sigma, r_{\mathrm{std}}, u) \\ = \quad & \frac{1}{(1-\chi_1(\pi)u)(1-\chi_2(\pi)u)(1-\chi_1^{-1}(\pi)u)(1-\chi_2^{-1}(\pi)u)(1-u)}. \end{aligned}$$

Analogous to the case of $PGL_3(F)$, one verifies that the zeta function $Z(X_\Gamma, u)$ is related to the Langlands L-function of $\Gamma\backslash PGSp_4(F)$ with respect to r_{spin} as follows:

$$L(\Gamma\backslash PGSp_4(F), r_{\mathrm{spin}}, q^{\frac{3}{2}}u) = \frac{1}{\det(I - A_1u + qA_2u^2 - q^3A_1u^3 + q^6Iu^4)}.$$

It is natural to ask whether there is a combinatorial zeta function attached to X_Γ which is related to the Langlands L-function with respect to the standard representation r_{std} of the dual group $\mathrm{Spin}_5(\mathbb{C})$. To answer this, introduce another zeta function

$$Z'(X_\Gamma, u) = \prod_{i=1}^{2} \prod_{1-\dim' \mathrm{l\ prime\ } [C] \mathrm{\ of\ type\ } i} \frac{1}{(1-u^{\ell(C)})^{2/i}}.$$

In [**49**] Kang-Li-Wang obtained a closed form expression of this zeta function, which is more complicated than the theorem above. To describe it, let $A_2' = A_2 - 2q^2I$ and $A_2'' = A_2 - q^2I$.

THEOREM 8.7 (Kang-Li-Wang [**49**]).

$$\begin{aligned} & Z'(X_\Gamma, u) = \frac{1}{\det(I - L_{P_1}u)^2} \frac{1}{\det(I - L_{P_2}u)} \\ = \quad & \frac{(1-u)^2(1+u)^{2\chi(X_\Gamma)-2}(1-qu)^{t_1}(1+qu)^{t_2}}{\det(I - A_2'u + (qA_1^2 - 2q^2A_2'')u^2 - q^4A_2'u^3 + q^8Iu^4)\det(I + L_I'u)}. \end{aligned}$$

Here t_1 and t_2 are integers determined by representations of certain types occurring in $L^2(\Gamma\backslash PGSp_4(F)/I)$. They sum to $-2(q^2-1)N(\Gamma)$, where $2N(\Gamma)$ is the number of special vertices in X_Γ. For detailed expression, see [**49**, Theorem 6.3].

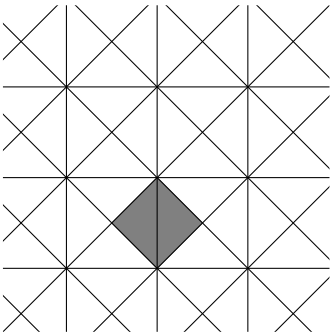
It is related to the Langlands L-function by

$$\begin{aligned} & (1-q^2u)^{2N(\Gamma)} L(\Gamma\backslash PGSp_4(F), r_{\mathrm{std}}, q^2u) \\ = \quad & \frac{1}{\det(I - A_2'u + (qA_1^2 - 2q^2A_2'')u^2 - q^4A_2'u^3 + q^8Iu^4)}. \end{aligned}$$

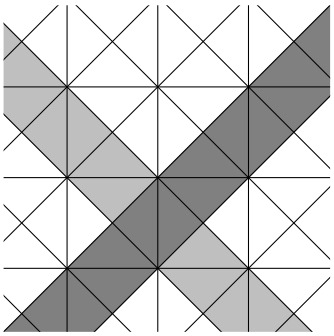
It should be pointed out that, like what we saw in §8.4 for the zeta functions attached to finite quotients of $\mathcal{B}_3$, the factors $\frac{1}{\det(I - L_I u)}$ occurring in $Z(X_\Gamma, u)$ in

Theorem 8.5 and $\frac{1}{\det(I-L'_I u)}$ in $Z'(X_\Gamma, u)$ in Theorem 8.6 each arise from counting closed geodesic galleries of spin type and of standard type, respectively. More precisely, a chamber of spin type is a "square"in an apartment obtained by gluing together two chambers which share a type 1 edge. Note that the boundaries of this square are four type 2 edges. A sequence of adjacent chambers of spin type with boundry a geodesic path of type 2 edges is called a geodesic gallery of chambers of spin type. Then $\frac{1}{\det(I-L_I u)}$ counts closed geodesic galleries of spin type in X_Γ. A chamber of standard type is a "square"in an apartment obtained by gluing together four chambers along type 2 edges which they share so that the boundaries of the square consist of four type 1 edges. Similarly $\frac{1}{\det(I-L'_I u)}$ counts closed geodesic galleries of standard type in X_Γ.

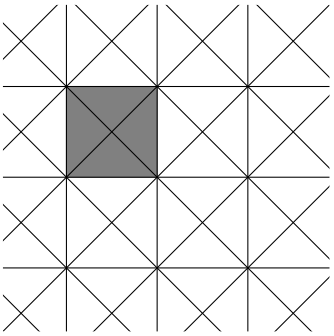
The figures below depict a chamber of spin/standard type and 2 geodesic galleries of chambers of the same type containing it:



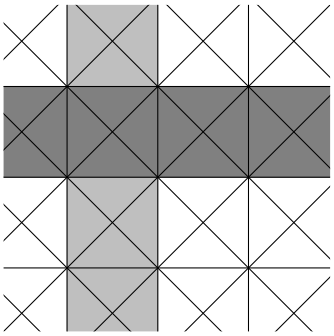
a spin type chamber



2 spin type geodesic galleries



a standard type chamber



2 standard type geodesic galleries

As before, call a 2-dimensional prime of spin/standard type an equivalence class of primitive closed geodesic galleries of spin/standard type. Then $\frac{1}{\det(I-L_I u)}$ (resp. $\frac{1}{\det(I-L'_I u)}$) is a product of $\frac{1}{1-u^{\ell(C)}}$ over 2-dimensional primes $[C]$ of spin (resp. standard) type. Here the length $\ell(C)$ of $[C]$ is the number of chambers of spin (resp. standard) type contained in C. The reader is referred to [**49**] for details.

8.6. Distribution of primes in finite quotients of $\mathcal{B}_3(F)$ and $\Delta(F)$

For a 2-dimensional complex X obtained as a finite quotient of the building $\mathcal{B}_3(F)$ or $\Delta(F)$, we defined 1-dimensional (resp. 2-dimensional) primes which are equivalence classes of primitive closed geodesic cycles (resp. galleries) in X of a

given type. In each case there is a zeta function defined as a product over the primes in question, and this zeta function is the reciprocal of a polynomial, of the form $\frac{1}{\det(I-Tu)}$ for a suitable adjacency operator T. Parallel to the case of graphs, very recently Li and Matias [**64**] obtain the "Prime Geodesic Theorem" for 1- and 2-dimensional primes of the given type by studying the analytic behavior of the associated zeta function, or equivalently, the eigenvalues of T. The key is to prove that given two directed edges/chambers of given type, there is a geodesic path of the same type and dimension in X connecting them. This result leads to the determination of the greatest common divisor δ of the length of primes of given type and dimension. Then using the Perron-Frobenius theorem, they determine the largest eigenvalue in absolute value, λ, of T, show that there are δ distinct eigenvalues of T with absolute value λ, and each occurs with multiplicity κ. Their result is summarized below.

THEOREM 8.8 (Prime geodesic theorem for 2-dimensional complexes [**64**]). *Let X be a finite quotient of the building $\mathcal{B}_3(F)$ of $PGL_3(F)$ or $\Delta(F)$ of $PGSp_4(F)$ by a discrete torsion-free cocompact subgroup as before. Given $i \in \{1,2\}$ and a type, let δ, λ and κ be as described above. Then for integers n large, we have*

$$\#\{i-dim'l\ prime\ [C]\ of\ X\ of\ given\ type : \ell(C) = n\delta\} \sim \frac{\kappa\lambda^{n\delta}}{n},$$

and

$$\#\{i-dim'l\ prime\ [C]\ of\ X\ of\ given\ type : \ell(C) < n\delta\} \sim \frac{\kappa\lambda^{n\delta}}{(\lambda^{\delta}-1)n},$$

where the values of λ, δ and κ are as follows:

building	$\mathcal{B}_3(F)$	$\mathcal{B}_3(F)$	$\Delta(F)$	$\Delta(F)$	$\Delta(F)$	$\Delta(F)$
dim i	1	2	1	1	2	2
type	1 *or* 2	1 *or* 2	1	2	*spin*	*standard*
λ	q^2	q	q^3	q^4	q^2	q^3
δ	3	3	2	2	2	1
κ	1	1	1	2	1	2

The Prime Geodesic Theorem provides an estimate of the number of primes of X of given dimension and type with given length. In the case that X is a Ramanujan complex as discussed in §3, this estimate has the best possible error term $O(\frac{\lambda^{n\delta/2}}{n})$, following [**48**, Theorem 2] reviewed in §4.

Given a finite unramified Galois cover $Y \to X$ with Galois group G, to each i-dimensional prime $[C]$ of X of a given type, where $i = 1$ or 2, we associate a conjugacy class in G, called $\mathrm{Frob}_{[C]}$, the same way as we did for graphs. Define, for each conjugacy class $\mathcal{C}$ of G, the set

$$S_{\mathcal{C}} = \{i-\text{dimensional prime } [C] \text{ of } X \text{ of given type} : \mathrm{Frob}_{[C]} = \mathcal{C}\}.$$

In other words, using the conjugacy classes of G we partition the i-dimensional primes of X of a given type, and we may ask a finer distribution question about these primes, namely whether each set $S_{\mathcal{C}}$ has a density. By studying the analytic behavior of the Artin L-functions $L(X,\rho,u)$ attached to finite-dimensional unitary irreducible representations ρ of G, Li and Matias proved in [**64**] that $L(X,\rho,u)$ is holomorphic on the closed disk $|u| \le 1/\lambda$ for all nontrivial ρ. Here λ is as in

Theorem 8.8. From this they conclude that the Chebotarev density theorem in natural density holds for the cover $Y \to X$.

THEOREM 8.9 (Chebotarev density theorem for 2-dimensional complexes [**64**]). *Let X be as in Theorem 8.8. Let $Y \to X$ be a finite unramified Galois cover with Galois group G. Then the Chebotarev density theorem holds in natural density for i-dimensional primes ($i \in \{1, 2\}$) of X of a given type. Specifically, for each conjugacy class $\mathcal{C}$ of G, the set $S_{\mathcal{C}}$ defined above has natural density $\frac{|\mathcal{C}|}{|G|}$.*

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